

On the face which appears in the orb of the moon



920 1 . . . These were Sulla's words. "For it concerns my story and that is its source; but I think that I should first like to learn whether there is any need to put back for a fresh start to those opinions concerning the face of the moon which are current and on the lips of everyone. "What else would you expect us to have done," I said, "since it was the difficulty in these opinions that drove us from our course upon those others? As people with chronic diseases when they have despaired of ordinary remedies and customary regimens turn to expiations and amulets and dreams, just so in obscure and perplexing speculations, when the ordinary and reputable and customary accounts are not persuasive, it is necessary to try those that are more out of the way and not scorn them but literally to chant over ourselves the charms of the ancients and use every means to bring the truth to test.

2 1 Well, to begin with, you see that it is absurd to call the figure seen in the moon an affection of vision in its feebleness giving way to brilliance, a condition which we call bedazzlement. Anyone who asserts this does not observe that this phenomenon should rather have occurred in relation to the sun, since the sun lights upon us keen and violent (as Empedocles too somewhere not infelicitously renders the difference of the two:

The sun keen-shafted and the gentle moon,

referring in this way to her allurements and cheerfulness and harmlessness), and moreover does not explain why dull and weak eyes discern no distinction of shape in the moon but her orb for them has an even and full light, whereas those of keen and robust vision make out more precisely and distinctly the pattern of facial features and more clearly perceive the variations. In fact the contrary, I think, should have been the case if the image resulted from an affection of the eye when it is overpowered: the weaker the subject affected, the clearer should be the appearance of the image. The unevenness also entirely refutes the hypothesis, for the shadow that one sees is not continuous and confused but is not badly depicted by the words of Agesianax:

She gleams with fire encircled, but within
Bluer than lapis show a maiden's eye
And dainty brow, a visage manifest.

In truth, the dark patches submerge beneath the bright ones which they encompass and confine them, being confined and curtailed by them in turn; and they are thoroughly

intertwined with each other so as to make the delineation of the figure resemble a painting. This, Aristotle, seemed to be a point not without cogency against your Clearchus also. For the man is yours, since he was an associate of the ancient Aristotle, although he did pervert many doctrines of the School.”

31 Apollonides broke in and inquired what the opinion of Clearchus was. “You are the last person,” I said, “who has any right not to know a theory of which geometry is, as it were, the very hearth and home. The man, you see, asserts that what is called the face consists of mirrored likenesses, that is images of the great ocean reflected in the moon, ⁹²¹ for the visual ray when reflected naturally reaches from many points objects which are not directly visible and the full moon is itself in uniformity and lustre the finest and clearest of all mirrors. Just as you think, then, that the reflection of the visual ray to the sun accounts for the appearance of the rainbow in a cloud where the moisture has become somewhat smooth and condensed, so Clearchus thought that the outer ocean is seen in the moon, not in the place where it is but in the place whence the visual ray has been deflected to the ocean and the reflection of the ocean to us. So Agesianax again has somewhere said:

Or swell of ocean surging opposite
Be mirrored in a looking-glass of flame.”

41 Apollonides was delighted. “What an original and absolutely novel contrivance the hypothesis is,” he said, “the work of a man of daring and culture; but how did you proceed to bring your counter-argument against it?” “In the first place,” I said, “in that, although the outer ocean is a single thing, a confluent and continuous sea, the dark spots in the moon do not appear as one but having something like isthmuses between them, the brilliance dividing and delimiting the shadow. Hence, since each part is separated and has its own boundary, the layers of light upon shadow, assuming the semblance of height and depth, have produced a very close likeness of eyes and lips. Therefore, one must assume the existence of several outer oceans separated by isthmuses and mainlands, which is absurd and false; or, if the ocean is single, it is not plausible that its reflected image be thus discontinuous. Tell me whether — for in your presence it is safer to put this as a question than as an assertion — whether it is possible, though the inhabited world has length and breadth, that every visual ray when reflected from the moon should in like manner reach the ocean, even the visual rays of those who are sailing in the great ocean itself, yes and who dwell in it as the Britons do, and that too even though the earth, as you say, does not have the relation of centre to the orbit of the moon. Well, this,” I said, “it is your business to consider; but the reflection of vision either in respect to the moon or in general is beyond your province and that of Hipparchus too. Although Hipparchus was industrious, still many find him unsatisfactory in his explanation of the nature of vision itself, which is more likely to involve a sympathetic compound and fusion than any impacts and rebounds such as those of the atoms that Epicurus invented.

Moreover, Clearchus, I think, would refuse to assume with us that the moon is a body of weight and solidity instead of an ethereal and luminiferous star as you say; and such a moon ought to shatter and divert the visual ray so that reflection would be out of the question. But if anyone dismisses our objections, we shall ask how it is that the reflection of the ocean exists as a face only in the moon and is seen in none of all the many other stars, although reason requires that all or none of them should affect the visual ray in this fashion. But let us have done with this; and do you," I said with a glance at Lucius, "recall to me what part of our position was stated first."

5 1 Whereat Lucius said: "Nay, lest we give the impression of flatly insulting Pharnaces by thus passing over the Stoic opinion unnoticed, do now by all means address some remark to the gentleman who, supposing the moon to be a mixture of air and gentle fire, then says that what appears to be a figure is the result of the blackening of the air as when in a calm water there runs a ripple under the surface." "You are very nice, Lucius," I said, "to dress up the absurdity in respectable language. Not so our comrade; but he said what is true, that they blacken the Moon's eye defiling her with blemishes and bruises, 922 at one and the same time addressing her as Artemis and Athena and making her a mass compounded of murky air and smouldering fire neither kindling nor shining of herself, an indiscriminate kind of body, forever charred and smoking like the thunderbolts that are darkling and by the poets called lurid. Yet a smouldering fire, such as they suppose that of the moon to be, cannot persist or subsist at all unless it get solid fuel that shelters and at the same time nourishes it; this some philosophers, I believe, see less clearly than do those who say in jest that Hephaestus is said to be lame because fire without wood, like the lame without a stick, makes no progress. If the moon really is fire, whence came so much air in it? For the region that we see revolving above us is the place not of air but of a superior substance, the nature of which is to rarefy all things and set them afire; and, if air did come to be there, why has it not been etherealized by the fire and in this transformation disappeared but instead has been preserved as a housemate of fire this long time, as if nails had fixed it forever to the same spots and riveted it together? Air is tenuous and without configuration, and so it naturally slips and does not stay in place; and it cannot have become solidified if it is commingled with fire and partakes neither of moisture nor of earth by which all air can be solidified. Moreover, velocity ignites the air in stones and in cold lead, not to speak of the air enclosed in fire that is whirling about with such great speed. Why, they are vexed by Empedocles because he represents the moon to be a hail-like congelation of air encompassed by the sphere of fire; but they themselves say that the moon is a sphere of fire containing air dispersed about it here and there, and a sphere moreover that has neither clefts nor depths and hollows, such as are allowed by those who make it an earthy body, but has the air evidently resting upon its convex surface. That it should so remain is both contrary to reason and impossible to square with what is observed when the moon is full. On that assumption there should have been no distinction of dark and shadowy air; but all the air should become dark when

occulted, or when the moon is caught by the sun it should all shine out with an even light. For with us too, while the air in the depths and hollows of the earth, wherever the sun's rays do not penetrate, remains shadowy and unlit, that which suffuses the earth outside takes on brilliance and a luminous colour. The reason is that air, because of its subtilty, is delicately attuned to every quality and influence; and, especially if it touches light or, to use your phrase, merely is tangent to it, it is altered through and through and entirely illuminated. So this same point seems right handsomely to re-enforce those who pack the air on the moon into depths of some kind and chasms, even as it utterly refutes you who make her globe an unintelligible mixture or compound of air and fire, — for it is not possible that a shadow remain upon the surface when the sun casts his light upon all of the moon that is within the compass of our vision.”

61 Even while I was still speaking Pharnaces spoke: “Here we are faced again with that stock manoeuvre of the Academy: on each occasion that they engage in discourse with others they will not offer any accounting of their own assertions but must keep their interlocutors on the defensive lest they become the prosecutors. Well, me you will not to day entice into defending the Stoics against your charges until I have called you people to account for turning the universe upside down.” Thereupon Lucius laughed and said: 923”Oh sir, just don't bring suit against us for impiety as Cleanthes thought that the Greeks ought to lay an action for impiety against Aristarchus the Samian on the ground that he was disturbing the hearth of the universe because he sought to save the phenomena by assuming that the heaven is at rest while the earth is revolving along the ecliptic and at the same time is rotating about its own axis. We express no opinion of our own now; but those who suppose that the moon is earth, why do they, my dear sir, turn things upside down any more than you do who station the earth here suspended in the air? Yet the earth is a great deal larger than the moon according to the mathematicians who during the occurrence of eclipses and the transits of the moon through the shadow calculate her magnitude by the length of time that she is obscured. For the shadow of the earth grows smaller the further it extends, because the body that cast the light is larger than the earth; and that the upper part of the shadow itself is taper and narrow was recognized, as they say, even by Homer, who called night ‘nimble’ because of the ‘sharpness’ of the shadow. Yet captured by this part in eclipses the moon barely escapes from it in a space thrice her own magnitude. Consider then how many times as large as the moon the earth is, if the earth casts a shadow which at its narrowest is thrice as broad as the moon. All the same, you fear for the moon lest it fall; whereas concerning the earth perhaps Aeschylus has persuaded you that Atlas

Stands, staying on his back the prop of earth
And sky no tender burden to embrace.

Or, while under the moon there stretches air unsubstantial and incapable of supporting a solid mass, the earth, as Pindar says, is encompassed by ‘steel-shod

pillars'; and therefore Pharnaces is himself without any fear that the earth may fall but is sorry for the Ethiopians or Taprobanians, who are situated under the circuit of the moon, lest such a great weight fall upon them. Yet the moon is saved from falling by its very motion and the rapidity of its revolution, just as missiles placed in slings are kept from falling by being whirled around in a circle. For each thing is governed by its natural motion unless it be diverted by something else. That is why the moon is not governed by its weight: the weight has its influence frustrated by the rotatory motion. Nay, there would be more reason perhaps to wonder if she were absolutely unmoved and stationary like the earth. As it is, while the moon has good cause for not moving in this direction, the influence of weight alone might reasonably move the earth, since it has no part in any other motion; and the earth is heavier than the moon not merely in proportion to its greater size but still more, inasmuch as the moon has, of course, become light through the action of heat and fire. In short, your own statements seem to make the moon, if it is fire, stand in greater need of earth, that is of matter to serve it as a foundation, as something to which to adhere, as something to lend it coherence, and as something that can be ignited by it, for it is impossible to imagine fire being maintained without fuel, but you people say that earth does abide without root or foundation." "Certainly it does," said Pharnaces, "in occupying the proper and natural place that belongs to it, the middle, for this is the place about which all weights in their natural inclination press against one another and towards which they move and converge from every direction, whereas all the upper space, even if it receive something earthy which has been forcibly hurled up into it, straightway extrudes it into our region or rather lets it go where its proper inclination causes it naturally to descend."

71 At this — for I wished Lucius to have time to collect his thoughts — I called to Theon. "Which of the tragic poets was it, Theon," I asked, "who said that physicians
With bitter drugs the bitter bile purge?"

Theon replied that it was Sophocles. "Yes," I said, "and we have of necessity to allow them this procedure; but to philosophers one should not listen if they desire to repulse paradoxes with paradoxes and in struggling against opinions that are amazing fabricate others that are more amazing and outlandish, as these people do in introducing their 'motion to the centre.' What paradox is not involved in this doctrine? Not the one that the earth is a sphere although it contains such great depths and heights and irregularities? Not that people live on the opposite hemisphere clinging to the earth like wood-worms or geckos turned bottomsides up? — and that we ourselves in standing remain not at right angles to the earth but at an oblique angle, leaning from the perpendicular like drunken men? Not that incandescent masses of forty tons falling through the depth of the earth stop when they arrive at the centre, though nothing encounter or support them; and, if in their downward motion the impetus should carry them past the centre, they swing back again and return of themselves? Not that pieces of meteors burnt out on either side of the earth do not move downwards continually but falling upon the surface of the earth force their way into it from the outside and conceal

themselves about the centre? Not that a turbulent stream of water, if in flowing downwards should reach the middle point, which they themselves call incorporeal, stops suspended or moves round about it, oscillating in an incessant and perpetual see-saw? Some of these a man could not even mistakenly force himself to conceive as possible. For this amounts to 'upside down' and 'all things topsy-turvy,' everything as far as the centre being 'down' and everything under the centre in turn being 'up.' The result is that, if a man should so coalesce with the earth that its centre is at his navel, the same person at the same time has his head up and his feet up too. Moreover, if he dig through the further side, his bottom in emerging is up, and the man digging himself 'up' is pulling himself 'down' from 'above'; and, if someone should then be imagined to have gone in the opposite direction to this man, the feet of both of them at the same time turn out to be 'up' and are so called.

81 Nevertheless, though of tall tales of such a kind and number they have shouldered and lugged in — not a wallet-full, by heaven, but some juggler's pack and hotchpotch, still they say that others are playing the buffoon by placing the moon, though it is earth, on high and not where the centre is. Yet if all heavy body converges to the same point and is compressed in all its parts upon its own centre, it is no more as centre of the sum of things than as a whole that the earth would appropriate to herself the heavy bodies that are parts of herself; and the downward tendency of falling bodies proves not that the earth is in the centre of the cosmos but that those bodies which when thrust away from the earth fall back to her again have some affinity and cohesion with her. For as the sun attracts to itself the parts of which it consists so the earth too accepts as her own the stone that has properly a downward tendency, and consequently every such thing ultimately unites and coheres with her. If there is a body, however, that was not originally allotted to the earth or detached from it but has somewhat independently a constitution and nature of its own, as those men would say of the moon, what is to hinder it from being permanently separate in its own place, compressed and bound together by its own parts? For it has not been proved that the earth is the centre of the sum of things, and the way in which things in our region press together and concentrate upon the earth suggests how in all probability things in that region converge upon the moon and remain there. The man who drives together into a single region all earthy and heavy things and makes them part of a single body — I do not see for what reason he does not apply the same compulsion to light objects in their turn but allows so many separate concentrations of fire and, since he does not collect all the stars together, clearly does not think that there must also be a body common to all things that are fiery and have an upward tendency.

91 Now," said I, "my dear Apollonides, you mathematicians say that the sun is an immense distance from the upper circumference and that above the sun Venus and Mercury and the other planets revolved lower than the fixed stars and at great intervals from one another; but you think that in the cosmos there is provided no scope and extension for heavy and earthy objects. You see that it is ridiculous for us to deny that

the moon is earth because she stands apart from the nether region and yet to call her a star although we see her removed so many thousands of miles from the upper circumference as if plunged into a pit. So far beneath the stars is she that the distance cannot be expressed, but you mathematicians in trying to calculate it run short of numbers; she practically grazes the earth and revolving close to it

Whirls like a chariot's axle-box about,
Empedocles says,
That skims the post in passing.

Frequently she does not even surmount the earth's shadow, though it extends but a little way because the illuminating body is very large; but she seems to revolve so close, almost within arm's reach of the earth, as to be screened by it from the sun unless she rises above this shadowy, terrestrial, and nocturnal place which is the earth's estate. Therefore we must boldly declare, I think, that the moon is within the confines of the earth inasmuch as she is occulted by its extremities.

¹⁰ ¹ Dismiss the fixed stars and the other planets and consider the demonstrations of Aristarchus in his treatise, *On Sizes and Distances*, that 'the distance of the sun is more than ¹⁸ times and less than ²⁰ times the distance of the moon,' that is its distance from us. According to the highest estimate, however, the moon's distance from us is said to be ⁵⁶ times the radius of the earth. Even according to the mean calculations this radius is 40,000 stades; and, if we reckon from this, the sun is more than 40,300,000 stades distant from the moon. She has migrated so far from the sun on account of her weight and has moved so close to the earth that, if properties are to be determined by locations, the lot, I mean the position, of earth lays an action against the moon and she is legally assignable by right of propinquity and kinship to the chattels real and personal of earth. We do not err at all, I think, if granting such altitude and extension to the things called 'upper' we leave what is 'down below' also some room to move about in and so much latitude as there is from earth to moon. For as he is immoderate who calls only the outermost surface of the heaven 'up' and all else 'down,' so is he intolerable who restricts 'down' to the earth or rather to the centre; but both there and here some extension must be granted since the magnitude of the universe permits it. The claim that everything away from the earth is *ipso facto* 'up' and 'on high' is answered by a counter-claim that what is away from the circuit of the fixed stars is *ipso facto* 'down.'

¹¹ After all, in what sense is earth situated in the middle and in the middle of what? The sum of things is infinite; and the infinite, having neither beginning nor limit, cannot properly have a middle, for the middle is a kind of limit too but infinity is a negation of limits. He who asserts that the earth is in the middle not of the sum of things but of the cosmos is naïve if he supposes that the cosmos itself is not also involved in the same difficulties. In fact, in the sum of things no middle has been left for the cosmos either, but it is without hearth and habitation, moving in infinite void to nothing of its own; or,

if it has come to rest because it has found some other reason for abiding, not because of the nature of its location, similar inferences are permissible in the cases of both earth and moon, that the former is stationary here and the latter is in motion there by reason of a different soul or nature rather than a difference of location. Besides this, consider whether they have not overlooked an important point. If anything in any way at all off the centre of the earth is ‘up,’ no part of the cosmos is ‘down’; but it turns out that the earth and the things on the earth and absolutely all body surrounding or enclosing the centre are ‘up’ and only one thing is ‘down,’ that incorporeal point which must be in opposition to the entire nature of the cosmos, if in fact ‘down’ and ‘up’ are natural opposites. This, moreover, does not exhaust the absurdity. The cause of the descent of heavy objects and of their motion to this region is also abolished, for there is no body that is ‘down’ towards which they are in motion and it is neither likely nor in accordance with the intention of these men that the incorporeal should have so much influence as to attract all these objects and keep them together around itself. On the contrary, it proves to be entirely unreasonable and inconsistent with the facts for the whole cosmos to be ‘up’ and nothing but an incorporeal and unextended limit to be ‘down’; but that statement of ours is reasonable, that ample space and broad has been divided between ‘up’ and ‘down.’

121 All the same, let us assume, if you please, that the motions of earthy objects in the heaven are contrary to nature; and then let us calmly observe without any histrionics and quite dispassionately that this indicates not that the moon is not earth but that she is earth in an ‘unnatural’ location. For the fire of Aetna too is below earth ‘unnaturally,’ but it is fire; and the air confined in skins, though by nature it is light and has an upward tendency, has been constrained to occupy an ‘unnatural’ location. As to the soul herself,” I said, “by Zeus, is her confinement in body not contrary to nature, swift as she is and fiery, as you say, and invisible in a sluggish, cold, and sensible vehicle? Shall we then on this account deny that there is a soul in body or that mind, a divine thing, though it traverses instantaneously in its flight all heaven and earth and sea, has passed into flesh and sinew and marrow under the influence of weight and density and countless qualities that attend liquefaction? This Zeus of yours too, is it not true that, while in his own nature he is single, a great and continuous fire, at present he is slackened and subdued and transformed, having become and continuing to become everything in the course of his mutations? So look out and reflect, good sir, lest in rearranging and removing each thing to its ‘natural’ location you contrive a dissolution of the cosmos and bring upon things the ‘Strife’ of Empedocles — or rather lest you arouse against nature the ancient Titans and Giants and long to look upon that legendary and dreadful disorder and discord when you have separated all that is heavy and all that is light.

The sun’s bright aspect is not there descried,
No, nor the shaggy might of earth, nor sea

as Empedocles says. Earth had no part in heat, water no part in air; there was not anything heavy above or anything light below; but the principles of all things were untempered and unamiable and solitary, not accepting combination or association with one another, but avoiding and shunning one another and moving with their own peculiar and arbitrary motions they were in the state in which, according to Plato, everything is from which God is absent, that is to say in which bodies are when mind or soul is wanting. So they were until desire came over nature providentially, for Affection arose or Aphrodite or Eros, as Empedocles says and Parmenides and Hesiod, in order that by changing position and interchanging functions and by being constrained some to motion and some to rest and compelled to give way and shift from the 'natural' to the 'better' the bodies might produce a universal concord and community.

^{13 1} If not a single one of the parts of the cosmos ever got into an 'unnatural' condition but each one is 'naturally' situated, requiring no transposition or rearrangement and having required none in the beginning either, I cannot make out what use there is of providence or of what Zeus, 'the master-craftsman' is maker and father-creator. In an army, certainly, tacticians are useless if each one of the soldiers should know of himself his post and position and the moment when he must take and keep them. Gardeners and builders are useless too if here water all of itself 'naturally' moves to the things that require it and irrigates them with its stream, and there bricks and timbers and stones by following their 'natural' inclinations and tendencies assume of themselves their appropriate position and arrangement. If, however, this notion eliminates providence forthwith and if the arrangement of existing things pertains to God and the distributing of them too, what wonder is there that nature has been so marshalled and disposed that here in our region there is fire but the stars are yonder and again that earth is here but the moon is established on high, held fast by the bonds of reason which are firmer than the bonds of nature? For, if all things really must follow their 'natural' inclinations and move with their 'natural' motions, you must order the sun not to revolve and Venus too and every other star as well, for light and fiery bodies move 'naturally' upwards and not in a circle. If, however, nature includes such variation in accordance with location that fire, though it is seen to move upwards here, as soon as it has reached the heavens revolves along with their rotation, what wonder is there that the same thing has happened to heavy and earthy bodies that have got there and that they too have been reduced by the environment to a different kind of motion? For it certainly cannot be that heaven 'naturally' deprives light objects of their upward motion but is unable to master objects that are heavy and have a downward inclination; on the contrary, by whatever influence it rearranged the former it rearranged the latter too and employed the nature of both of them for the better.

^{14 1} What is more, if we are finally to throw off the habits and opinions that have held our minds in thrall and fearlessly to say what really appears to be the case, no part of a whole all by itself seems to have any order, position, or motion of its own which could be called unconditionally 'natural.' On the contrary, each and every such part, whenever

its motion is usefully and properly accommodated to that for the sake of which the part has come to be and which is the purpose of its growth or production, and whenever it acts or is affected or disposed so that it contributes to the preservation or beauty or function of that thing, then, I believe, it has its 'natural' position and motion and disposition. In man, at any rate, who is the result of 'natural' process if any being is, the heavy and earthy parts are above, chiefly in the region of the head, and the hot and fiery parts are in the middle regions; some of the teeth grow from above and some from below, and neither set is 'contrary to nature'; and it cannot be said that the fire which flashes in the eyes above is 'natural' whereas that in the bowels and heart is 'contrary to nature,' but each has been assigned its proper and useful station. Observe, as Empedocles says, the nature of

Tritons and tortoises with hides of stone

and of all testaceans,

Thou'lt see earth there established over flesh;

and the stony matter does not oppress or crush the constitution on which it is superimposed, nor on the other hand does the heat by reason of lightness fly off to the upper region and escape, but they have been somehow intermingled and organically combined in accordance with the nature of each.

15 1 Such is probably the case with the cosmos too, if it really is a living being: in many places it has earth and in many fire and water and breath as the result not of forcible expulsion but of rational arrangement. After all, the eye has its present position in the body not because it was extruded thither as a result of its lightness, and the heart is in the chest not because its heaviness has caused it to slip and fall thither but because it was better that each of them should be so located. Let us not then believe with regard to the parts of the cosmos that earth is situated here because its weight has caused it to subside or that the sun, as Metrodorus of Chios once thought, was extruded into the upper region like an inflated skin by reason of its lightness or that the other stars got into their present positions because they tipped the balance, as it were, at different weights. On the contrary, the rational principle is in control; and that is why the stars revolve fixed like 'radiant eyes' in the countenance of the universe, the sun in the heart's capacity transmits and disperses out of himself heat and light as if it were blood and breath, and earth and sea 'naturally' serve the cosmos to the ends that bowels and bladder do an animal. The moon, situate between sun and earth as the liver or another of the soft viscera is between heart and bowels, transmits hither the warmth from above and sends upwards the exhalations from our region, refining them in herself by a kind of concoction and purification. It is not clear to us whether her earthiness and solidity have any use suitable to other ends also. Nevertheless, in everything the better has control of the necessary. Well, what probability can we thus conceive in the statements of the Stoics? They say that the luminous and tenuous part of the ether by reason of its subtilty became sky and the part which was condensed or compressed became stars, and that of these the most sluggish and turbid is the moon. Yet all the same anyone can see that the

moon has not been separated from the ether but that there is still a large amount of it about her in which she moves and much of it beneath her in which they themselves as that the bearded stars and comets whirl. So it is not the inclinations consequent upon weight and lightness that have circumscribed the precincts of each of the bodies, but their arrangement is the result of a different principle.”

181 With these remarks I was about to yield the floor to Lucius, since the proofs of our position were next in order; but Aristotle smiled and said: “The company is my witness that you have directed your entire refutation against those who suppose that the moon is for her part semi-igneous and yet assert of all bodies in common that of themselves they incline either upwards or downwards. Whether there is anyone, however, who says that the stars move naturally in a circle and are of a substance far superior to the four substances here did not even accidentally come to your notice, so that I at any rate have been spared trouble.” And Lucius broke in and said” “. . . good friend, probably one would not for the moment quarrel with you and your friends, dispute the countless difficulties involved, when you ascribe to the other stars and the whole heaven a nature pure and undefiled and free from qualitative change and moving in a circle whereby it is possible to have the nature of endless revolution too; but let this doctrine descend and touch the moon, and in her it no longer preserves the impassivity and beauty of that body. Not to mention her other irregularities and divergencies, this very face which she displays is the result of some alteration of her substance or of the admixture somehow of another substance. That which is subjected to mixture, however, is the subject of some affection too, for it loses its purity, since it is perforce infected by what is inferior to it. The moon’s sluggishness and slackness of speed and the feebleness and faintness of her heat which, in the words of Ion,

ripes not the grape to duskiness,

to what shall we ascribe them except to her weakness and alteration, if an eternal and celestial body can have any part in alteration? The fact is in brief, my dear Aristotle, that regarded as earth the moon has the aspect of a very beautiful, august, and elegant object; but as a star or luminary or a divine and heavenly object she is, I am afraid, misshapen, ugly, and a disgrace to the noble title, if it is true that of all the host in heaven she alone goes about in need of alien light, as Parmenides says

Fixing her glance forever on the sun.

Our comrade in his discourse won approval by his demonstration of this very proposition of Anaxagoras’s that ‘the sun imparts to the moon her brilliance’; for my part, I shall not speak about these matters that I learned from you or in your company but gladly proceed to what remains. Well then, it is plausible that the moon is illuminated not by the sun’s irradiating and shining through her in the manner of glass or ice nor again as the result of some sort of concentration of brilliance or aggregation of rays, the light increasing as in the case of torches. Were that true, we should see the moon at the

full on the first of the month no less than in the middle of the month, if she does not conceal and obstruct the sun but because of her subtlety let his light through or as a result of combining with it flashes forth and joins in kindling the light in herself. Certainly her deviations or aversions cannot be alleged as the cause of her invisibility when she is in conjunction, as they are when she is at the half and gibbous or crescent; then, rather, ‘standing in a straight line with her illuminant,’ says Democritus, ‘she sustains and receives the sun,’ so that it would be reasonable for her to be visible and to let him shine thru. Far from doing this, however, she is at that time invisible herself and often has concealed and obliterated him

His beams she put to flight,
as Empedocles says,
From heaven above as far as to the earth,
Whereof such breadth as had the bright-eyed moon
She cast in shade,

just as if the light had fallen into night and darkness and not upon another star. As for the explanation of Posidonius that the profundity of the moon prevents the light of the sun from passing through her to us, this is obviously refuted by the fact that the air, though it is boundless and has many times the profundity of the moon, is in its entirety illumined and filled with sunshine by the rays. There remains then the theory of Empedocles that the moonlight which we see comes from the moon’s reflection of the sun. That is why there is neither warmth nor brilliance in it when it reaches us, as we should expect there to be if there had been a kindling or mixture of the lights of sun and moon. To the contrary, just as voices when they are reflected produce an echo which is fainter than the original sound and the impact of missiles after a ricochet is weaker,

Thus, having struck the moon’s broad disk, the ray
comes to us in a reflue weak and faint because the deflection slackens its force.”

171 Sulla then broke in and said: “No doubt this position has its plausible aspects; but what tells most strongly on the other side, did our comrade explain that away or did he fail to notice it?” “What’s that?” said Lucius, “or do you mean the difficulty with respect to the half-moon?” “Exactly,” said Sulla, “for there is some reason in the contention that, since all reflection occurs at equal angles, whenever the moon at the half is in mid-heaven the light cannot move earthwards from her but must glance off beyond the earth. The ray that then touches the moon comes from the sun on the horizon and therefore, being reflected at equal angles, would be produced to the point on the opposite horizon and would not shed its light upon us, or else there would be great distortion and aberration of the angle, which is opposite.” “Yes, by Heaven,” said Lucius, “there was talk of this too”; and, looking at Menelaus the mathematician as he

spoke, he said: "In your presence, my dear Menelaus, I am ashamed to confute a mathematical proposition, the foundation, as it were, on which rests the subject of catoptrics. Yet it must be said that the proposition, 'all reflection occurs at equal angles,' is neither self-evident nor an admitted fact. It is refuted in the case of convex mirrors when the point of incidence of the visual ray produces images that are magnified in one respect; and it is refuted by folding mirrors, either plane of which, when they have been inclined to each other and have formed an inner angle, exhibits a double image, so that four likenesses of a single object are produced, two reversed on the outer surfaces and two dim ones not reversed in the depth of the mirrors. The reason for the production of these images Plato explains, for he has said that when the mirror is elevated on both sides the visual rays interchange their reflection because they shift from one side to the other. So, if of the visual rays some revert straight to us from the plane surfaces while others glance off to the opposite sides of the mirrors and thence return to us again, it is not possible that all reflections occur at equal angles. Consequently some people take direct issue with the mathematicians and maintain that they confute the equality of the angles of incidence and reflection by the very streams of light that flow from the moon upon the earth, for they deem this fact to be much more credible than that theory. Nevertheless, suppose that this must be conceded as a favour to geometry, the dearly beloved! In the first place, it is likely of the occur only in mirrors that have been polished to exact smoothness; but the moon is very uneven and rugged, with the result that the rays from a large body striking against considerable heights which receive reflections and diffusions of light from one another are multifariously reflected and intertwined and the refulgence itself combines with itself, coming to us, as it were, from many mirrors. In the second place, even if we assume that the reflections on the surface of the moon occur at equal angles, it is not impossible that the rays as they travel through such a great interval get fractured and deflected so as to be blurred and to bend their light. Some people even give a geometrical demonstration that the moon sheds many of her beams upon the earth along a line extended from the surface that is bent away from us; but I could not construct a geometrical diagram while talking, and talking to many people too.

^{18 1} Speaking generally," he said, "I marvel that they adduce against us the moon's shining upon the earth at the half and at the gibbous and the crescent phases too. After all, if the mass of the moon that is illumined by the sun were ethereal or fiery, the sun would not leave her a hemisphere that to our perception is ever in shadow and unilluminated; on the contrary, if as he revolves he grazed her ever so slightly, she should be saturated in her entirety and altered through and through by the light proceeding easily in all directions. Since wine that just touches water at its surface or a drop of blood fallen into liquid at the moment of contact stains all the liquid red, and since they say that the air itself is filled with sunshine not by having any effluences or rays commingled with it but by an alteration and change that results from impact or contact of the light, how do they imagine that a star can come in contact with a star or

light with light and instead of blending and producing a thorough mixture and change merely illumine those portions of the surface which it touches? In fact, the circle which the sun in its revolution describes and causes to turn about the moon now coinciding with the circle that divides her visible and invisible parts and now standing at right angles to it so as to intersect it and be intersected by it, by different inclinations and relations of the bright part to the dark producing in her the gibbous and crescent phases, conclusively demonstrates that her illumination is the result not of combination but of contact, not of a concentration of light within her but of light shining upon her from without. In that she is not only illumined herself, however, but also transmits to us the semblance of her illumination, she gives us all the more confidence in our theory of her substance. There are no reflections from anything rarefied or tenuous in texture, and it is not easy even to imagine light rebounding from light or fire from fire; but whatever is to cause a repercussion or a reflection must be compact and solid, in order that it may stop a blow and repel it. At any rate, the same sunlight that the air lets pass without impediment or resistance is widely reflected and diffused from wood and stone and clothing exposed to its rays. The earth too we see illumined by the sun in this fashion. It does not let the light penetrate its depths as water does or pervade it through and through as air does; but such as is the circle of the sun that moves around the moon and so great as is the part of her that it intercepts, just such a circle in turn moves around the earth, always illuminating just so much and leaving another part unlimited, for the illumined portion of either body appears to be slightly greater than a hemisphere. Give me leave then to put it in geometrical fashion in terms of a proportion. Given three things approached by the light from the sun: earth, moon, air; if we see that the moon is illumined not as the air is rather than as the earth, the things upon which the same agent produces the same effects must be of a similar nature.”

191 When all had applauded Lucius, I said: “Congratulations upon having added to an elegant account an elegant proportion, for you must not be defrauded of what belongs to you.” He smiled thereat and said: “Well then proportion must be used a second time, in order that we may prove the moon to be like the earth not only because the effects of the same agent are the same on both but also because the effects of have on the same patient are the same. Now, grant me that nothing that happens to the sun is so like its setting as a solar eclipse. You will if you call to mind this conjunction recently which, beginning just after noonday, made many stars shine out from my parts of the sky and tempered the air in the manner of twilight. If you do not recall it, Theon here will cite us Mimnermus and Cydias and Archilochus and Stesichorus besides and Pindar, who during eclipses bewail ‘the brightest star bereft’ and ‘at midday falling’ and say that the beam of the sun ‘is sped the path of shade’; and to crown all he will cite Homer, who says ‘the faces of men are covered with night and gloom’ and ‘the sun has perished out of heaven’ speaking with reference to the moon and hinting that this naturally occurs

When waning month to waxing month gives way.

For the rest, I think that it has been reduced by the precision of mathematics to the clear and certain formula that night is the shadow of earth and the eclipse of the sun is the shadow of the moon whenever the visual ray encounters it. The fact is that in setting the sun is screened from our vision by the earth and in eclipse by the moon; but are cases of occultation, but the vespertine is occultation by the earth and the ecliptic by the moon with her shadow intercepting the visual ray. What follows from this is easy to perceive. If the effect is similar, the agents are similar, for it must be the same agents that cause the same things to happen to the same subject. Nor should we marvel if the darkness of eclipses is not so deep or so oppressive of the air as night is. The reason is that the body which produces night and that which produces the eclipse while the same in substance are not equal in size. In fact the Egyptians, I think, say that the moon is one seventy-second part (of the earth), and Anaxagoras that it is the size of the Peloponnesus; and Aristarchus demonstrates that the ratio of the earth's diameter to the diameter of the moon is smaller than 60 to 19 and greater than 108 to 43. Consequently the earth because of its size removes the sun from sight entirely, for the obstruction is large and its duration is that of the night. even if the moon, however, does sometimes cover the sun entirely, the eclipse does not have duration or extension; but a kind of light is visible about the rim which keeps the shadow from being profound and absolute. The ancient Aristotle gives this as a reason besides some others why the moon is observed in eclipse more frequently than the sun, saying that the sun is eclipse by interposition of the moon but the moon by that of the earth, which is much larger. Posidonius gave this definition: 'The following condition is an eclipse of the sun, conjunction of the moon's shadow with whatever parts of the earth it may obscure, for there is an eclipse only for those whose visual ray the shadow of the moon intercepts and screens from the sun'; — since he concedes then that a shadow of the moon falls upon us, he has left himself nothing to say that I can see. Of a star there can be no shadow, for shadow means the unlighted and light does not produce shadow but naturally destroys it.

201 Well now," he said, "which of the proofs came for this?" And I replied, "That the moon is subject to the same eclipse." "Thank you," he said, "for reminding me; but now shall I assume that you have been persuaded and do hold the moon to be eclipsed by being caught in the shadow and so turn straightway to my argument, or do you prefer that I give you a lecture and demonstration in which each of the arguments is enumerated?" "By heaven," said Theon, "do give these gentlemen a lecture. As for me, I want some persuasion as well, since I have only heard it put this way: when the three body, earth and sun and moon, get into a straight line, eclipses take place because the earth deprives the moon or the moon, on the other hand, deprives the earth of the sun, the sun being eclipsed when the moon and the moon when the earth takes the middle position of the three, the former of which cases occurs at conjunction and the latter at the middle of the month." Whereupon Lucius said, "Those soldier roughly the main points, though, of what is said on the subject. Add thereto first, if you will, the argument from the shape of the shadow. It is a cone, as is natural when a large fire or light that is spherical

circumfuses a smaller but spherical mass. This is the reason why in eclipses of the moon the darkened parts are outlined against the bright in segments that are curved, for whenever two round bodies come into contact the lines by which either intersects the other turn out to be circular since they have everywhere a uniform tendency. Secondly, I think that you are aware that of the moon the eastward parts are first eclipsed and of the sun the westward parts and that, while the shadow of the earth moves from east to west, the sun and the moon move contrariwise towards the east. This is made visible to sense-perception by the phenomena and needs no very lengthy explanations to be understood, and these Philadelphia confirm the cause of the eclipse. Since the sun is eclipsed by being overtaken and the moon by encountering that which produces the eclipse, it is reasonable or rather it is necessary that the sun be caught first from behead and the moon from the front, for the obstruction begins from that point which the intercepting body first assails. The sun is assailed from the west by the moon that is striving after him, and she is assailed from the east by the earth's shadow that is sweeping down as it were in the opposite direction. Thirdly, moreover, consider the matter of the duration and the magnitude of lunar eclipses. If the moon is eclipsed when she is high and far from the earth, she is concealed for a little time; but, if the survey thing happens to her when she is low and near the earth, she is strongly curbed and is slow to get out of the shadow, although when she is low her exertions of motion are greatest and when she is high they are least. The reason for the difference lies in the shadow, which being broadest at the base, as cones are, and gradually contracting terminates at the vertex in a sharp and fine tip. Consequently the moon, if she has met the shadow when she is low, is involved by it in its largest circles and traverses its deep and darkest part; but above as it were in shallow water by reason of the fineness of the shadow she is just grazed and quickly gets clean away. I pass over all that was said besides with particular reference to the phases and variations, for these too, in so far as is possible, admit the cause alleged; and instead I revert to the argument before us which has its basis in the evidence of the senses. We see that from a shadowy place fire glows and shines forth more intensely, whether because the dark air being dense does not admit its effluences and diffusions but confines and concentrates the substance in a single place or because this is an affection of our senses that as hot things appear to be hotter in comparison with cold and pleasures more intense in comparison with pains so bright things appear conspicuous when compared with dark, their appearance being intensified by contrast to the different impressions. The former explanation seems to be the more plausible, for in sunlight fire of every kind not only loses its brilliance but by giving way becomes ineffective and less keen, the reason being that the heat of the sun disperses and dissipates its potency. If, then, as the Stoics themselves assert, the moon, being a rather turbid star, has a faint and feeble fire of her own, she ought to have none of the things happen to her that now obviously do but the very opposite; she ought to be revealed when she is hidden and hidden whenever she is now revealed, that is hidden all the rest of the time when she is bedimmed by the circumambient ether but shining forth and

becoming brilliantly clear at intervals of six months or again at intervals of five when she sinks under the shadow of the earth, since of ⁴⁶⁵ ecliptic full moons ⁴⁰⁴ occur in cycles of six months and the rest in cycles of five months. It ought to have been at such intervals of time then that the moon is revealed resplendent in the shadow, whereas in the shadow she is eclipsed and loses her light but regains it again as soon as she escapes the shadow and is revealed often even by day, which implies that she is anything but a fiery and star-like body.”

²¹¹ When Lucius said this, almost while he was speaking Pharnaces and Apollonides sprang forth together. Then, Apollonides having yielded, Pharnaces said that this very point above all proves the moon to be a star or fire, since she is not entirely invisible in her eclipses but displays a colour smouldering and grim which is peculiar to her. Apollonides raised an objection concerning the “shadow” on the ground that the scientists always give this name to the region that is without light and the heaven does not admit shadow. “This,” I said, “is the objection of one who speaks captiously to the name rather than like a natural scientist and mathematician to the fact. If one refuses to call the region screened by the earth ‘shadow’ and insists upon calling it ‘lightless space,’ nevertheless when the moon gets into it she must be obscured since she is deprived of the solar light. Speaking generally too, it is silly,” I said, “to deny that the shadow of the earth reaches that point from which on its part the shadow of the moon by impinging upon the sight and extending to the earth produces an eclipse of the sun. Now I shall turn to you, Pharnaces. That smouldering and glowing colour of the moon which you say is peculiar to her is characteristic of a body that is compact and a solid, for no remnant or trace of flame will remain in tenuous things nor is incandescence possible unless there is a hard body that has been ignited through and through and sustains the ignition. So Homer too has somewhere said:

But when fire’s bloom had flown and flame had ceased
He smoothed the embers. . . .

The reason probably is that what is igneous is not fire but body that has been ignited and subjected to the action of fire, which adheres to a solid and stable mass and continues to occupy itself with it, whereas flames are the kindling and flux of tenuous nourishment or matter which because of its feebleness is swiftly dissolved. Consequently there would be no other proof of the moon’s earthy and compact nature so manifest as the smouldering colour, if it really were her own. But it is not so, my dear Pharnaces, for as she is eclipsed she exhibits many changes of colour which scientists have distinguished as follows, delimiting them according to time or hour. If the eclipse occurs between eventide and half after the third hour, she appears terribly black; if at midnight, then she gives of this reddish and fiery colour; from half after the seventh hour a blush arises on her face; and finally, if she is eclipsed when the dawn is already near, she takes on a bluish or azure hue, from which especially it is that the poets and Empedocles give her the epithet ‘bright-eyed.’ Now, when one sees the moon take on so many hues in the shadow, it is a mistake to settle upon the smouldering colour alone, the

very one that might especially be called alien to her and rather an admixture or remnant of the light shining round about through the shadow, while the black or earthy colour could be called her own. Since here on earth places near lakes and rivers open to the sun take on the colour and brilliance of the purple and red awnings that shade them, by reason of the reflections giving off many various effulgences, what wonder if a great flood of shade debouching as it were into a heavenly sea of light, not calm or at rest but undergoing all sorts of combinations and alterations as it is churned about by countless stars, takes from the moon at different times the stain of different hues and presents them to our sight? A star or fire could not in shadow shine out black or glaucous or bluish; but over mountains, plains, and sea flit many kinds of colours from the sun, and blended with the shadows and mists his brilliance induces such tints as brilliance does when blended with a painter's pigments. Those of the sea Homer has endeavoured somehow or other to designate, using the terms 'violet' and 'wine-dark deep' and again 'purple swell' and elsewhere 'glaucous sea' and 'white calm'; but he passed over as being an endless multitude the variations of the colours that appear differently at different times about the land. It is likely, however, that the moon has not a single plane surface like the sea but closely resembles in constitution the earth that the ancient Socrates made the subject of a myth, whether he really was speaking in riddles about this earth or was giving a description of some other. It is in fact not incredible or wonderful that the moon, if she has nothing corrupted or slimy in her but garners pure light from heaven and is filled with width, which is fire not glowing or raging but moist and harmless and in its natural state, has got open regions of marvellous beauty and mountains flaming bright and has zones of royal purple with gold and silver not scattered in her depths but bursting forth in abundance on the plains or openly visible on the smooth heights. If through the shadow there comes to us a glimpse of these, different at different times because of some variation and difference of the atmosphere, the honourable repute of the moon is surely not impaired nor is her divinity because she is held by men to be a celestial and holy earth rather than, as the Stoics say, a fire turbid and dreggish. Fire, to be sure, is given barbaric honours among the Medes and Assyrians, who from fear by way of propitiation worship the maleficent rather than the reverend; but to every Greek, of course, the name of earth is dear and honourable, and it is our ancestral tradition to revere her like any other god. As men we are far from thinking that the moon, because she is a celestial earth, is a body without soul and mind and without share in the first-fruits that it beseems us to offer to the gods according to custom requiting them for the goods we have received and naturally revering what is better and more honourable in virtue and power. Consequently let us not think it an offence to suppose that she is earth and that for this which appears to be her face, just as our earth has certain great gulfs, so that earth yawns with great depths and clefts which contain water or murky air; the interior of these the light of the sun does not plumb or even touch, but it fails and the reflection which it sends back here is discontinuous."

22 1 Here Apollonides broke in. "Then by the moon herself," he said, "do you people think it possible that any clefts and chasms cast shadows which from the moon reach our sight here' or do you not reckon the consequence and shall I tell you what it is? Please listen then, though it is not anything unknown to you. The diameter of the moon measures twelve digits in apparent size at her mean distance; and each of the black and shadowy spots appears greater than half a digit and consequently would be greater than one twenty-fourth of her diameter. Well then, if we should suppose that the circumference of the moon is only thirty thousand stades and her diameter ten thousand each of the shadowy spots on her would in accordance with the assumption measure not less than five hundred stades. Consider now in the first place whether it is possible for the moon to have depths and corrugations so great as to cast such a large shadow; in the second place why, if they are of such great magnitude, we do not see them." Then I said to him with a smile: "Congratulations for having discovered such a demonstration, Apollonides. It would enable you to prove that both you and I are taller than the famous sons of Aloeus, not at every time of day to be sure but early in the morning particularly and in late afternoon if, when the sun makes our shadows enormous, you intend to supply sensation with this lovely reasoning that, if the shadow cast is large, what casts the shadow is immense. I am well aware that neither of us has been in Lemnos; we have both, however, often heard this line that is on everyone's lips:

Athos will veil the Lemnian heifer's flank.

The point of this apparently is that the shadow of the mountain, extending not less than seven hundred stades over the sea, falls upon a left bronze heifer; but it is not necessary, I presume, that what casts the shadow be size stades high, for the reason that shadows are made many times the size of the objects that cast them by the remoteness of the light from the objects. Come then, observe that, when the moon is at the full and because of the shadow's depth exhibits most articulately the appearance of the face, the sun is at his maximum distance from her. The reason is that the remoteness of the light alone and not the magnitude of the irregularities on the surface of the moon has made the shadow large. Besides, even in the case of mountains the dazzling beams of the sun prevent their crags from being discerned in broad daylight, although their depths and hollows and shadowy parts are visible from afar. So it is not at all strange that in the case of the moon too it is not possible to discern accurately the reflection and illumination, whereas the juxtapositions of the shadowy and brilliant parts by reason of the contrast do not escape our sight.

23 1 There is this, however," I said, "which seems to be a stronger objection to the alleged reflection from the moon. It happens that those who have placed themselves in the path of reflected rays see not only the object illumined but also what illuminates it. For example, if when a ray rebounds from water to a wall the eye is situated in the place that is itself illumined by the reflection, the eye discerns all three things, the reflected ray and the water that causes the reflection and the sun itself, the source of the light which has been reflected by impinging upon the water. On the basis of these

admitted and apparent facts those who maintain that the moon illuminates the earth with reflected light are bidden (by their adversaries) to point out in the moon at night an appearance of the sun such as there is in water by day whenever there is a reflection of the sun from it. Since there is no such appearance, (these adversaries) think that the illumination comes about in another way and not by reflection and that, if there is not reflection, neither is the moon an earth.” “What response must be made to them then?” said Apollonides, “for the characteristics of reflection seem to present us with a problem in common.” “In common in a way certainly,” said I, “but in another way not in common either. In the first place consider the matter of the image, how topsy-turvy and like ‘rivers flowing uphill’ they conceive it. The fact is that water is on earth and below, and the moon above the earth and on high; and hence the angles produced by the reflected rays are the converse of each other, the one having its apex above at the moon, the other below at the earth. So they must not demand that every kind of mirror or a mirror at every distance produce a similar reflection, since (in doing so) they are at variance with the manifest facts. Those, on the other hand, who declare that the moon is not a tenuous or smooth body as water is but a heavy and earthy one, I do not under why it is required of them that the sun be manifest to vision in her. For milk does not return such mirrorings either or produce reflections of the visual ray, and the reason is the irregularity and roughness of its particles; how in the world then is it possible for the moon to cast the visual ray back from herself in the way that the smoother mirrors do? Yet even these, of course, are occluded if a scratch or speck of dirt or roughness covers the point from which the visual ray is naturally reflected, and while the mirrors themselves are seen they do not return the customary reflection. One who demands that the moon either reflect our vision from herself to the sun as well or else not reflect the sun from herself to us either is naïve, for he is demanding that the eye be a sun, the vision light, and the human being a heaven. Since the light of the sun because of its intensity and brilliance arrives at the moon with a shock, it is reasonable that its reflection should reach to us; but the visual ray, since it is weak and tenuous and many times slighter, what wonder if it does not have an impact that produces recoil or if in rebounding it does not maintain its continuity but is dispersed and exhausted, not having light enough to keep it from being scattered about the irregularities and corrugations (of the moon)? From water, to be sure, and from mirrors of other kinds it is not impossible for the reflection (of the visual ray) to rebound to the sun, since it is still strong because it is near to its point of origin; but from the moon, even if the visual rays do in some cases glance off, they will be weak and dim and prematurely exhausted because of the magnitude of the distance. What is more too, whereas mirrors that are concave make the ray of light more intense after reflection than it was before so as often even to send off flames, convex and spherical mirrors by not exerting counter-pressure upon it from all points give it off weak and faint. You observe, I presume, whenever two rainbows appear, as one cloud encloses another, that the encompassing rainbow produces colours that are faint and indistinct. The reason for this is that the outer cloud, being situated

further off from the eye, returns reflection that is not intense or strong. Nay, what need of further arguments? When the light of the sun by being reflected from the moon loses all its heat and of its brilliance there barely reaches us a slight and feeble remnant, is it really possible that of the visual ray travelling the same double-course any fraction of a remnant should from the moon arrive at the sun? For my part, I think not; and do you too," I said, "consider this. If the visual ray were affected in the same way by warm water and by the moon, the full moon ought to show such reflections of the earth and plants and human beings and stars as all other mirrors do; but, if there occur no reflections of the visual ray to these objects either because of the weakness of the ray or the ruggedness of the moon, let us not require that there be such reflection to the sun either.

241 So we for our part," said I, "have now reported as much of that conversation as has not slipped our mind; and it is high time to summon Sulla or rather to demand his narrative as the agreed condition upon which he was admitted as a listener. So, if it is agreeable, let us stop our promenade and sit down upon the benches, that we may provide him with a settled audience." To this then they agreed; and, when we had sat down, Theon said: "Though, as you know, Lamprias, I am as eager as any of you to hear what is going to be said, I should like before that to hear about that beings that are said to dwell on the moon — not whether any really do inhabit it but whether habitation there is possible. If it is not possible, the assertion that the moon is an earth is itself absurd, for she would then appear to have come into existence vainly and to no purpose, neither bringing forth fruit nor providing for men of some kind an origin, an abode, and a means of life, the purposes for which this earth of ours came into being, as we say with Plato, 'our nurse, strict guardian and artificer of day and night.' You see that there is much talk about these things both in jest and seriously. It is said that those who dwell under the moon have her suspended overhead like the stone of Tantalus and on the other hand that those who dwell upon her, fast bound like so many Ixions by such great velocity, Fare kept from falling by being whirled round in a circle. Yet it is not with a single motion that she moves; but she is, as somewhere she is in fact called, the goddess of three ways, for she moves on the zodiac against the signs in longitude and latitude and in depth at the same time. Of these movements the mathematicians call the first 'revolution,' the second 'spiral,' and the third, I know not why 'anomaly,' although they see that she has no motion at all that is uniform and fixed by regular recurrences. There is reason to wonder then not that the velocity caused a lion to fall on the Peloponnesus but how it is that we are not forever seeing countless

Men falling headlong and lives spurned away,

tumbling off the moon, as it were, and turned head over heels. It is moreover ridiculous to raise the question how the inhabitants of the moon remain there, if they cannot come to be or exist. Now, when Egyptians and Troglodytes, for whom the sun stands in the zenith one moment of one day at the solstice and then departs, are all but burnt to a cinder by the dryness of the atmosphere, is it really likely that the men on the

moon endure twelve summers every year, the sun standing fixed vertically above them each month at the full moon? Yet winds and clouds and rains, without which plants can neither arise nor having arisen be preserved, because of the heat and tenuousness of the atmosphere cannot possibly be imagined as forming there, for not even here on earth do the lofty mountains admit fierce and continual storms but the air, being tenuous already and having a rolling swell as a result of its lightness, escapes this compaction and condensation. otherwise, by Heaven, we shall have to say that, as Athena when Achilles was taking no food instilled into him some nectar and ambrosia, so the moon, which is Athena in name and fact, nourishes her men by sending up ambrosia for them day by day, the food of the gods themselves as the ancient Pherecydes believes. For even the Indian root which according to Megasthenes the Mouthless Men, who neither eat nor drink, kindle and cause to smoulder and inhale for their nourishment, how could it be supposed to grow there if the moon is not moistened by rain?"

25 1 When Theon had so spoken, I said "Bravo, you have most excellently smoothed our brows by the sport of your speech, wherefore we have been inspired with boldness to reply, since we anticipate no very sharp or bitter scrutiny. It is, moreover, a fact that there really is no difference between those who in such matters are firm believers and those who are violently annoyed by them and firmly disbelieve and refuse to examine calmly what can be and what might be. So, for example, in the first place, if the moon is not inhabited by men, it is not necessary that she have come to be in vain and to no purpose, for we see that this earth of ours is not productive and inhabited throughout its while extent either but only a small part of it is fruitful of animals and plants on the peaks, as it were, and peninsulas rising out of the dee, while of the rest some parts are desert and from less with winter-storms and summer-droughts and the most are sunk in the great sea. You, however, because of your constant fondness and admiration for Aristarchus, give no heed to the text that Crates read:

Ocean, that is the universal source
Of men and gods, spreads over most of earth.

Yet it is by no means for nothing that these parts have come to be. The sea gives off gentle exhalations, and the most pleasant winds when summer is at its height are released and dispersed from the uninhabited and frozen region by the snows that are gradually melting there. 'A strict guardian and artificer of day and night' has according to Plato been stationed in the centre. Nothing then prevents the moon too, while destitute of living beings, from providing reflections for the light that is diffused about her and for the rays of the stars a point of confluence in herself and a blending whereby she digests the exhalations from the earth and at the same time slackens the excessive torridity and harshness of the sun. Moreover, conceding a point perhaps to ancient tradition also, we shall say that she was held to be Artemis on the ground that she is a virgin and sterile but is helpful and beneficial to other females. In the second place,

my dear Theon, nothing that has been said proves impossible the alleged inhabitation of the moon. As to the rotation, since it is very gentle and serene, it smooths the air and distributes it in settled order, so that there is no danger of falling and slipping off for those who stand there. And if it is not simple either, even this complication and variation of the motion is not attributable to irregularity or confusion; but in them astronomers demonstrate a marvellous order and progression, making her revolve with circles that unroll about other circles, some assuming that she is herself motion less and others that she retrogresses smoothly and regularly with ever constant velocity, for these superpositions of the circles and their rotations and relations to one another and to us combine most harmoniously to produce the apparent variations of her motion in altitude and the deviations in latitude at the same time as her revolutions in longitude. As to the great heat and continual scorching of the sun, you will cease to fear it, if first of all you set the conjunctions over against the universe summery full-moons and suppose that the continuousness of the change produces in the extremes, which do not last a long time, a suitable tempering and removes the excess from either. Between these then, as is likely, they have a season most nearly approaching spring. In the second place, upon us the sun sends, through air which is turbid and nourished by the exhalations, whereas there the air being tenuous and translucent scatters and diffuses the sun's light, which has no tinder or body to sustain it. The fruits of tree and field here in our region are nourished by rains; but elsewhere, as up in your home around Thebes and Syene, the land drinking water that springs from earth instead of rain-water and enjoying breezes and dews would refuse, I think, to adapt itself to the fruitfulness that attends the most abundant rainfall, and that because of a certain excellence and temperament that it has. Plants of the same kind, which in our region if sharply nipped in by winter bear good fruit in abundance, in Libya and in your home in Egypt are very sensitive to cold and afraid of winter. And, while Gedrosia and Ethiopia which comes down to the ocean is barren and entirely treeless because of the aridity, in the adjacent and surrounding sea there grow and thrive down in the deep plants of great magnitude, some of which are called olives, some laurels, and some tresses of Isis; and the plants here called 'love-restorers' when lifted out of the earth and hung up not only live as long as you wish but sprout < . . >. Some plants are sown towards winter, and some at the height of summer as sesame and millet. Thyme or centaury, if sown in good, rich soil and wetted and watered, departs from its natural quality and loses its strength, whereas drought delights it and causes it to reach its proper stature; and some plants, as they say, cannot stand even dew, as is true of the majority of Arabian plants, but are blighted and destroyed by being moistened. What wonder then if on the moon there grow roots and seeds and trees that have no need of rain nor yet of snow but are naturally adapted to a summery and rarefied air? And why is it unlikely that winds arise warmed by the moon and that breezes steadily accompany the rolling swell of her revolution by scattering off and diffusing dews and light moisture suffice for the vegetation and that she herself is not fiery or dry in temperament but soft and humidifying? After all, no influence of dry and

comes to us from her but much of moistness and femininity: the growth of plants, the decay of meats, the souring and flattening of wine, the softening of timbers, the easy delivery of women. Now that Pharnaces is quiet I am afraid of provoking and arousing him again if I cite, in the words of his own school, the flood-tides of Ocean and the swelling of the straits when they are increased and poured abroad by the liquefying action of the moon. Therefore I shall rather turn to you, my dear Theon, for when you expound these words of Alcman's,

Such as are nourished by Dew, daughter of Zeus and of divine Selene,
you tell us that at this point he calls the air 'Zeus' and says that it is liquefied by the moon and turns to dew-drops. It is in fact probable, my friend, that the moon's nature is contrary to that of the sun, if of herself she not only naturally softens and dissolves all that he condenses and dries but liquefies and cools even the heat that he casts upon her and imbues her with. They err then who believe the moon to be a fiery and glowing body; and those who demand that living beings there be equipped just as those here are for generation, nourishment, and livelihood seem blind to the diversities of nature, among which one can discover more and greater differences and dissimilarities between living beings than between them and inanimate objects. Let there not be mouthless men nourished by odours who Megasthenes thinks do exist; yet the Hungerbane, the virtue of which he was himself trying to explain to us, Hesiod hinted at when he said

Nor what great profit mallow has and squill
and Epimenides made manifest in fact when he showed that with a very little fuel nature kindles and sustains the living creature, which needs no further nourishment if it gets as much as the size of an olive. It is plausible that the men on the moon, if they do exist, are slight of body and capable of being nourished by whatever comes their way. After all, they say that the moon herself, like the sun which is an animate being of fire many times as large as the earth, is nourished by the moisture on the earth, as are the rest of the stars too, though they are countless; so light and frugal of requirements they do conceive the creatures to be that inhabit the upper region. We have no comprehension of these beings, however, nor of the kind that a different place and nature and temperature are suitable to them. Just as, assuming that we were unable to approach the sea or touch it but only had a view of it from afar and the information that it is bitter, unpotable, and salty water, if someone said that it supports in its depths many large animals of multifarious shapes and is full of beasts that use water for all the ends that we use air, his statements would seem to us like a tissue of myths and marvels, such appears to be our relation to the moon and our attitude towards her is apparently the same when we disbelieve that any men dwell there. Those men, I think, would be much more amazed at the earth, when they look out at the sediment and dregs of the universe, as it were, obscurely visible in moisture, mists, and clouds as a light less, low, and motionless spot, to think that it engenders and nourishes animate beings which partake of motion, breath, and warmth. If they should chance to hear somewhere these Homeric words,

Dreadful and dank, which even gods abhor
and

Deep under Hell as far as Earth from Heaven,

these you would say are simply a description of this place and Hell and Tartarus have been relegated hither while the moon alone is earth, since it is equally distant from those upper regions and these lower ones.”

(940 F) 26 1 Almost before I had finished, Sulla broke in. “Hold on, Lamprias,” he said, “and put to the wicket of your discourse lest you unwittingly run the myth aground, as it were, and confound my drama, which has a different setting and a different disposition. 941 Well, I am but the actor of the piece, but first I shall say that its author began for our sake — if there be no objection — with a quotation from Homer:

An isle, Ogygia, lies far out at sea,

run of five days off from Britain as you sail westward; and three other islands equally distant from it and from one another lie out from it in the general direction of the summer sunset. In one of these, according to the tale told by the natives, Cronus is confined by Zeus, and the antique Briareus, holding watch and ward over those islands and the sea that they call the Cronian main, has been settled close beside him. The great mainland, by which the great ocean is encircled, while not so far from the other islands, is about five thousand stades from Ogygia, the voyage being made by oar, for the main is slow to traverse and muddy as a result of the multitude of streams. The streams are discharged by the great land-mass and produce alluvial deposits, thus giving density and earthiness to the sea, which has been thought actually to be congealed. On the coast of the mainland Greeks dwell about a gulf which is not smaller than the Maeotis and the mouth of the Caspian sea. These people consider and call themselves continentals and the inhabitants of this land islanders because the sea flows around it on all sides; and they believe that with the peoples of Cronus there mingled at a later time those who arrived in the train of Heracles and were left behind by him and that these latter so to speak rekindled again to a strong, high flame the Hellenic spark there which was already being quenched and overcome by the tongue, the laws, and the manners of the barbarians. Therefore Heracles has the highest honours and Cronos the second. Now when at intervals of thirty years the star of Cronus, which we call ‘Splendent’ but they, our author said, call ‘Night-watchman,’ enters the sign of the Bull, they, having spent a long time in preparation for the sacrifice and the expedition, choose by lot and send forth a sufficient number of envoys in a correspondingly sufficient number of ships, putting aboard a large retinue and the provisions necessary for men who are going to cross so much sea by oar and live such a long time in a foreign land. Now when they have put to sea the several voyagers meet with various fortunes as one might expect; but those who survive the voyage first put in at the outlying islands, which are inhabited by Greeks, and see the sun pass out of sight for less than an hour over a period of thirty days, — and this is night, though it has a darkness that is slight and twilight glimmering from the west. There they spend ninety days regarded with honour and friendliness as

holy men and so addressed, and then winds carry them across to their appointed goal. Nor do any others inhabit it but themselves and those who have been dispatched before them, for, while those who have served the god together for the stint of thirty years are allowed to sail off home, most of them usually choose to settle in the spot, some out of habit and others because without toil or trouble they have all things in abundance while they constantly employ their time in sacrifices and celebrations or with various discourse and philosophy, for the nature of the island is marvellous as is the softness of the circumambient air. Some when they intend to sail away are even hindered by the divinity which presents itself to them as to intimates and friends not in dreams only or by means of omens, but many also come upon the visions and the voices of spirits manifest. For Cronus himself sleeps confined in a deep cave of rock that shines like gold — the sleep that Zeus has contrived like a bond for him — , and birds flying in over the summit of the rock bring ambrosia to him, and all the island is suffused with fragrance ⁹⁴² scattered from the rock as from a fountain; and those spirits mentioned before tend and serve Cronus, having been his comrades what time he ruled as king over gods and men. Many things they do foretell of themselves, for they are oracular; but the prophecies that are greatest and of the greatest matters they come down and report as dreams of Cronus, for all that Zeus premeditates Cronus sees in his dreams and the titanic affections and motions of his soul make him rigidly tense until sleep restores his repose once more and the royal and divine element is all by itself, pure and unalloyed. Here then the stranger was conveyed, as he said, and while he served the god became at his leisure acquainted with astronomy, in which he made as much progress as one can by practising geometry, and with the rest of philosophy by dealing with so much of it as is possible for the natural philosopher. Since he had a strange desire and longing to observe the Great Island (for so, it seems, they call our part of the world), when the thirty years had elapsed, the relief-party having arrived from home, he saluted his friends and sailed away, lightly equipped for the rest but carrying a large viaticum in golden beakers. Well, all his experiences and all the men whom he visited, encountering sacred writings and being initiated in all rites — to recount all this as he reported it to us, relating it thoroughly and in detail, is not a task for a single day; but listen to so much as is pertinent to the present discussion. He spent a great deal of time in Carthage inasmuch as Cronus receives great honour in our country, and he discovered certain sacred parchments that had been secretly spirited off to safety when the earlier city was being destroyed and had lain unnoticed in the ground for a long time. Among the visible gods he said that one should especially honour the moon, and so he kept exhorting me to do, inasmuch as she is sovereign over life and death, bordering as she does upon the meads of Hades.

^{27 1} When I expressed surprise at this and asked for a clearer account, Dhe said: ‘Many assertions about the gods, Sulla, are current among the Greeks, but not all tom are right. So, for example, although they give the right names to Demeter and Cora, they are wrong in believing that both are together in the same region. The fact is that the

former is in the region of earth and is sovereign over terrestrial things, and the latter is in the moon and mistress of lunar things. She has been called both Cora and Phersephonê, the latter as being a bearer of light and Cora because that is what we call the part of the eye in which is reflected the likeness of him who looks into it as the light of the sun is seen in the moon. The tales told of the wandering and the quest of these goddesses Econtain the truth <spoken covertly>, for they long for each other when they are apart and they often embrace in the shadow. The statement concerning Cora that now she is in the light of heaven and now in darkness and night is not false but has given rise to error in the computation of the time, for not throughout six months but every six months we see her being wrapped in shadow by the earth as it were by her mother, and infrequently we see this happen to her at intervals of five months, for she cannot abandon Hades since she is the boundary of Hades, as Homer too has rather well put it in veiled terms:

But to Elysium's plain, the bourne of earth.

Where the range of the earth's shadow ends, this he set as the term and boundary of the earth. To this point rises no one who is evil or unclean, but the good are conveyed thither after death and there continue to lead a life most easy to be sure though not blessed or divine until their second death.

281 And what is this, Sulla? Do not ask about these things, for I am going to give a full explanation myself. 943 Most people rightly hold man to be composite but wrongly hold him to be composed of only two parts. The reason is that they suppose mind to be somehow part of soul, thus erring no less than those who believe soul to be part of body, for in the same degree as soul is superior to body so is mind better and more divine than soul. The result of soul and body commingled is the irrational or the affective factor, whereas of mind and soul the conjunction produces reason; and of these the former is source of pleasure and pain, the latter of virtue and vice. In the composition of these three factors earth furnishes the body, the moon the soul, and the sun furnishes mind to man for the purpose of his generation even as it furnishes light to the moon herself. As to the death we die, one death reduces man from three factors to two and another reduces him from two to one; and the former takes place in the earth that belongs to Demeter (wherefore "to make an end" is called "to render one's life to her" and Athenians used in olden times to call the dead "Demetrians"), the latter in the moon that belongs to Phersephonê, and associated with the former is Hermes the terrestrial, with the latter Hermes the celestial. While the goddess here dissociates the soul from the body swiftly and violently, Phersephonê gently and by slow degrees detaches the mind from the soul and has therefore been called "single-born" because the best part of man is "born single" when separated off by her. Each of the two separations naturally occurs in this fashion: All soul, whether without mind or with it, when it has issued from the body is destined to wander in the region between earth and moon but not for an equal time. Unjust and licentious souls pay penalties for their offences; but the good souls must in the gentlest part of the air, which they call "the meads of Hades,"

pass a certain set time sufficient to purge and blow away the pollutions contracted from the body as from an evil odour. Then, as if brought home from banishment abroad, they savour joy most like that of initiates, which attended by glad expectation is mingled with confusion and excitement. For many, even as they are in the act of clinging to the moon, she thrusts off and sweeps away; and some of those souls too that are on the moon they see turning upside down as if sinking again into the deep. Those that have got up, however, and have found a firm footing first go about like victors crowned with wreaths of feathers called wreaths of steadfastness, because in life they had made the irrational or affective element of the soul orderly and tolerably tractable to reason; secondly, in appearance resembling a ray of light but in respect of their nature, which in the upper region is buoyant as it is here in ours, resembling the ether about the moon, they get from it both tension and strength as edged instruments get a temper, for what laxness and diffuseness they still have is strengthened and becomes firm and translucent. In consequence they are nourished by any exhalation that reaches them, and Heraclitus was right in saying: "Souls employ the sense of smell in Hades."

29 1 First they behold the moon as she is in herself: her magnitude and beauty and nature, which is not simple and unmixed but a blend as it were of star and earth. Just as the earth has become soft by having been mixed with breath and moisture and as blood gives rise to sense-perception in the flesh with which it is commingled, so the moon, they say, because it has been permeated through and through by ether is at once animated and fertile and at the same time has the proportion of lightness to heaviness in equipoise. In fact it is in this way too, they say, that the universe itself has entirely escaped local motion, because it has been constructed out of the things that naturally move upwards and those that naturally move downwards. This was also the conception of Xenocrates who, taking his start from Plato, seems to have reached it by a kind of superhuman reasoning. Plato is the one who declared that each of the stars as well was constructed of earth and fire bound together in a proportion by means of the two intermediate natures, for nothing, as he said, attains perceptibility that does not contain an admixture of earth and light; but Xenocrates says that the stars and the sun are composed of fire and the first density, the moon of the second density and air that is proper to her, and the earth of water and air and the third kind of density and that in general neither density all by itself nor subtilty is receptive of soul. So much for the moon's substance. As to her breadth or magnitude, it is not what the geometers say but many times greater. She measures off the earth's shadow with few of her own magnitudes not because it is small but she more ardently hastens her motion in order that she may quickly pass through the gloomy place bearing away the souls of the good which cry out and urge her on because when they are in the shadow they no longer catch the sound of the harmony of heaven. At the same time too with wails and cries the souls of the chastised then approach through the shadow from below. That is why most people have the custom of beating brasses during eclipses and of raising a din and clatter against the souls, which are frightened off also by the so called face when they

get near it, for it has a grim and horrible aspect. It is no such thing, however; but just as our earth contains gulfs that are deep and extensive, one here pouring in towards us through the Pillars of Heracles and outside the Caspian and the Red Sea with its gulfs, so those features are depths and hollows of the moon. The largest of them is called "Hecatê's Recess," where the souls suffer and exact penalties for whatever they have endured or committed after having already become Spirits; and the two long ones are called "the Gates", for through them pass the souls now to the side of the moon that faces heaven and now back to the side that faces earth. The side of the moon towards heaven is named "Elysian plain," the hither side "House of counter-terrestrial Phersephonê."

30 1 Yet not forever do the Spirits tarry upon the moon; they descend hither to take charge of oracles, they attend and participate in the highest of the mystic rituals, they act as warders against misdeeds and chastisers of them, and they flash forth as saviour a manifest in war and on the sea. For any act that they perform in these matters not fairly but inspired by wrath or for an unjust end or out of envy they are penalized, for they are cast out upon earth again confined in human bodies. To the former class of better Spirits the attendants of Cronos said that they belong themselves as did aforetime the Idaean Dactyls in Crete and the Corybants in Phrygia as well as the Boeotian Trophoniads in Udora and thousands of others in many parts of the world whose rites, honours, and titles persist but whose powers tended to another place as they achieved the ultimate alteration. They achieve it, some sooner and some later, once the mind has been separated from the soul. It is separated by love for the image in the sun through which shines forth manifest the desirable and fair and divine and blessed towards which all nature in one way or another yearns, for it must be out of love for the sun that the moon herself goes her rounds and gets into conjunction with him in her yearning to receive from him what is most fructifying. The substance of the soul is left upon the moon and retains certain vestiges and dreams of life as it were; it is this that you must properly take to be the subject of the statement

Soul like a dream has taken wing and sped,

for it is not straightway nor once it has been released from the body that it reaches this state but later when, divorced from the mind, it is deserted and alone. Above all else that Homer said his words concerning those in Hades appear to have been divinely inspired

Thereafter marked I mighty Heracles —

His shade; but he is with the deathless god. . . .

In fact the self of each of us is not anger or fear or desire just as it is not bits of flesh or fluids either but is that which we reason and understand; and the soul receives the impression of its shape through being moulded by the mind and moulding in turn and enfolding the body on all sides, so that, even if it be separated from either one for a long time, since it preserves the likeness and the imprint it is correctly called an image. Of these, as has been said, the moon is the element, for they are resolved into it as the

bodies of the dead are resolved into earth. This happens quickly to the temperate souls who had been fond of a leisurely, unmeddlesome, and philosophical life, for abandoned by the mind and no longer exercising the passions for anything they quickly wither away. Of the ambitious and the active, the irascible and those who are enamoured of the body, however, some pass their time as it were in sleep with the memories of their lives for dreams as did the soul of Endymion; but, when they are excited by restlessness and emotion and drawn away from the moon to another birth, she forbids them <to sink towards earth> and keeps conjuring them back and binding them with charms, for it is no slight, quiet, or harmonious business when with the affective faculty apart from reason they seize upon a body. Creatures like Tityus and Typho and the Python that with insolence and violence occupied Delphi and confounded the oracle belonged to this class of souls, void of reason and subject to the affective element gone astray through delusion; but even these in time the moon took back to herself and reduced to order. Then when the sun with his vital force has again sowed mind in her she receives it and produces new souls, and earth in the third place furnishes body. In fact, the earth gives nothing in giving back after death all that she takes for generation, and the sun takes nothing but takes back the mind that he gives, whereas the moon both takes and gives and joins together and divides asunder in virtue of her different powers, of which the one that joins together is called Ilithyia and that which divides asunder Artemis. Of the three Fates too Atropos enthroned in the sun initiates generation, Clotho in motion on the moon mingles and binds together, and finally upon the earth Lachesis too puts her hand to the task, she who has the largest share in chance. For the inanimate is itself powerless and susceptible to alien agents, and the mind is impassable and sovereign; but the soul is a mixed and intermediate thing, even as the moon has been created by god a compound and blend of the things above and below and therefore stands to the sun in the relation of earth to moon.'

This," said Sulla, "I heard the stranger relate; and he had the account, as he said himself, from the chamberlains and servitors of Cronus. You and your companions, Lamprias, may make what you will of the tale."

On the principle of cold



(945 F) 1 1 Is there, then, Favorinus, an active principle or substance of Cold (as fire is of Heat) through the presence of which and through participation in which everything else becomes cold? Or is coldness rather a negation of warmth, as they say darkness is of light and rest of motion? ⁹⁴⁶ Cold, indeed, seems to have the quality of being stationary, as heat has that of motion; while the cooling off of hot things is not caused by the presence of any force, but merely by the displacement of heat, for it can be seen to depart completely at the same time as the remainder cools off. The steam, for example, which boiling water emits, is expelled in company with the departing heat; that is why the amount becomes less by cooling off; for this removes the heat and nothing else takes place.

2 1 First of all, must we not be wary of one point in this argument? It eliminates many obvious forces by considering them not to be qualities or properties, but merely the negation of qualities or properties, weight being the negation of lightness and hardness that of softness, black that of white, and bitter that of sweet, and so in any other case where there is a natural opposition of forces rather than a relation of positive and negative. Another point is that all negation is inert and unproductive: blindness, for example, and deafness, silence or death. Here you have the defection of a definite form and the annihilation of a reality, not something that is in itself a part of nature or reality. It is the nature of coldness, however, to produce affects and alterations in bodies that it enters no less than those caused by heat. Many objects can be frozen solid, or become condensed or made viscous, by cold. Moreover, the property whereby coldness promotes rest and resists most is not inert, but acts by pressure and resistance, being constrictive and preservative because of its strength. This explains how, though negation is a disappearance and departure of the contrary force, many things may yet become cold while all the time containing within themselves considerable warmth. There are even some objects which cold solidifies and consolidates the more readily the hotter they are: steel, for example, plunged in water. The Stoics also affirm that in the bodies of infant children the breath is tempered by cooling and, from being a physical substance, becomes a soul. This, however, is debatable; yet since there are many other effects which may be seen to be produced through the agency of cold, we are not justified in regarding it as a negation.

3 1 Besides, a negation does not permit degrees of less or more. Surely nobody will affirm that one blind man is blinder than another, or one dumb man more silent than another, or one corpse deader than its fellow; but among cold things there is a wide range of deviation from much to little, from very cold to not very, and, generally

speaking, in degrees of intensity and remission, just as there is in hot things. This occurs because the matter involved is in different cases acted upon by the opposing forces with more or less intensity; it thus exhibits degrees of one or the other, and so of hot and cold. There is, in fact, no such thing as a blending of positive qualities with negative ones, nor may any positive force accept the assault of the negation that corresponds to it or take it into partnership; instead it gives place to it. Now hot things do admit a blending with cold up to a point, just as do black with white, high notes with low, sweet tastes with sour; and this harmonious association of colours and sounds, drugs and sauces, produces many combinations that are pleasant and grateful to the senses.

For the opposition of a negation to a positive quality is an irreconcilable hostility, since the existence of the one is the annihilation of the other. The other opposition, however, of positive forces, if it occurs in due measure, is often operative in the arts, and very often indeed in various phenomena of nature, especially in connexion with the weather and the seasons and those matters from which the god derives his title of harmonizer and musician, because he organizes and regulates them. He does not receive these names merely for bringing sounds of high and low pitch, or black and white colours, into harmonious fellowship, but because he has authority over the association and disunion of heat and cold in the universe, to see that they observe due measure in their combination and separation, and because, by eliminating the excess of either, he brings both into proper order.

4 1 947 Furthermore, we find that cold can be perceived as well as heat; but mere negation cannot be seen or heard or touched or recognized by the other senses. Perception, in fact, must be of something existent; but where nothing existent is observed, privation may be inferred, being the negation of existence, as blindness is of sight, silence of sound, void and emptiness of matter. We cannot perceive a void by touch; but where no matter can be touched, void is inferred. Nor can we hear silence; yet, even though we hear nothing, we infer silence. Nor, in the same way, is sense active when things are unseen or bare; there is, rather, inference from the negation of perception. If, therefore, cold were a privation of warmth, we ought not to be able to feel it, but only to infer it from the deficiency in warmth; but if cold is perceived by the contraction and condensation of our flesh (just as heat is by the warming and loosening of it), clearly there is some special first principle and source of coldness, just as there is of heat.

5 1 And yet another point: privation of any sort is something simple and uncomplicated, whereas substances have many differences and powers. Silence, for example, is of only one kind, while sound varies, sometimes annoying, at other times delighting, the perception. Both colours and figures show the same variation, for they produce different effects on different occasions when they meet the eye; but that which cannot be touched and is without colour or any quality whatever, admits no difference, but is always the same.

61 Is cold, then, so like this sort of privation that it produces no effects that differ? Or is the contrary true: Do not great and useful pleasures accrue to our bodies from the presence of cold, as well as mighty detriments and pains and depressions, before which the heat does not always depart and quit the field? Often, rather, though cut off within, it makes a stand and gives battle. This struggle of hot and cold is called shivering or shaking; and if heat is overcome, freezing and torpor set in; but if cold is defeated, there is diffused through the body a relaxed and pleasantly warm sensation which Homer calls “to be aglow.” Surely these facts are obvious to everyone; and it is chiefly by these effects that cold is shown to be in opposition to heat, not as a negation or privation, but as one substance or one state to another: it is not a mere destruction or abolition of heat, but a positive substance or force. Otherwise we might just as well exclude winter from the list of seasons or the northerly blasts from that of winds, on the pretext that they are only a deficiency of hot weather or southerly gales and have no proper origin of their own.

71 Furthermore, given four primary bodies in the universe which, because of their quantity, simplicity, and potentiality, most judges regard as being the elements or first principles of everything else — I mean fire, water, air, and earth — the number of primary, simple qualities must be the same. And what should these be but warmth and cold, dryness and moisture, which by their very nature cause all the elements to act and be acted upon? Just as in grammar we have elements long and short and in music elements high and low in pitch — and in neither case is one element merely a negation of the other — so also in physical bodies we must assume an elementary opposition of wet to dry and cold to hot, and in this way we shall be faithful both to logic and to experience. Or are we, as old Anaximenes maintained, to leave neither hot nor cold in the realm of being, but to treat them as states belonging equally to any matter and occurring as a result of changes within it? He affirms, in fact, that anything which undergoes contraction and condensation of matter is cold, while anything that suffers rarefaction and distention — this comes close to his phrasing — is hot. So there is no contradiction in the remark that the man blew both hot and cold, for breath grows cold when it is compressed and condensed by the lips; 948 but when it is expelled from the mouth left slack, it becomes hot through rarefaction. Aristotle, however, holds that in this Anaximenes were mistaken: when the mouth is slack, what is exhaled is warm air from our own bodies; but when we compress the lips and blow, it is not air from ourselves, but the cold air in front of the mouth that is propelled forward and makes contact.

81 Perhaps we should now leave the question whether heat and cold are substances; if so, let us advance the argument to the next point and inquire what sort of substance coldness has, and what is its first principle and nature. Now those who affirm that there are certain uneven, triangular formations in our bodies and that shivering and trembling, shuddering and the like manifestations, proceed from this rough irregularity, even if they are wrong in the particulars, at least derive the first principle from the proper place; for

the investigation should begin as it were from the very hearth, from the substance of all things. This is, it would seem, the great difference between a philosopher and a physician or a farmer or a flute-player; for the latter are content to examine the causes most remote from the first cause, since as soon as the most immediate cause of an effect is grasped — that fever is brought about by exertion or an overflow of blood, that rusting of grain is caused by days of blazing sun after a rain, that a low note is produced by the angle and construction of the pipes — that is enough to enable a technician to do his proper job. But when the natural philosopher sets out to find the truth as a matter of speculative knowledge, the discovery of immediate causes is not the end, but the beginning of his journey to the first and highest causes. This is the reason why Plato and Democritus, when they were inquiring into the causes of heat and heaviness, were right not to stop their investigation with earth and fire, but to go on carrying back sensible phenomena to rational origins until they reached, as it were, the minimum number of seeds.

91 Nevertheless it is better for us first to attack things perceptible to the senses, in which Empedocles and Strato and the Stoics locate the substances that underlie the qualities, the Stoics ascribing the primordially cold to the air, Empedocles and Strato to water; and someone else may, perhaps, be found to affirm that earth is the original substance of coldness. But let us examine Stoic doctrine before the others.

Since fire is not only warm but bright, the opposite natural entity (they say) must be both cold and dark: as gloomy is the opposite of bright, so is cold of hot. Besides, as darkness confounds the sight, so cold confuses the sense of touch. Heat, on the other hand, transmits the sensation of touching, as brightness does that of seeing. It follows, then, that in nature the primordially dark is also the primordially cold; and that it is air which is primordially dark does not, in fact, escape the notice of the poets since they use the term “air” for “darkness”:

Thick air lay all about the ships, nor could

The moon shine forth from heaven.

and another instance:

So clad in air they visit all the earth.

And another:

The air at once he scattered and dispelled the mist;

The sun shone forth and all the battle came in view.

They also call the lightless air *knephas*, being as it were, *kenon phaos* “void of light”; and collected and condensed air has been termed *nephos* “cloud” because it is a negation of light. Flecks in the sky and mist and fog and anything else that does not provide a transparent medium for light to reach our senses are merely variations of air; and its invisible and colourless part is called *Hades* and *Acheron*. In the same way, then, as air is dark when light is gone, so when heat departs the residue is cold air and nothing else. And this is the reason why it has been termed *Tartarus* because of its

coldness. Hesiod makes this obvious when he writes “murky Tartarus”; and to shake and shiver with cold is to “tartarize.” Such, then, is the reason for these names.

10 1 949 Since corruption, in each case, is a change of the things that are corrupted into their opposites, let us see whether the saying holds good that “the death of fire is the birth of air.” Fire, indeed, perishes like a living creature, being either extinguished by main force or dying out of itself. Now if it is extinguished, that makes the change of fire into air more conspicuous. Smoke, in fact, is a form of air, as is reek and exhalation, which, to quote Pindar,

Stabs at the air with unctuous smoke.

Nevertheless, even when fire goes out for lack of nourishment, one may see, as for instance in the case of lamps, the apex of the flame passing off into murky, dusky air. Moreover, the vapour ascending from our bodies when, after a bath or sweat, cold water is poured on them, sufficiently illustrates the change of heat, as it were, into the air; and this implies that it is the natural opposite of fire. From this the Stoics drew the conclusion that air was primordially dark and cold.

11 Moreover, freezing, which is the most extreme and violent effect of cold in bodies, is a condition of water, but a function of air. For water of itself is fluid, uncongealed and not cohesive; but when it is compressed by air because of its cold state, it becomes taut and compact. This is the reason for the saying

If Southwind challenges North, instantly snow will appear.

For after the Southwind has collected the moisture as raw material, the Boreal air takes over and congeals it. This is particularly evident in snowfields: when they have discharged a preliminary exhalation of air that is thin and cold, they melt. Aristotle also declares that whetstones of lead will melt and become fluid in the wintertime through excess of cold when no water is anywhere near them; it seems probable that the air with its coldness forces the bodies together until it crushes and breaks them.

12 1 Furthermore, portions of water will freeze sooner than the spring from which they are drawn, for the air more readily masters the smaller amount. If you will draw from a cold well cold water in a jar and let it down again into the well in such a way that the jar does not touch the water, but is suspended in the air, and if you wait a short time, you will find that the water has become colder. This is very good evidence that the First Cause of coldness is not water but air. Certainly, none of the great rivers freezes through its entire depth; for the air does not penetrate down into the whole, but merely renders stationary as much as, by contact and proximity, it includes within the range of its coldness. And this is the reason why barbarians do not cross frozen rivers until they have tried them out with foxes: if the ice is not thick, but merely superficial, the foxes perceive this by the sound of the current running underneath and return to the bank. Some even catch fish by weakening and softening the ice with hot water — enough of the ice,

at least, to admit their lines; so the cold has no effect at a depth. Yet the water near the surface undergoes so great a change through freezing that ships are crushed by it when it is forced in on itself and squeezed tight, as those relate who recently passed the winter with Caesar on the Danube. Nevertheless, what happens in our own case is ample testimony: after warm baths and sweats we are cooler, since our bodies are relaxed and porous, so that we take in a good deal of cold along with the air. The same thing happens to water, too: it freezes faster when it has first been heated, thus becoming more susceptible to air; and those who draw off boiling water and suspend it in the air do this, surely, only to secure the admixture of great quantities of air. So now, Favorinus, the argument that attributes the primal force of cold to the air depends on such plausibilities as these.

13 1 But the argument which attributes it to water finds in the same way facts to support it; Empedocles says something like this:

Behold the sun, everywhere bright and warm;
And then the rain, to all men dark and cold.

By thus setting cold against hot, as he does dark against bright, he has given us to understand that dark and cold belong to the same substance, as do also bright and hot. ⁹⁵⁰ And our senses bear witness that darkness is an attribute of water, not of air, since nothing, to put it simply, is blackened by air and everything is by water. For if you throw the whitest wool or the whitest garment into the water, it will come out black and it will remain black until the moisture is evaporated by heat or is squeezed out by some sort of wringing or pressure. When a patch of ground is sprinkled, the pots which are covered by the drops turn black, but the rest remains as it was. In fact, of water itself the deepest looks the darkest because there is so much of it, while those parts that lie near the air flash and sparkle; and of the other liquids oil is the most transparent, as containing the most air. A proof of this is its lightness, by reason of which it maintains itself on the surface of all other things, buoyed up by the air. If it is sprinkled upon the waves, it will calm the sea, not because it is so smooth that the winds slip off it, as Aristotle affirmed; but because the waves are dissipated when they are struck by any moist substance. But it is peculiar to oil that it provides light and sight at the bottom since the moist elements are interspersed with air; it is, in fact, not only on the surface that it provides light for those who pass the night at sea; Cit does so also for sponge-divers below the surface when it is blown out of their mouths. Air, therefore, has no greater proportion of darkness than water has, and it has less cold. Certainly oil, which has more air than any other moist substance, is least cold; and when it freezes, it forms a soft jelly: the air that is intermixed does not permit it to freeze hard. They dip needles, iron clasps, and all delicate artifacts in oil rather than in water, fearing that the water's excessive frigidity may distort them. It is, in fact, fairer to judge the argument by this evidence than by that of colour, since snow and hail and ice are at their brightest when they are coldest. Moreover, pitch is both hotter and darker than honey.

141 I am surprised, nevertheless, when those who maintain that the air is cold because it is dark do not perceive that others think it must be hot because it is light. For darkness is not so closely connected and akin to cold as heaviness and stability are; many things, in fact, which have no heat are bright, but nothing cold is buoyant, light, and soaring. Why, the very clouds, as long as they are akin to the substance of air, float aloft; but as soon as they change to moisture, they fall at once and lose their lightness no less than their warmth as coldness grows within them. Contrariwise, when heat supervenes, they reverse the movement again, for their substance begins to soar as soon as it has changed to air.

Nor is the argument from destruction true either; for when anything is destroyed, it does not perish by becoming its opposite, though it does perish by the action of its opposite, as fire, for instance, is changed by water into air. For of water Aeschylus speaks in tragic style, but accurately, as

The riot-quelling justicer of fire.

And when Homer matched Hephaestus against the river and Apollo against Poseidon in the battle, he did it rather as a philosopher than as a poet. And Archilochus expressed himself well on a woman who was of two minds:

With guileful thoughts she bore

In one hand water, in the other fire.

Among the Persians it was the most compelling plea to gain an end, one which would admit no refusal, if the suppliant took fire, stood in a river, and threatened that if he lost his suit, he would drop the fire into the water. Now he got how he asked, but though he did so, he was punished for the threat, on the ground that it was contrary to law and against nature. Again, the familiar proverb that is on everyone's lips, "to mix fire with water," as an example of the impossible, seems to bear witness that water is hostile to fire, which is destroyed by it and so is punished by being extinguished; 951 it is not so affected by air, which, on the contrary, supports fire and welcomes it in its changed form. For if anything into which the thing destroyed changes is its opposite, why will fire, any more than water, seem opposite to air? For air changes into water by condensation, and into fire by rarefaction just as, on the other hand, water vanishes into air by rarefaction, but into earth by condensation. Now these processes take place, in my opinion, not because these elements are contrary or hostile to one another, but because they are in close affinity and relationship. But my opponents, whichever way they state their case, ruin their proof. Certainly it is perfectly absurd for them to say that water is frozen by air when they have never seen air itself freezing. For clouds, mists, and flecks in the sky are not congelations, but condensations and thickenings of air that is moist and vaporous. But waterless, dry air never admits loss of heat to the point where such a change might occur. There are, in fact, mountains which do not know clouds or dew or mist because their peaks reach a region of pure air that has no humidity at all. From this fact it is especially obvious that it is the condensation and

density below that contribute to air the cold, moist element that is found in combination with it.

15 1 It is reasonable that the lower portion of large rivers should not freeze; for the upper portion, being frozen, does not transmit the exhalation which is, accordingly, shut in and turned back, and so provides heat for the deep waters. A demonstration of this is the fact that when the ice melts again a great quantity of vapour rises from the waters. This is also the reason why the bodies of animals are warmer in the winter, because the heat is driven inwards by the cold from without and they keep it within them.

Now drawing off water and suspending it in the air not only takes away its warmth, but its coldness also; those, therefore, who want a very cold drink take care not to disturb the snowpacks or the wet matter that is formed from them by compression, for movement expels both heat and cold.

That such a function of cold belongs not to air, but to water, may be demonstrated as follows from a fresh start. In the first place, it is improbable that air, which lies adjacent to the aether and touches and is touched by the revolving fiery substance, should have a force that is contrary to that of aether. For one thing, it is impossible for two substances whose boundaries touch and are contiguous not to be acted upon by each other — and if acted upon, for the weaker not to be contaminated by the force that resides in the stronger. Nor is it reasonable to suppose that Nature has placed side by side destroyer and victim, as though she were the author of strife and dissension, not of union and harmony. She does, indeed, make use of opposites to constitute the universe; yet she does not employ them without a tempering element, or where they will collide. She disposes them rather so that a space is skipped and an inserted strip duly assigned whereby they will not destroy one another, but may enjoy communication and co-operation. And this strip is occupied by air, suffused as it is through a space under the fire between it and water. It makes distribution both ways and receives contributions from both, being itself neither hot nor cold, but a blending and union of the two. When these are so fused, they meet without injury and the fused matter sends forth or takes to itself the opposing extremes without violence.

16 1 Then, too, air is everywhere equal, though neither winter nor cold is identical everywhere. It is no accident that some parts of the world are cold and damp, while others are hot and dry; it is due to the existence of a single substance that includes coldness and wetness in one. The greater part of Africa is hot and without water; while those who have travelled through Scythia, Thrace, and Pontus report that these regions have great lakes or marshes and are traversed by many deep rivers. As for the regions that lie between, those that are near lakes and marshes are especially cold because of the exhalations from the water. Posidonius, then, in affirming that the freshness and moistness of marsh air is the reason for the cold, has done nothing to disturb the plausibility of the argument; he has, rather, made it more plausible. For fresh air would not always seem colder if cold did not take its origin from moisture. So Homer spoke more truly when he affirmed

952 The river-air blows chill before the dawn,
thereby indicating the source of coldness.

Our senses, moreover, often deceive us and we imagine, when we touch cold garments or cold wool, that we are touching moist objects: this is because wet and cold have a common substance and their natures have a close affinity and relationship. In very cold climates the low temperature often breaks vessels whether they are of bronze or of clay — not, of course, when they are empty, but only when they are full and the water exerts pressure by means of its coldness. Theophrastus, to be sure, declares that the air breaks these vessels, using the liquid as a spike. But take care that there isn't more wit than truth in such a remark! For if it were so, vessels full of pitch or of milk would more readily be broken by the air.

Water, however, seems to be cold of itself, and primordially so. It is the antithesis, in its coldness, to the heat of fire, just as in its wetness to the dryness of fire, and in its heaviness to the other's lightness. To sum up: fire is of a disintegrating and separative nature, while water is adhesive and retentive, holding and gluing together by means of its moistness. Empedocles alluded to this, when, as often as he mentioned them, he termed Fire a "Destructive Strife" and Water "Tenacious Love." For the nourishment of fire is that which can be changed into fire and only things that have affinity and a close relationship to it can be so changed; while its opposites, like water, are not easily changed to fire. Water itself is practically incombustible, and it renders matter such as damp grass and moist timber very hard to consume; the greenness in them produces a dusky, dull flame because, by dint of cold, it struggles against heat as against its natural enemy.

171 Now you must pursue the subject by comparing these arguments with those of my opponents. For Chrysippus, thinking that the air is primordially cold because it is also dark, merely mentioned those who affirm that water is at a greater distance from the aether than is air; and, wishing to make them some answer, he said, "If so, we might as well declare that even earth is primordially cold because it is at the greatest distance from the aether" — tossing off this argument as if it were utterly inadmissible and absurd. But I have a mind to maintain the thesis that earth too is not destitute of probable and convincing arguments, and I shall start with the one that Chrysippus has found most serviceable for air. And what is this? Why, that it is primordially dark and cold. For if he takes these two pairs of opposing forces and assumes that one must of necessity accompany the other, there are, surely, innumerable oppositions and antipathies between the aether and the earth with which one might suppose this to be consistent. For it is not only opposed as heavy to light and as moving by gravity downwards, not upwards, or as dense to rare or as slow and stable to mobile and active, but as heaviest to lightest and as densest to rarest and, finally, as immovable of itself to self-moving, and as occupying the central position in the universe to revolving forever around a centre. It is not absurd, then, if oppositions so numerous and important carry with them the opposition of cold and heat as well. "Yes," Chrysippus may say, "but fire is bright." Is not the earth, then,

dark? Why, it is the darkest and most unilluminated of all things. Certainly air is first of all to participate in light; it is instantly altered and when it is saturated, it distributes illumination everywhere, lending itself to light as a body in which to reside. For when the sun arises, as one of the dithyrambic writers has said,

It straightway fills the mighty home of the air-borne winds.

Next the air, moving downward, infuses a part of its brightness into the lakes and the sea, and the depths of the rivers flash brightly, to the extent that air is able to penetrate them. Of all bodies only the earth remains constantly without light, impenetrable to the illumination of sun or moon; yet it is warmed by them and permits the heat to sink in and warm it up to a slight depth.⁹⁵³ But because it is solid, earth does not give passage to light, but is encircled by light on its surface only, while the inner parts are called Darkness and Chaos and Hades — so that Erebus turns out to be the subterranean and interior darkness. Then, too, the poets tell us that Night was born of Earth and mathematicians demonstrate that night is the shadow of Earth blocking the light of the sun. The air, indeed, is saturated with darkness by the earth, just as it is with light by the sun. The unlighted portion of the air is the area of night, amounting to the space occupied by the earth's shadow. This is the reason why men make use of the air out of doors even when it is night, as well as many beasts which do their pasturing in the darkness, since it retains some vestiges of light and dispersed glimmerings of radiance; but the house-bound man who is under a roof is utterly blind and without light inasmuch as there the earth envelops him from all directions. Whole skins, furthermore, and horns of animals do not let light pass through them because of their solidity; yet if sections are sawed off and polished, they become translucent when once the air has been mixed with them. It is also my opinion that the earth is called black by the poets, whenever they have occasion to do so, because of its murky and lightless characteristics. The result, then, of these considerations is that the much-prized antithesis of light and darkness belongs to earth rather than to air.

¹⁸¹ This, however, has no relevance to the question under discussion; for it has been shown that there are many cold objects which are bright and many hot which are dull and dark. Yet there are qualities more closely connected that belong to coldness; heaviness, stability, solidity, and resistance to change. Air has no part at all in them, while earth has a greater share in all of them than water has. Cold, moreover, is perceptibly one of the hardest of things and it makes things hard and unyielding. Theophrastus, for instance, tells us that when frozen fish are dropped on the ground, they are broken and smashed to bits just like objects of glass or earthenware. And at Delphi you yourself heard, in the case of those who climbed Parnassus to rescue the Thyiades when they were trapped by a fierce gale and snowstorm, that their capes were frozen so stiff and wooden that when they were opened out, they broke and split apart. Excessive cold, because of its hardness and immobility, also stiffens the muscles and renders the tongue speechless, for it congeals the moist and tender parts of the body.

19 1 In view of these considerations, regard the facts in the following light: every force, presumably, whenever it prevails, by a law of nature changes and turns into itself whatever it overcomes. What is mastered by heat is reduced to flames, what is mastered by wind turns to air; and anything that falls into the water, unless it gets out quickly, dissolves and liquefies. It follows, then, that whatever is completely frozen must turn into primordial cold. Now freezing is extreme refrigeration that terminates in a complete alteration and petrification when, since the cold has obtained complete mastery, the moist elements are frozen solid and the heat is squeezed out. This is the reason why the earth at its bottom-most point is practically all solid frost and ice. For there undiluted and unmitigated cold abides at bay, thrust back to the point farthest removed from the flaming aether. As for these features that are visible, cliffs and crags and rocks, Empedocles thinks that they have been fixed in place and are upheld by resting on the fire that burns in the depths of the earth; but the indications are rather that all these things from which the heat was squeezed out and evaporated were completely frozen by the cold; and for this reason they are called pagoi. So also the peaks of many of them have a black crust where the heat has been expelled and have the appearance of debris from a conflagration. For the cold freezes substances to a varying degree, but hardest those of which it is naturally a primary constituent. ⁹⁵⁴ Thus, if it is the nature of heat to lighten, the lightest object will have most heat, and if it is the nature of humidity to soften, the softest will have the most humidity; so, if it is also true that the nature of cold is to harden, then it must also follow that the hardest object will have the most cold — that is to say, just as the earth has. But what is coldest by nature is surely also primarily cold, so that the earth is in fact cold both primordially and naturally; and, of course, this is obvious even to the senses. Mud, in fact, is a colder thing than water; and men extinguish a fire by dumping earth upon it. Blacksmiths, when their iron becomes fiery and begins to melt, sprinkle on it marble chips and gypsum to check and cool it off before it melts too much. It is also true that dust cools the bodies of athletes and dries up their sweat.

20 1 And what is the meaning of our demand for a yearly change of habitation? In winter we retreat to the loftiest parts of our homes, those farthest from the earth, while in summer we require the lowest parts, submerging ourselves and going in quest of comfortable retreats, as we make the best of a life in the embrace of mother earth. Since we do this, are we not guided to the earth by our perception of its coldness? Do we not acknowledge it as the natural seat of primordial cold? And surely our living by the sea in the winter is, in a way, an escape from the earth, since we abandon the land as far as possible because of the frost and wrap ourselves in salt sea air because it is warm. Then again, in the summer by reason of the heat, we long for the earth-born, upland air, not because it is itself chilly, but because it has sprung from the naturally and primordially cold and has been imbued with its earthy power, as steel is tempered by being plunged in water. And of flowing waters, also, the coldest are those that fall from rocks or mountains, and of well waters the deepest are the coldest; the air from outside

does not, in the case of these wells, affect the water, so deep are they, while any such streams burst forth through pure unmixed earth, like the one at Taenarum, which they call the water of Styx: it flows from the rock in a trickle, but so cold that no vessel except an ass's hoof can contain it — all others it bursts and breaks apart.

21 1 We are, further, informed by physicians that generically earth is by nature astringent and cold, and they enumerate many metals that provide a styptic, staying effect for medicinal use. The element of earth is not sharp or mobile or slender or prickly or soft or ductile, but solid and compact like a cube. This is how it came to have weight; and the cold, which is its true power, by thickening, compressing, and squeezing out the humidity of bodies, induces shivering and shaking through its inequality; and if it becomes complete master and expels or extinguishes all the heat, it fixes the body in a frozen and corpselike condition. This is the reason why earth does not burn at all, or burns only grudgingly and with difficulty. Air, on the other hand, often shoots forth flames from itself and, turning into fire, makes streams and flashes of lightning. Heat feeds on moisture, for it is not the solid part of wood, but the damp part, that is combustible; and when this is distilled, the solid, dry part remains behind, reduced to ashes. Those who emulously strive to prove that this too is changed and consumed, sprinkling it, perhaps, with oil or kneading it with suet and setting it alight, accomplish nothing; for when the oily part is consumed, the earthy remains as a permanent residue, do what they may. Not only, therefore, because the earth is physically immovable from its station, but also because it is unalterable in essence, it was quite appropriately called Hestia by the ancients — in as much as she “remains in the home of the gods” — because of its stationary and compact nature; and coldness is what binds it together, as Archelaüs the natural philosopher declared, since nothing can relax or soften it, as a substance that is subject to heating or warming might be loosened.

955 As for those who suppose that they feel cold air and water, but are less sensible of earth's coldness, what they perceive is that portion of earth which is closest to them and has come to be a medley, a congeries, abounding in air and water, sun and heat. There is no difference between such people and those who declare that the aether is not naturally and primordially hot, but rather that scalding water or red hot iron ore are — because they can feel and touch these, but are unable to touch and feel the primordially pure and heavenly fire. Nor likewise are these persons able to touch and feel the earth at its bottom-most, which is what we particularly mean by earth — earth set off alone by itself, without admixture of any other element. But we can see a sample of such earthiness in that statement about the cliffs that display from deep down so intense a cold that it can scarcely be endured. Then, too, those who want a colder drink throw pebbles into the water, which becomes thicker and denser through the coldness the streams upward, fresh and undiluted, from the stones.

22 1 We must, therefore, believe that the reason why ancient learned men held that there is no commerce between earthly and celestial things was not that they distinguished up and down by relative position, as we do in the case of scales; but

rather it was the difference in powers that led them to assign such things as are hot and bright, swift and buoyant, to the eternal and imperishable part of nature, while darkness and cold and slowness they considered the unhappy heritage of transitory and submerged beings. Then too, the body of a living creature, as long as it breathes and flourishes, does, as the poets say, enjoy both warmth and life; but when these forsake it and it is abandoned in the realm of earth alone, immediately frigidity and congelation seize upon it, since warmth naturally resides in anything else rather than in the earthy.

²³ ¹ Compare these statements, Favorinus, with the pronouncements of others; and if these notions of mine are neither less probable nor much more plausible than those of others, say farewell to dogma, being convinced as you are that it is more philosophic to suspend judgement when the truth is obscure than to take sides.

Whether fire or water is more useful



955) ^{1 1} Water is best, but gold is a flaming fire,

says Pindar. He, therefore, bluntly assigns the second place to fire; and Hesiod agrees with him in the words

And first of all came Chaos into being;

for most people believe that this is his name for water because it flows (chysis). Yet the balance of witnesses on both sides seems to be equal. There are, in fact, some who state that fire is the first principle of the universe and, like a seed, creates everything out of itself and receives all things into itself when the conflagration occurs. Ignoring the authors, let us examine the arguments on both sides and see where they will lead us.

^{2 1} Is not that element the more useful of which most of all, everywhere, invariably, we stand in need as a household tool and, I swear, a friend, ready to help us at any time, in any emergency? Yet fire is not always useful; sometimes, indeed, we find it too much and interrupt our use of it. But water is used both winter and summer, sick and well, night and day: ⁹⁵⁶ there is no time when a man does not need it. That, of course, is the reason why the dead are called alibantes, meaning that they are without libas, “moisture,” and for lack of that deprived of life. Man has often existed without fire, but without water never. Besides, that which, from the beginning, was coincidental with the inception of man is more useful than that which was discovered later; for it is obvious that Nature bestowed the one as vitally necessary, while the other was brought to light by luck or contrivance for a superfluous use. Now, none may tell of a time when water was unknown to man, nor is any god or hero said to be its discoverer; it was, in fact, at hand instantly when man appeared and was itself the cause of his appearance. But the use of fire, they say, was discovered only a day or two ago by Prometheus; <consequently all our preceding life was deprived of> fire, though it was not without water. And that this is no poetic fiction is proved by present modes of living; for there are certain races of man who live without fire, with no house or hearth, under the open sky. And Diogenes the Cynic reduced the use of fire to a minimum, so that he even swallowed a squid raw, remarking, “Thus, gentlemen, do I risk my life for you.” But without water no one ever thought it good, or even possible, to live.

^{3 1} And why do I split hairs by discussing merely human nature? For though there are many, or rather countless, sorts of creatures, man is practically the only one that knows the use of fire, while all the others live and feed without it: they subsist, whether they

range abroad or fly or crawl, upon roots or produce or flesh, all without fire; but without water no creature of the sea or land or air ever existed. For even flesh-eating animals, some of which Aristotle says do not drink, nevertheless keep alive by using the fluids in the flesh. That element, therefore, without which no living nature can subsist or endure is more useful.

4 1 Let us pass from the people who use fire to the things that we use, namely plants and produce, of which some are completely devoid of heat, while others have an infinitesimal and uncertain amount. Moisture, however, is the element in nature that makes them all burgeon, growing and bearing fruit. And why should I enumerate honey and wine and oil and all the rest that come to us from the vintage, the milking of herds, or taking off of honey — and it is obvious where they belong — when even wheat itself, though it is classed as a dry food, moves into the category of liquids by alteration, fermentation, and deliquescence?

5 1 Moreover, what is never detrimental is more useful. Now fire, when it forms a stream, is most destructive; but the nature of water is never harmful. Then again, of two elements that is more beneficial which is cheaper and provides its help without any preparation. Now the use of fire requires a supply of fuel, for which reason rich people have more of it than poor, and kings than private persons; but water has another merit in service to man, that of equality, with no discrimination. For it needs no tools or implements, being a self-sufficient, self-fulfilling good.

6 1 Then, too, that which by multiplication destroys its own contribution is the less useful. Such a thing is fire which, like an all-devouring beast, consumes everything near, so that it is useful rather by skilful handling and craft and moderation in use than by its own nature; but water is never dangerous. Further, of two things the one which may be joined with its fellow is more useful. Now fire does not admit moisture and is of no use when in conjunction with it; but water is of service when combined with fire, for hot water is healing and well adapted to medicinal purposes. A watery fire you will never see; but water is as useful to mankind when hot as when cold.

7 1 Furthermore, though there are but four elements, water provides from itself a fifth, so to say, the sea, ⁹⁵⁷ one no less beneficial than the others, especially for commerce among other things. This element, therefore, when our life was savage and unsociable, linked it together and made it complete, redressing defects by mutual assistance and exchange and so bringing about co operation and friendship. Now Heraclitus declares, “If there were no sun, it would be perpetual night”; in the same way we may say that if there were no sea, man would be the most savage and destitute of all creatures. But as it is, the sea brought the Greeks the vine from India, from Greece transmitted the use of grain across the sea, from Phoenicia imported letters. Was a memorial against forgetfulness, thus preventing the greater part of mankind from being wineless, grainless, and unlettered. How, then, should water not be more useful when it has the advantage over fire of one more element?

81 What could anyone find to say on the other side from this point on? This, that God, the master workman, had as material four elements from which to construct the universe. Among these, again, there is a simple mutual distinction, namely, that earth and water are a foundation at the bottom of the universe, being, like raw material, the substance of which things are constructed and moulded, having just so much form and organization, and indeed of capacity for growth and procreation, as is imparted to them by the other elements, air and fire, which are makers and artisans and rouse them, lying lifeless as they were until then, to the act of creation. Between these two, again, fire and air, there is the distinction that fire assumes the rule and leadership. This is clear by induction: earth without warmth is barren and unfruitful, but fire, when it takes possession and inflames, causes it to swell to the point of generation; and it is impossible to find any other reason why rocks and the bare bones of mountains are barren except that they have either no part at all, or very little share, in fire.

91 And, in general, water is so far from being self-sufficient for the preservation or generation of other things that the want of fire is water's destruction. For heat maintains everything in its proper being and keeps it in its proper substance, water itself as well as everything else. When fire withdraws and fails, water putrefies: the dearth of heat is the death and destruction of water. It is, of course, marsh waters and such as are stagnant, some too that have drained into depressions with no outlet, that are bad and finally putrefy because they have very little motion, which preserves everything by stirring up its heat. This is the reason why we commonly say that those waters are "living" which have most motion and the strongest current; the heat is maintained by their motion. How, then, should that not be the more useful of two things which has provided what is necessary for the other's existence, as fire does for water? And surely that is the more useful, the lack of which, if it be entirely taken away, causes the living creature to die. For it is obvious that anything without which a creature cannot live must have been a necessary cause of its existence, while it did exist. Now even corpses have moisture which does not entirely vanish; otherwise dead bodies would not putrefy, since putrefaction is not a change from dry to moist, but rather a corruption of the moisture in flesh. Death, then, is nothing but the total disappearance of heat and so dead men are extremely cold; if you attack them with a razor-blade, you will blunt the edge of it through excess of cold. In the living creature itself, too, the parts that have the least heat are the least sensitive, like bones and hair and the parts that are a long way from the heart. And, in general, the presence of fire makes a greater difference than that of moisture; for it is not mere moisture that produces plants and fruits, but warm moisture; cold water, of course, is either less productive or not productive at all. Yet if by its own nature water were fruitful, it would always bear fruit by itself; 958 but on the contrary it is even harmful.

101 To begin again: for the use of fire as fire we do not need water; on the contrary, it would be in our way since it extinguishes and destroys it. But in most circumstances it is impossible to use water without fire. When water is heated, it is more useful;

otherwise it is harmful. And it is heat which has made the sea more beneficial, its waters being warmer, since it differs from other waters in no other respect. So that of two things, that is better which of itself lends us its use without need of the other. Besides, water is solely beneficial to the touch, when you wash or bathe in it; but fire is profitable to all the senses. It can, in fact, both be touched and seen from a distance, so that in addition to its other uses, there is also its variegated character.

¹¹ For to say that man ever exists without fire is absurd, nor can he exist at all without it; but there are differences in kind as in other things. As for men who have no need of fire from without, they have this experience not because they do not need it, but because their own heat more than suffices. This must be predicated also of other animals which do not need fire. So that in this respect, too, the use of fire is probably superior. Water is never in such a condition as to need no external support, but fire is self-sufficient because of its great excellence. As, then, a general is better who manages the affairs of his city so that it needs no allies from without, so also an element is superior which does not often need external assistance.

Yet, to take the opposite point of view, that is more useful which we alone make great use of, since by the powers of our reason we are able to choose what is better. For what is more useful and more profitable to man than reason? But brute beasts do not have it. What then? Is what has been discovered by the foresight of our better part for this reason less useful?

^{12 1} And since we have arrived at this point in our argument: What is more profitable to life than Art? And it was fire that discovered and still preserves all the arts. That is why they make Hephaestus the first of artificers. Man has been granted but a little time to live and, as Ariston says, sleep, like a tax-collector, takes away half of that. But I would rather say that it is a question of darkness; for although a man might stay awake all night, yet no good would come of his wakefulness if fire did not give him the benefits of day and remove the difference between day and night. If, then, there is nothing more advantageous to man than life and life is many times increased by fire, how should fire not be the most useful of all things?

^{13 1} And, to be sure, will not that be the most advantageous of which each of the senses has the greatest proportion? Do you not perceive, then, that there is no one of the senses which uses moisture by itself without an admixture of air or fire; and that every sense partakes of fire inasmuch as it supplies the vital energy; and especially that sight, the keenest of the physical senses, is an ignited mass of fire and is that which has made us believe in the gods? And further, through sight, as Plato says, we are able to conform our souls to the movements of the celestial bodies.

Whether land or sea animals are cleverer



(The speakers in the dialogue are Autobulus, Soclarus, Optatus, Aristotimus, Phaedimus, and Heracleon.)

959 1 1 Autobulus. When Leonidas was asked what sort of a person he considered Tyrtaeus to be, he replied, “A good poet to whet the souls of young men,” on the ground that by means of verses the poet inspired in young men keenness, accompanied by ardour and ambition whereby they sacrificed themselves freely in battle. And I am very much afraid, my friends, that the *Praise of Hunting* which was read aloud to us yesterday may so immoderately inflame our young men who like the sport that they will come to consider all other occupations as of minor, or of no, importance and concentrate on this. As a matter of fact, I myself caught the old fever all over again in spite of my years and longed, like Euripides’ Phaedra,

To halloo the hounds and chase the dappled deer;
so moved was I by the discourse as it brought its solid and convincing arguments to bear.

Soclarus. Exactly so, Autobulus. That reader yesterday seems to have roused his rhetoric from its long disuse to gratify the young men and share their vernal mood. I was particularly pleased with his introduction of gladiators and his argument that it is as good a reason as any to applaud hunting that after diverting to itself most of our natural or acquired pleasure in armed combats between human beings it affords an innocent spectacle of skill and intelligent courage pitted against witless force and violence. It agrees with that passage of Euripides:

Slight is the strength of men;
But through his mind’s resource
He subdues the dread
Tribes of the deep and races
Bred on earth and in the air.

2 1 Autobulus. Yet that is the very source, my dear Soclarus, from which they say insensibility spread among men and the sort of savagery that learned the taste of slaughter on its hunting trips and has grown accustomed to feel no repugnance for the wounds and gore of beasts, but to take pleasure in their violent death. The next step is like what happened at Athens: the first man put to death by the Thirty was a certain

informer who was said to deserve it, and so was the second and the third; but after that they went on, step by step, until they were laying hands on honest men and eventually did not spare even the best of the citizens. Just so the first man to kill a bear or a wolf won praise; and perhaps some cow or pig was condemned as suitable to slay because it had tasted the sacred meal placed before it. So from that point, as they now went on to eat the flesh of deer and hare and antelope, men were introduced to the consumption of sheep and, in some places, of dogs and horses.

The tame goose and the dove upon the hearth,
as Sophocles says, were dismembered and carved for food — not that hunger compelled men as it does weasels and cats, but for pleasure and as an appetizer. Thus the brute and the natural lust to kill in man were fortified and rendered inflexible to pity, while gentleness was, for the most part, deadened. It was in this way, on the contrary, that the Pythagoreans, to inculcate humanity and compassion, made a practice of kindness to animals; for habituation has a strange power to lead men onward by ⁹⁶⁰ a gradual familiarization of the feelings.

Well, we have somehow fallen unawares into a discussion not unconnected with what we said yesterday nor yet with the argument that is presently to take place to day. Yesterday, as you know, we proposed the thesis that all animals partake in one way or another of reason and understanding, and thereby offered our young hunters a field of competition not lacking in either instruction or pleasure: the question whether land or sea animals have superior intelligence. This argument, it seems, we shall to day adjudicate if Aristotimus and Phaedimus stand by their challenges; for Aristotimus put himself at his comrades' disposal to advocate the land as producer of animals with superior intelligence, while the other will be pleader for the sea.

Soclarus. They'll stand by their word, Autobulus; they'll be here any minute now. Early this morning I observed them both preparing for the fray. But, if you like, before the contest begins, let us review the discussion of whatever topics are germane to our conversation of yesterday, but were not then discussed, either because no occasion offered, or, since we were in our cups, were treated too lightly. I thought, in fact, that I caught the reverberation of a material objection from the Stoa: just as the immortal is opposed to the mortal and the imperishable to the perishable, and, of course, the incorporeal to the corporeal; just so, if there is rationality, the irrational must exist as its opposite and counterpart. This alone, among all these pairings, must not be left incomplete and mutilated.

^{3 1} Autobulus. But who ever, my dear Soclarus, maintained that, while rationality exists in the universe, there is nothing irrational? For there is a plentiful abundance of the irrational in all things that are not endowed with a soul; we need no other sort of counterpart for the rational: everything that is soulless, since it has no reason or intelligence, is by definition in opposition to that which, together with a soul, possesses also reason and understanding. Yet suppose some were to maintain that nature must not be left maimed, but that that part of nature which is endowed with a soul should have its

irrational as well as its rational aspect, someone else is bound to maintain that nature endowed with a soul must have both an imaginative and an unimaginate part, and both a sentient part and an insentient. They want nature, they say, to have these counteractive and contraposed positives and negatives of the same kind counterbalanced, as it were. But if it is ridiculous to require an antithesis of sentient and insentient within the class of living things, or an antithesis of imaginative and unimaginate, seeing that it is the nature of every creature with a soul to be sentient and imaginative from the hour of its birth, so he, also, is unreasonable who demands a division of the living into a rational and an irrational part — and that, too, when he is arguing with men who believe that nothing is endowed with sensation which does not also partake of intelligence and that there is no living thing which does not naturally possess both opinion and reason, just as it has sensation and appetite. For nature, which, they rightly say, does everything with some purpose and to some end, did not create the sentient creature merely to be sentient when something happens to it. No, for there are in the world many things friendly to it, many also hostile; and it could not survive for a moment if it had not learned to give the one sort a wide berth while freely mixing with the other. It is, to be sure, sensation that enables each creature to recognize both kinds; but the acts of seizing or pursuing that ensue upon the perception of what is beneficial, as well as the eluding or fleeing of what is destructive or painful, could by no means occur in creatures naturally incapable of some sort of reasoning and judging, remembering and attending. Those beings, then, which you deprive of all expectation, memory, design, or preparation, and of all hopes, fears, desires, or griefs — they will have no use for eyes or ears either, even though they have them. Indeed, it would be better to be rid of all sensation and imagination that has nothing to make use of it, ⁹⁶¹ rather than to know toil and distress and pain while not possessing any means of averting them.

There is, in fact, a work of Strato, the natural philosopher, which proves that it is impossible to have sensation at all without some action of the intelligence. Often, it is true, while we are busy reading, the letters may fall on our eyes, or words may fall on our ears, which escape our attention since our minds are intent on other things; but later the mind recovers, shifts its course, and follows up every detail that had been neglected; and this is the meaning of the saying:

Mind has sight and Mind has hearing;

Everything else is deaf and blind,

indicating that the impact on eyes and ears brings no perception if the understanding is not present. For this reason also King Cleomenes, when a recital made at a banquet was applauded and he was asked if it did not seem excellent, replied that the others must judge, for his mind was in the Peloponnesus. So that, if we are so constituted that to have sensation we must have understanding, then it must follow that all creatures which have sensation can also understand.

But let us grant that sensation needs no help of intelligence to perform its own function; nevertheless, when the perception that has caused an animal to distinguish

between what is friendly and what is hostile is gone, what is it that from this time on remembers the distinction, fears the painful, and wants the beneficial? And, if what it wants is not there, what is there in animals that devises means of acquiring it and providing lairs and hiding-places — both traps for prey and places of refuge from attackers? And yet those very authors rasp our ears by repeatedly defining in their *Introductions* “purpose” as “an indication of intent to complete,” “design” as “an impulse before an impulse,” “preparation” as “an act before an act,” and “memory” as “an apprehension of a proposition in the past tense of which the present tense has been apprehended by perception.” For there is not one of these terms that does not belong to logic; and the acts are all present in all animals as, of course, are cognitions which, while inactive, they call “notions,” but when they are once put into action, “concepts.” And though they admit that emotions one and all are “false judgements and seeming truths,” it is extraordinary that they obviously fail to note many things that animals do and many of their movements that show anger or fear or, so help me, envy or jealousy. They themselves punish dogs and horses that make mistakes, not idly but to discipline them; they are creating in them through pain a feeling of sorrow, which we call repentance.

Now pleasure that is received through the ears is a means of enchantment, while that which comes through the eyes is a kind of magic: they use both kinds against animals. For deer and horses are bewitched by pipes and flutes, and crabs are involuntarily lured from their holes by lotus pipes; it is also reported that shad will rise to the surface and approach when there is singing and clapping. The horned owl, again, can be caught by the magic of movement, as he strives to twist his shoulders in delighted rhythm to the movements of men dancing before him.

As for those who foolishly affirm that animals do not feel pleasure or anger or fear or make preparations or remember, but that the bee “as it were” remembers and the swallow “as it were” prepares her nest and the lion “as it were” grows angry and the deer “as it were” is frightened — I don’t know what they will do about those who say that beasts do not see or hear, but “as it were” hear and see; that they have no cry but “as it were”; nor do they live at all but “as it were.” For these last statements (or so I believe) are no more contrary to plain evidence than those that they have made.

41 Soclarus. Well, Autobulus, you may count me also as one who believes your statements; yet on comparing the ways of beasts with human customs and lives, ⁹⁶² with human actions and manner of living, I find not only many other defects in animals, but this especially: they do not explicitly aim at virtue, for which purpose reason itself exists; nor do they make any progress in virtue or have any bent for it; so that I fail to see how Nature can have given them even elementary reason, seeing that they cannot achieve its end.

Autobulus. But neither does this, Soclarus, seem absurd to those very opponents of ours; for while they postulate that love of one’s offspring is the very foundation of our social life and administration of justice, and observe that animals possess such love in a

very marked degree, yet they assert and hold that animals have no part in justice. Now mules are not deficient in organs; they have, in fact, genitals and wombs and are able to use them with pleasure, yet cannot attain the end of generation. Consider another approach: is it not ridiculous to keep affirming that men like Socrates and Plato are involved in vice no less vicious than that of any slave you please, that they are just as foolish and intemperate and unjust, and at the same time to stigmatize the alloyed and imprecise virtue of animals as absence of reason rather than as its imperfection or weakness? And this, though they acknowledge that vice is a fault of reason and that all animals are infected with vice: many, in fact, we observe to be guilty of cowardice and intemperance, injustice and malice. He, then, who holds that what is not fitted by nature to receive the perfection of reason does not even receive any reason at all is, in the first place, no better than one who asserts that apes are not naturally ugly or tortoises naturally slow for the reason that they are not capable of possessing beauty or speed. In the second place, he fails to observe the distinction which is right before his eyes: mere reason is implanted by nature, but real and perfect reason is the product of care and education. And this is why every living creature has the faculty of reasoning; but if what they seek is true reason and wisdom, not even man may be said to possess it. For as one capacity for seeing or flying differs from another (hawks and cicadas do not see alike, nor do eagles and partridges fly alike), so also not every reasoning creature has in the same way a mental dexterity or acumen that has attained perfection. For just as there are many examples in animals of social instincts and bravery and ingenuity in ways and means and in domestic arrangements, so, on the other hand, there are many examples of the opposite: injustice, cowardliness, stupidity. And the very factor which brought about our young men's contest to day provides confirmation. It is on an assumption of difference that the two sides assert, one that land animals, the other that sea animals, are naturally more advanced toward virtue. This is clear also if you contrast hippopotamuses with storks: the latter support their fathers, while the former kill them in order to consort with their mothers. The same is true if you compare doves with partridges; for the partridge cock steals the eggs and destroys them since the female will not consort with him while she is sitting, whereas male doves assume a part in the care of the nest, taking turns at keeping the eggs warm and being themselves the first to feed the fledglings; and if the female happens to be away for too long a time, the male strikes her with his beak and drives her back to her eggs or squabs. And while Antipater was reproaching asses and sheep for their neglect of cleanliness, I don't know how he happened to overlook lynxes and swallows; for lynxes dispose of their excrement by concealing and doing away with it, while swallows teach their nestlings to turn tail and void themselves outward.

Why, moreover, do we not say that one tree is less intelligent than another, as a sheep is by comparison with a dog; or one vegetable more cowardly than another, as a stag is by comparison with a lion? ⁹⁶³ Is the reason not that, just as it is impossible to call one immovable object slower than another, so among all creatures to whom Nature has not

given the faculty of understanding, we cannot say that one is more cowardly or more slothful or more intemperate? Whereas it is the presence of understanding, of one kind in one animal, of another kind in another, and in varying degree, that has produced the observable differences.

51 Soclarus. Yet it is astonishing how greatly man surpasses the animals in his capacity for learning and in sagacity and in the requirements of justice and social life.

Autobulus. There are in fact, my friend, many animals which surpass all men, not only in bulk and swiftness, but also in keen sight and sharp hearing; but for all that man is not blind or crippled or earless. We can run, if less swiftly than deer; and see, if less keenly than hawks; nor has Nature deprived us of strength and bulk even though, by comparison with the elephant and the camel, we amount to nothing in these matters. In the same way, then, let us not say of beasts that they are completely lacking in intellect and understanding and do not possess reason even though their understanding is less acute and their intellect inferior to ours; what we should say is that their intellect is feeble and turbid, like a dim and clouded eye. And if I did not expect that our young men, learned and studious as they are, would very shortly present us here, one with a large collection of examples drawn from the land, the other with his from the sea, I should not have denied myself the pleasure of giving you countless examples of the docility and native capacity of beasts — of which fair Rome has provided us a reservoir from which to draw in pails and buckets, as it were, from the imperial spectacles. Let us leave this subject, therefore, fresh and untouched for them to exercise their art upon in discourse.

There is, however, one small matter which I should like to discuss with you quietly. It is my opinion that each part and faculty has its own particular weakness or defect or ailment which appears in nothing else, as blindness in the eye, lameness in the leg, stuttering in the tongue. There can be no blindness in an organ which was not created to see, or lameness in a part which was not designed for walking; nor would you ever describe an animal without a tongue as stuttering, or one voiceless by nature as inarticulate. And in the same way you would not call delirious or witless or mad anything that was not endowed by Nature with reason or intelligence or understanding; for it is impossible to ail where you have no faculty of which the ailment is a deficiency or loss or some other kind of impairment. Yet certainly you have encountered mad dogs, and I have also known of mad horses; and there are some who say that cattle and foxes also go mad. But dogs will do, since no one questions the fact in their case, which provides evidence that the creature possesses reason and a by no means despicable intellectual faculty. What is called rabies and madness is an ailment of that faculty when it becomes disturbed and disordered. For we observe no derangement either of the dogs' sight or of their hearing; yet, just as when a human being suffers from melancholy or insanity, anyone is absurd who does not admit that it is the organ that thinks and reasons and remembers which has been displaced or damaged (we habitually say, in fact, of madmen that they "are not themselves," but have "fallen out of their wits"), just

so, whoever believes that rabid dogs have any other ailment than an affliction of their natural organ of judgement and reason and memory so that, when this has become infected with disorder and insanity, they no longer recognized beloved faces and shun their natural haunts — such a man, I say, either must be disregarding the evidence or, if he does take note of the conclusion to which it leads, must be quarrelling with the truth.

61 Soclarus. Your inference seems quite justified. For the Stoics and Peripatetics strenuously agree on the other side, to the effect that justice could not then come into existence, but would remain completely without form or substance, 964 if all the beasts partake of reason. For either we are necessarily unjust if we do not spare them; or, if we do not take them for food, life becomes impracticable or impossible; in a sense we shall be living the life of beasts once we give up the use of beasts. I omit the numberless hosts of Nomads and Troglodytes who know no other food but flesh. As for us who believe our lives to be civilized and humane, it is hard to say what pursuit on land or sea, what aerial art, what refinement of living, is left to us if we are to learn to deal innocently and considerately with all creatures, Bas we are bound to if they possess reason and are of one stock with us. So we have no help or cure for this dilemma which either deprives us of life itself or of justice, unless we do preserve that ancient limitation and law by which, according to Hesiod, he who distinguished the natural kinds and gave each class its special domain:

To fish and beasts and winged birds allowed
Licence to eat each other, for no right
Exists among them; right, he gave to men

for dealing with each other. Those who know nothing of right action toward us can receive no wrong from us either. For those who have rejected this argument have left no path, either broad or narrow, by which any may slip in.

71 Autobulus. This, my friend, has been spoken “from the heart.” We certainly must not allow philosophers, as though they were women in difficult labour, to put about their necks a charm for speedy delivery so that they may bring justice to birth for us easily and without hard labour. For they themselves do not concede to Epicurus, for the sake of the highest considerations, a thing so small and trifling as the slightest deviation of a single atom — which would permit the stars and living creatures to slip in by chance and would preserve from destruction the principle of free will. But, seeing that they bid him demonstrate whatever is not obvious or take as his starting-point something that is obvious, how are they in any position to make this statement about animals a basis of their own account of justice, when it is neither generally accepted nor otherwise demonstrated by them? For justice has another way to establish itself, a way which is neither so treacherous nor so precipitous, nor is it a route lined with the wreckage of obvious truths. It is the road which, under the guidance of Plato, my son and your companion, Soclarus, points out to those who have no love of wrangling, but are willing

to be led and to learn. For certain it is that Empedocles and Heraclitus accept as true the charge that man is not altogether innocent of injustice when he treats animals as he does; often and often do they lament and exclaim against Nature, declaring that she is “Necessity” and “War,” that she contains nothing unmixed and free from tarnish, that her progress is marked by many unjust inflictions. As an instance, say, even birth itself springs from injustice, since it is a union of mortal with immortal, and the offspring is nourished unnaturally on members torn from the parent.

These strictures, however, seem to be unpalatably strong and bitter; for there is an alternative, an inoffensive formula which does not, on the one hand, deprive beasts of reason, yet does, on the other, preserve the justice of those who make fit use of them. When the wise men of old had introduced this, gluttony joined luxury to cancel and annul it; Pythagoras, however, reintroduced it, teaching us how to profit without injustice. There is no injustice, surely, in punishing and slaying animals that are anti-social and merely injurious, while taming those that are gentle and friendly to man and making them our helpers in the tasks for which they are severally fitted by nature:

Offspring of horse and ass and seed of bulls

which Aeschylus’ Prometheus says that he bestowed on us

⁹⁶⁵ To serve us and relieve our labours;

and thus we make use of dogs as sentinels and keep herds of goats and sheep that are milked and shorn. For living is not abolished nor life terminated when a man has no more platters of fish or *pâté de foie gras* or mincemeat of beef or kids’ flesh for his banquets — or when he no longer, idling in the theatre or hunting for sport, compels some beasts against their will to stand their ground and fight, while he destroys others which have not the instinct to fight back even in their own defence. For I think sport should be joyful and between playmates who are merry on both sides, not the sort Bof which Bion spoke when he remarked that boys throw stones at frogs for fun, but the frogs don’t die for “fun,” but in sober earnest. Just so, in hunting and fishing, men amuse themselves with the suffering and death of animals, even tearing some of them piteously from their cubs and nestlings. The fact is that it is not those who make use of animals who do them wrong, but those who use them harmfully and heedlessly and in cruel ways.

⁸¹ Soclarus. Restrain yourself, Autobulus, and turn off the flow of these accusations. I see a good many gentlemen approaching who are all hunters; you will hardly convert them and you needn’t hurt their feelings.

Autobulus. Thanks for the warning. Eubiotus, however, I know quite well and my cousin Ariston, and Aeacides and Aristotimus here, the sons of Dionysius of Delphi, and Nicander, the son of Euthydamus, all of them “expert,” as Homer expresses it, in the chase by land — and for this reason they will be on Aristotimus’ side. So too yonder comes Phaedimus with the islanders and coast-dwellers about him, Heracleon from Megara and the Euboean Philostratus,

Whose hearts are on deeds of the sea.

And here is my contemporary Optatus: like Diomedes, it is

Hard to tell the side on which he ranges,

for “with many a trophy from the sea, many likewise from the chase on the mountain, he has glorified” the goddess who is at once the Huntress and Dictynna. It is evident that he is coming to join us with no intention of attaching himself to either side. Or am I wrong, my dear Optatus, in supposing that you will be an impartial and neutral umpire between the young men?

Optatus. It is just as you suppose, Autobulus. Solon’s law, which used to punish those who adhered to neither side in a factious outbreak, has long since fallen into disuse.

Autobulus. Come over here, then, and take your place beside us so that, if we need evidence, we shall not have to disturb the tomes of Aristotle, but may follow you as expert and return a true verdict on the arguments.

Soclarus. Well then, my young friends, have you reached any agreement on procedure?

Phaedimus. We have, Soclarus, though it occasioned considerable controversy; but at length, as Euripides has it,

The lot, the child of chance,

made arbiter, admits into court the case of the land animals before that of creatures from the sea.

Soclarus. The time has come, then, Aristotimus, for you to speak and us to hear.

965 9 1 Aristotimus. The court is open for the litigants . . . And there are some fish that waste their milt by pursuing the female while she is laying her eggs.

There is also a type of mullet called the grayfish which feeds on its own slime; and the octopus sits through the winter devouring himself,

In fireless home and domicile forlorn,

so lazy or insensible or gluttonous, or guilty of all of these charges, is he. So this also is the reason, again, why Plato in his *Laws* enjoined, For rather prayed, that his young men might not be seized by “a love for sea hunting.” For there is no exercise in bravery or training in skill or anything that contributes to strength or fleetness or agility when men endure toil in contests with bass or conger or parrot-fish; 966 whereas, in the chase on land, brave animals give play to the courageous and danger-loving qualities of those matched against them, crafty animals sharpen the wits and cunning of their attackers, while swift ones train the strength and perseverance of their pursuers. These are the qualities which have made hunting a noble sport, whereas there is nothing glorious about fishing. And, and there’s not a god, my friend, who has allowed himself to be called “conger-killer,” as Apollo is “wolf-slayer,” or “surmullet-slayer,” as Artemis is “deer-slaying.” And what is surprising in this when it’s a more glorious thing for a man to have caught a boar or a stag, or, so help me, a gazelle or a hare than to have bought one? As for your tunny and your mackerel and your bonito! They’re more

honourable to buy than to catch oneself. B For their lack of spirit or of any kind of resource or cunning has made the sport dishonourable, unfashionable, and illiberal.

10 1 In general, then, the evidence by which the philosophers demonstrate that beasts have their share of reason is their possession of purpose and preparation and memory and emotions and care for their young and gratitude for benefits and hostility to what has hurt them; to which may be added their ability to find what they need and their manifestations of good qualities, such as courage and sociability and continence and magnanimity. Let us ask ourselves if marine creatures exhibit any of these traits, or perhaps some suggestion of them, that is extremely faint and difficult to discern (the observer only coming at long last to the opinion that it may be descried); whereas in the case of terrestrial and earth-born animals it is easy to find remarkably plain and unanswerable proofs of every one of the points I have mentioned.

In the first place, then, behold the purposeful demonstrations and preparations of bulls stirring up dust when intent on battle, and wild boars whetting their tusks. Since elephants' tusks are blunted by wear when, by digging or chopping, they fell the trees that feed them, they use only one tusk for this purpose and keep the other always pointed and sharp for defence. Lions always walk with paws clenched and claws retracted so that these may not be dulled by wear at the point or leave a plain trail for trackers; for it is not easy to find any trace of a lion's claw; on the contrary, any sign of a track that is found is so slight and obscure that hunters lose the trail and go astray. You have heard, I am sure, how the ichneumon girds itself for battle as thoroughly as any soldier putting on his armour, such a quantity of mud does it don and plaster about its body when it plans to attack the crocodile. Moreover, we see house-martins preparing for procreation: how well they lay the solid twigs at the bottom to serve as a foundation, then mould the lighter bits about them; and if they perceive that the nest needs a lump of mud to glue it together, they skim over a pond or lake, touching the water with only the tips of their feathers to make them moist, yet not heavy with dampness; then they scoop up dust and so smear over and bind together any parts that begin to sag or loosen. As for the shape of their work, it has no angles nor many sides, but is as smooth and circular as they can make it; such a shape is, in fact, both stable and capacious and provides no hold on the outside for scheming animals.

There is more than one reason for admiring spiders' webs, the common model for both women's looms and fowlers' nets; for there is the fineness of the thread and the evenness of the weaving, which has no disconnected threads and nothing like a warp, but is wrought with the even continuity of a thin membrane and a tenacity that comes from a viscous substance inconspicuously worked in. Then too, there is the blending of the colours that gives it an airy, misty look, the better to let it go undetected; and most notable of all is the art itself, like a charioteer's or a helmsman's, with which the spinner handles her artifice. When a possible victim is entangled, she perceives it, and uses her wits, like a skilled handler of nets, to close the trap suddenly and make it tight.

967 Since this is daily under our eyes and observation, my account is confirmed.

Otherwise it would seem a mere fiction, as I formerly regarded the tale of the Libyan crows which, when they are thirsty, throw stones into a pot to fill it and raise the water until it is within their reach; but later when I saw a dog on board ship, since the sailors were away, putting pebbles into a half empty jar of oil, I was amazed at its knowing that lighter substances are forced upward when the heavier settle to the bottom.

Similar tales are told of Cretan bees and of geese in Cilicia. When the bees are going to round some windy promontory, they ballast themselves with little stones so as not to be carried out to sea; while the geese, in fear of eagles, take a large stone in their beaks whenever they cross Mt. Taurus, as it were reining in and bridling their gaggling loquacity that they may pass over in silence unobserved. It is well known, too, how cranes behave when they fly. Whenever there is a high wind and rough weather they do not fly, as on fine days, in line abreast or in a crescent-shaped curve; but they form at once a compact triangle with the point cleaving the gale that streams past, so that there is no break in the formation. When they have descended to the ground, the sentinels that stand watch at night support themselves on one foot and with the other grasp a stone and hold it firmly; the tension of grasping this keeps them awake for a long time; but when they do relax, the stone escapes and quickly rouses the culprit. So that I am not at all surprised that Heracles tucked his bow under his arm:

Embracing it with mighty arm he sleeps,
Keeping his right hand gripped about the club.

Nor, again, am I surprised at the man who first guessed how to open an oyster when I read of the ingenuity of herons. For they swallow a closed mussel and endure the discomfort until they know that it has been softened and relaxed by their internal heat; then they disgorge it wide open and unfolded and extract the meat.

¹¹ It is impossible to relate in full detail all the methods of production and storage practised by ants, but it would be careless to omit them entirely. Nature has, in fact, nowhere else so small a mirror of greater and nobler enterprises. Just as you may see greater things reflected in a drop of clear water, so among ants there exists the delineation of every virtue.

Love and affection are found, namely their social life. You may see, too, the reflection of courage in their persistence in hard labour. There are many seeds of temperance and many of prudence and justice. Now Cleanthes, even though he declared that animals are not endowed with reason, says that he witnessed the following spectacle: some ants came to a strange anthill carrying a dead ant. Other ants then emerged from the hill and seemed, as it were, to hold converse with the first party and then went back again. This happened two or three times until at last they brought up a grub to serve as the dead ant's ransom, whereupon the first party picked up the grub, handed over the corpse, and departed.

A matter obvious to everyone is the consideration ants show when they meet: those that bear no load always give way to those who have one and let them pass. Obvious also is the manner in which they gnaw through and dismember things that are difficult to

carry or to convey past an obstacle, in order that they may make easy loads for several. And Aratus takes it to be a sign of rainy weather when they spread out their eggs and cool them in the open:

When from their hollow nest the ants in haste
Bring up their eggs;

and some do not write “eggs” here, but “provisions,” in the sense of stored grain which, when they notice that it is growing mildewed and fear ⁹⁶⁸ that it may decay and spoil, they bring up to the surface. But what goes beyond any other conception of their intelligence is their anticipation of the germination of wheat. You know, of course, that wheat does not remain permanently dry and stable, but expands and lactifies in the process of germination. In order, then, to keep it from running to seed and losing its value as food, and to keep it permanently edible, the ants eat out the germ from which springs the new shoot of wheat.

I do not approve of those who, to make a complete study of anthills, inspect them, as it were, anatomically. But, be that as it may, they report that the passage leading downward from the opening is not at all straight or easy for any other creature to traverse; it passes through turns and twists with branching tunnels and connecting galleries and terminates in three hollow cavities. One of these is their common dwelling-place, another serves as storeroom for provisions, while in the third they deposit the dying.

¹²¹ I don't suppose that you will think it out of order if I introduce elephants directly on top of ants in order that we may concurrently scrutinize the nature of understanding in both the smallest and the largest of creatures, for it is neither suppressed in the latter nor deficient in the former. Let others, then, be astonished that elephants learn, or are taught, to exhibit in the theatre all the many postures and variations of movement that they do, these being so varied and so complicated to memorize and retain that they are not at all easy even for human artists. For my part, I find the beast's understanding better manifested in his own spontaneous and uninstructed feelings and movements, in a pure, as it were, and undiluted state.

Well, not very long ago at Rome, where a large number of elephants were being trained to assume dangerous stances and wheel about in complicated patterns, one of them, who was the slowest to learn and was always being scolded and often punished, was seen at night, alone by himself in the moonlight, voluntarily rehearsing his lessons and practising them.

Formerly in Syria, Hagnon tells us, an elephant was brought up in its master's house and every day the keeper, when he received a measure of barley, would filch away and appropriate half of it; but on one occasion, when the master was present and watching, the keeper poured out the whole measure. The elephant gave a look, raised its trunk, and made two piles of the barley, setting aside half of it and thus revealing as eloquently as

could be the dishonesty of its keeper. And another elephant, whose keeper used to mix stones and dirt in its barley ration, when the keeper's meat was cooking, scooped up some ashes and threw them into the pot. And another in Rome, being tormented by little boys who pricked its proboscis with their writing styluses, grabbed one of them and raised him into the air as first to dash him to death; but when the spectators cried out, it gently set the child down on the ground again and passed along, thinking it sufficient punishment for one so young to have been frightened.

Concerning wild elephants who are self-governing they tell many wonderful tales, particularly the one about the fording of rivers: the youngest and smallest volunteers his services to go first into the stream. The others wait on the bank and observe the result, for if his back remains above water, those that are larger than he will have a wide margin of safety to give them confidence.

¹³¹ At this point in my discourse, I imagine that I shall do well not to omit the case of the fox, since it is so similar. Now the story-books tell us that when Deucalion released a dove from the ark, as long as she returned, it was a certain sign that the storm was still raging; but as soon as she flew away, it was a harbinger of fair weather. So even to this day the Thracians, whenever they propose crossing a frozen river, make use of a fox as an indicator of the solidity of the ice. ⁹⁶⁹ The fox moves ahead slowly and lays her ear to the ice; if she perceives by the sound that the stream is running close underneath, judging that the frozen part has no great depth, but is only thin and insecure, she stands stock still and, if she is permitted, returns to the shore; but if she is reassured by the absence of noise, she crosses over. And let us not declare that this a nicety of perception unaided by reason; it is, rather, a syllogistic conclusion developed from the evidence of perception: "What makes noise must be in motion; what is in motion is not frozen; what is not frozen is liquid; what is liquid gives way." So logicians assert that a dog, at a point where many paths split off, makes use of a multiple disjunctive argument and reasons with himself: "Either the wild beast has taken this path, or this, or this. But surely it has not taken this, or this. Then it must have gone by the remaining road." Perception here affords nothing but the minor premiss, while the force of reason gives the major premisses and adds the conclusion to the premisses. A dog, however, does not need such a testimonial, which is both false and fraudulent; for it is perception itself, by means of track and spoor, which indicates the way the creature fled; it does not bother with disjunctive and copulative propositions. The dog's true capacity may be discerned from many other acts and reactions and the performance of duties, which are neither to be smelled out nor seen by the eye, but can be carried out or perceived only by the use of intelligence and reason. I should only make myself ridiculous if I described the dog's self-control and obedience and sagacity on hunting parties to you who see and handle these matters every day.

There was a Roman named Calvus slain in the Civil Wars, but no one was able to cut off his head until they encircled and stabbed to death the dog who guarded his master and defended him. And King Pyrrhus on a journey chanced upon a dog guarding the

body of a murdered man; in answer to his questions he was told that the dog had remained there without eating for three days and refused to leave. Pyrrhus gave orders for the corpse to be buried and the dog cared for and brought along in his train. A few days later there was an inspection of the soldiers, who marched in front of the king seated on his throne, while the dog lay quietly by his side. But when it saw its master's murderers filing past, it rushed at them with furious barking and, as it voiced its accusation, turned to look at the king so that not only he, but everyone present, became suspicious of the men. They were at once arrested and when put to the question, with the help of some bits of external evidence as well, they confessed the murder and were punished.

The same thing is said to have been done by the poet Hesiod's dog, which convicted the sons of Ganyctor the Naupactian, by whom Hesiod had been murdered. But a matter which came to the attention of our fathers when they were studying at Athens is even plainer than anything so far mentioned. A certain fellow slipped into the temple of Asclepius, took such gold and silver offerings as were not bulky, and made his escape, thinking that he had not been detected. But the watchdog, whose name was Capparus, when none of the sacristans responded to its barking, pursued the escaping temple-thief. First the man threw stones at it, but could not drive it away. When day dawned, the dog did not approach close, but followed the man, always keeping him in sight, and refused the food he offered. When he stopped to rest, the dog passed the night on guard; when he struck out again, the dog got up and kept following, fawning on the other people it met on the road and barking at the man and sticking to his heels. When those who were investigating the robbery learned this from men who had encountered the pair and were told the colour and size of the dog, they pursued all the more vigorously and overtook the man and brought him back from Crommyon. ⁹⁷⁰ On the return the dog led the procession, capering and exultant, as though it claimed for itself the credit for pursuing and capturing the temple-thief. The people actually voted it a public ration of food and entrusted the charge of this to the priests in perpetuity, thereby imitating the ancient Athenian kindness to the mule. For when Pericles was building the Hecatompedon on the Acropolis, stones were naturally brought by numerous teams of draught-animals every day. Now one of the mules who had assisted gallantly in the work, but had now been discharged because of old age, used to go down every day to the Ceramicus and meet the beasts which brought the stones, turning back with them and trotting along by their side, as though to encourage and cheer them on. So the people of Athens, admiring its enterprise, gave orders for it to be maintained at the public expense, voting it free meals, as though to an athlete who had succumbed to old age.

¹⁴ Therefore those who deny that there is any kind of justice owed to animals by us must be conceded to be right so far as marine and deep-sea creatures are concerned; for these are completely lacking in amiability, apathetic, and devoid of all sweetness of disposition. And well did Homer say

The gray-green sea bore you,

with reference to a man regarded as uncivilized and unsociable, implying that the sea produces nothing friendly or gentle. But a man who would use such speech in regard to land animals is himself cruel and brutal. Or perhaps you will not admit that there was a bond of justice between Lysimachus and the Hyrcanian dog which alone stood guard by his corpse and, when his body was cremated, rushed into the flames and hurled itself upon him. The same is reported to have been done by the eagle which was kept by Pyrrhus, not the king, but a certain private citizen; when he died, it kept vigil by his body; at the funeral it hovered about the bier and finally folded its wings, settled on the pyre and was consumed with its master's body.

The elephant of King Porus, when he was wounded in the battle against Alexander, gently and solicitously pulled out with its trunk many of the javelins sticking in its master. Though it was in a sad state itself, it did not give up until it perceived that the king had lost much blood and was slipping off; then, fearing that he would fall, it gently kneeled and afforded its master a painless glide.

Bucephalas unsaddled would permit his groom to mount him; but when he was all decked out in his royal accoutrements and collars, he would let no one approach except Alexander himself. If any others tried to come near, he would charge at them loudly neighing and rear and trample any of them who were not quick enough to rush far away and escape.

¹⁵ I am not unaware that you will think that my examples are rather a hodge-podge; but it is not easy to find naturally clever animals doing anything which illustrates merely one of their virtues. Their probity, rather, is revealed in their love of offspring and their cleverness in their nobility; then, too, their craftiness and intelligence is inseparable from their ardour and courage. Those, nevertheless, who are intent on classifying and defining each separate occasion will find that dogs give the impression of a mind that is at once civil and superior when they turn away from those who sit on the ground — which is presumably referred to in the lines

The dogs barked and rushed up, but wise Odysseus
Cunningly crouched; the staff slipped from his hand;

for dogs cease attacking those who have thrown themselves down and taken on an attitude that resembles humility.

They relate further that the champion of the Indian dogs, one greatly admired by Alexander, when a stag was let loose and a boar and a bear, lay quiet and still and disregarded them; but when a lion appeared, it sprang up at once to prepare for the fray, showing clearly ⁹⁷¹ that it chose to match itself with the lion and scorned all the others.

Hounds that hunt hares, if they themselves kill them, enjoy pulling them to pieces and eagerly lap up the blood; but if, as frequently happens, a hare in desperation exhausts all its breath in a final sprint and expires, the hounds, when they come upon it dead, will

not touch it at all, but stand there wagging their tails, as much as to say that they do not strive for food, but for victory and the honour of winning.

¹⁶¹ There are many examples of cunning, but I shall dismiss foxes and wolves and the tricks of crane and daw (for they are obvious), and shall take for my witness Thales, the most ancient of the Wise Men, not the least of whose claims to admiration, they say, was his getting the better of a mule by a trick. For one of the mules that were used to carry salt, on entering a river, accidentally stumbled and, since the salt melted away, it was free of its burden when it got up. It recognized the cause of this and bore it in mind. The result was that every time it crossed the river, it would deliberately lower itself and wet the bags, crouching and bending first to one side, then to the other. When Thales heard of this, he gave orders to fill the bags with wool and sponges instead of salt and to drive the mule laden in this manner. So when it played its customary trick and soaked its burden with water, it came to know that its cunning was unprofitable and thereafter was so attentive and cautious in crossing the river that the water never touched the slightest portion of its burden even by accident.

Partridges exhibit another piece of cunning, combined with affection for their young. They teach their fledglings, who are not yet able to fly, to lie on their backs when they are pursued and to keep above them as a screen some piece of turf or rubbish. The mothers meanwhile lure the hunters in another direction and divert attention to themselves, fluttering along at their feet and rising only briefly until, by making it seem that they are on the point of being captured, they draw them far away from their young.

When hares return for repose, they put to sleep their leverets in quite different places, often as much as a hundred feet apart, so that, if man or dog comes near, they shall not all be simultaneously in danger. The hares themselves run to and fro and leave tracks in many places, but last of all with a great leap they leave their traces far behind, and so to bed.

The she-bear, just prior to the state called hibernation, before she becomes quite torpid and heavy and finds it difficult to move, cleans out her lair and, when about to enter, approaches it as lightly and inconspicuously as possible, treading on tiptoe, then turns around and backs into the den.

Hinds are inclined to bear their young beside a public road where carnivorous animals do not come; and stags, when they observe that they have grown heavy by reason of their fat and surplus flesh, vanish and preserve themselves by hiding when they do not trust to their heels.

The way in which hedgehogs defend and guard themselves has occasioned the proverb:

The fox knows many tricks, but the hedgehog one big one;
for when the fox approaches, as Ion says, it,
Curling its spiny body in a coil,
Lies still, impregnable to touch or bite.

But the provision that the hedgehog makes for its young is even more ingenious. When autumn comes, it creeps under the vines and with its paws shakes down to the ground grapes from the bunches and, having rolled about in them, gets up with them attached to its quills. ⁹⁷² Once when I was a child I saw one, like a creeping or walking bunch of grapes! Then it goes down into its hole and delivers its load to its young for them to enjoy and draw rations from. Their lair has two openings, one facing the south, the other the north; when they perceive that the wind will change, like good skippers who shift sail, they block up the entrance which lies to the wind and open the other. And a man in Cyzicus observing this acquired a reputation for being able to predict unaided which way the wind would blow.

^{17 1} Elephants, as Juba declares, exhibit a social capacity joined with intelligence. Hunters dig pits for them, covering them with slender twigs and light rubbish; when, accordingly, any elephant of a number travelling together falls in, the others bring wood and stones and throw them in to fill up the excavation so that their comrade can easily get out. He also relates that, without any instruction, elephants pray to the gods, purifying themselves in the sea and, when the sun rises, worshipping it by raising their trunks, as if they were hands of supplication. For this reason they are the animal most loved of the gods, as Ptolemy Philopator has testified; for when he had vanquished Antiochus and wished to honour the gods in a really striking way, among many other offerings to commend his victory in battle, he sacrificed four elephants. Thereafter, since he had dreams by night in which the deity angrily threatened him because of that strange sacrifice, he employed many rites of appeasement and set up as a votive offering four bronze elephants to match those he had slaughtered.

Social usages are to be found no less among lions. For young lions take along with them to the hunt the old and slow; when the latter are tired out, they rest and wait, while the young lions hunt on. When they have taken anything, they summon the others by a raring like the bleat of a calf; the old ones hear it at once and come to partake in common of the prey.

^{18 1} The loves of some animals are wild and furious, while others have a refinement which is not far from human and an intercourse conducted with much grace. Such was the elephant which at Alexandria played the rival to Aristophanes the grammarian. They were, in fact, in love with the same flower-girl; nor was the elephant's love less manifest: as he passed by the market, he always brought her fruit and stood beside her for a long time and would insert his trunk, like a hand, within her garments and gently caress her fair breasts.

The serpent that fell in love with an Aetolian woman used to visit her at night and slip under some part of her body next the skin and coil about her without doing her any harm at all, either intentional or accidental; but always at daybreak it was decent enough to glide away. And this it did constantly until the kinsmen of the woman removed her to a house at some distance. The serpent did not come to her for three or four nights; but all the time, we may suppose, it was going about in search of her and missing its goal. At

last, when it had somehow found her with difficulty, it embraced her, not with that former gentleness it had used, but rather more roughly, its coils binding her hands to her body, and with the end of its tail it lashed the calves of her legs, displaying a light and tender anger that had in it more indulgence than punishment.

As for the goose in Aegium that loved a boy and the ram that set his heart on Glaucus the harp-player, since these are famous tales and I rather imagine you have had enough of such to spoil your appetite for more, I omit them.

19 1 As for starlings and crows and parrots which learn to talk and afford their teachers so malleable and imitative a vocal current to train and discipline, ⁹⁷³ they seem to me to be champions and advocates of the other animals in their ability to learn, instructing us in some measure that they too are endowed both with rational utterance and with articulate voice; for which reason it is quite ridiculous to admit a comparison of them with creatures who have not enough voice even to howl or groan. And what music, what grace do we not find in the natural, untaught warbling of birds! To this the most eloquent and musical of our poets bear witness when they compare their sweetest songs and poems to the singing of swans and nightingales. Now since there is more reason in teaching than in learning, we must yield assent to Aristotle when he says that animals do teach: a nightingale, in fact, has been observed instructing her young how to sing. A further proof that supports him is the fact that birds which have been taken young from the nest and bred apart from their mothers sing the worse for it; for the birds that are bred with their mothers are taught and learn, not for pay or glory, but for the joy of rivalling each other in song and because they cherish the beautiful in their utterance rather than the useful.

On this subject I have a story to tell you which I heard myself from many Greeks and Romans who were eye-witnesses. A certain barber at Rome had his shop directly opposite the precinct which they call the Market of the Greeks. He bred up a wonderful prodigy of a jay with a huge range of tones and expressions, which could reproduce the phrases of human speech and the cries of beasts and the sound of instruments — under no compulsion, but making it a rule and point of honour to let nothing go unrepeated or unimitated. Now it happened that a certain rich man was buried from that quarter to the blast of many trumpets and, as is customary, there was a halt in front of the barber-shop while the trumpeters, who were applauded and encored, played for a long time. From that day on the jay was speechless and mute, not letting out even a peep to request the necessities of life; so those who habitually passed the place and had formerly wondered at her voice, were now even more astonished at her silence. Some suspected that she had been poisoned by rival bird-trainers, but most conjectured that the trumpets had blasted her hearing and that her voice had been simultaneously extinguished. Now neither of these guesses was correct: it was self-discipline, it would seem, and her talent for mimicry that had sought an inner retreat as she refitted and prepared her voice like a musical instrument. For suddenly her mimicry returned and there blazed forth none of those old familiar imitations, but only the music of the trumpets, reproduced

with its exact sequences and every change of pitch and rhythm and tone. I conclude, as I said before, that self-instruction implies more reason in animals than does readiness to learn from others.

Still, I believe that I should not pass over one example at least of a dog's learning, of which I myself was a spectator at Rome. The dog appeared in a pantomime with a dramatic plot and many characters and conformed in its acting at all points with the acts and reactions required by the text. In particular, they experimented on it with a drug that was really soporific, but supposed in the story to be deadly. The dog took the bread that was supposedly drugged, swallowed it, and a little later appeared to shiver and stagger and nod until it finally sprawled out and lay there like a corpse, letting itself be dragged and hauled about, as the plot of the play prescribed. But when it recognized from the words and the action that the time had come, at first it began to stir slightly, as though recovering from a profound sleep, and lifted its head and looked about.⁹⁷⁴ Then to the amazement of the spectators it got up and proceeded to the right person and fawned on him with joy and pleasure so that everyone, and even Caesar himself (for the aged Vespasian was present in the Theatre of Marcellus), was much moved.

²⁰ Yet perhaps it is ridiculous for us to make a parade of animals distinguished for learning when Democritus declares that we have been their pupils in matters of fundamental importance: of the spider in weaving and mending, of the swallow in home-building, of the sweet-voiced swan and nightingale in our imitation of their song. Further, of the three divisions of medicine, we can discern in animals a generous portion of each; for it is not cure by drugs alone of which they make use. After devouring a serpent tortoises take a dessert of marjoram, and weasels of rue. Dogs purge themselves when bilious by a certain kind of grass. The snake sharpens and restores its fading sight with fennel. When the she-bear comes forth from her lair, the first thing she eats is wild arum; for its acidity opens her gut which has become constricted. At other times, when she suffers from nausea, she resorts to anthills and sits, holding out her tongue all running and juicy with sweet liquor until it is covered with ants; these she swallows and is alleviated. The Egyptians declare that they have observed and imitated the ibis' clyster-like purging of herself with brine; and the priests make use of water from which an ibis has drunk to purify themselves; for if the water is tainted or unhealthy in any way, the ibis will not approach it.

Then, too, some beasts cure themselves by a short fast, like wolves and lions who, when they are surfeited with flesh, lie still for a while, basking in the sun. And they say a tigress, if a kid is given her, will keep fasting for two days without eating; on the third, she grows hungry and asks for some other food. She will even pull her cage to pieces, but will not touch the kid which she has now come to regard as a fellow-boarder and room mate.

Yet again, they relate that elephants employ surgery: they do, in fact, bring aid to the wounded by easily and harmlessly drawing out spars and javelins and arrows without any laceration of the flesh. And Cretan goats, when they eat dittany, easily expel arrows

from their bodies and so have presented an easy lesson for women with child to take to heart, that the herb has an abortive property; for there is nothing except dittany that the goats, when they are wounded, rush to search for.

21 1 These matters, though wonderful, are less surprising than are those creatures which have cognition of number and can count, as do the cattle near Susa. At that place they irrigate the royal park with water raised in buckets by wheels, and the number of bucketfuls is prescribed. For each cow raises one hundred bucketfuls each day, and more you could not get from her, even if you wanted to use force. In fact, they often try to add to the number to see; but the cow balks and will not continue when once she has delivered her quota, so accurately does she compute and remember the sum, as Ctesias of Cnidus has related.

The Libyans laugh at the Egyptians for telling a fabulous tale about the oryx, that it lets out a cry at that very day and hour when the star rises that they call Sothis, which we call the Dog Star or Sirius. At any rate, when this star rises flush with the sun, practically all the goats turn about and look toward the east; and this is the most certain sign of its return and agrees most exactly with the tables of mathematical calculation.

22 1 975 But that my discourse may add its finishing touch and terminate, let me “make the move from the sacred line” and say a few words about the divine inspiration and the mantic power of animals. It is, in fact, no small or ignoble division of divination, but a great and very ancient one, which takes its name from birds; for their quickness of apprehension and their habit of responding to any manifestation, so easily are they diverted, serves as an instrument for the god, who directs their movements, their calls or cries, and their formations which are sometimes contrary, sometimes favouring, as winds are; so that he uses some birds to cut short, others to speed enterprises and inceptions to the destined end. It is for this reason that Euripides calls birds in general “heralds of the gods”; and, in particular, Socrates says that he considers himself a “fellow-slave of the swans.” So again, among monarchs Pyrrhus liked to be called an Eagle and Antiochus a Hawk. But when we deride, or rail at, stupid and ignorant people we call them “fish.” Really, we can produce cases by the thousand of signs and portents manifested to us by the gods through creatures of land and air, but not one such can the advocate for aquatic creatures name. No, they are all “deaf and blind” so far as foreseeing anything goes, and so have been cast aside into the godless and titanic region, as into a Limbo of the Unblessed, where the rational and intelligent part of the soul has been extinguished. Having, however, only a last remnant of sensation that is clogged with mud and deluged with water, they seem to be at their last gasp rather than alive.

975 23 1 Heracleon. Raise your brows, dear Phaedimus, and rouse yourself to defend us the sea folk, the island-dwellers! This bout of argument has become no child’s play, but a hard-fought contest, a debate which lacks only the actual bar and platform.

Phaedimus. Not so, Heracleon, but an ambush laid with malice aforethought has been disclosed. While we are still tipsy and soused from yesterday's bout, this gentleman, as you see, has attacked us with premeditation, cold sober. Yet there can be no begging off. Devotee of Pindar though I am, I do not want to be addressed with the quotation

To excuse oneself when combat is offered

Has consigned valour to deep obscurity;

for we have much leisure; and it is not our discourse that will be idle, but our dogs and horses, our nets and seines of all kinds, for a truce is granted for to day because of our argument to every creature both on land and sea. Yet do not fear: I shall use it with moderation, introducing no opinions of philosophers or Egyptian fables or unattested tales of Indians or Libyans. But those facts that may be observed everywhere and have as witnesses the men who exploit the sea and acquire their credit from direct observation, of these I shall present a few. Yet there is nothing to impede illustrations drawn from land animals: the land is wide open for investigation by the senses. The sea, on the other hand, grants us but a few dubious glimpses. She draws a veil over the birth and growth, the attacks and reciprocal defences, of most of her denizens. Among these there are no few feats of intelligence and memory and community spirit that remain unknown to us and so obstruct our argument. Then too, land animals by reason of their close relationship and their cohabitation have to some extent been imbued with human manners; they have the advantage of their breeding and teaching and imitation, which sweetens all their bitterness and sullenness, like fresh water mixed with brine, while their lack of understanding and dullness are roused to life by human contacts. Whereas the life of sea creatures, being set apart by mighty bounds from intercourse with men and having nothing adventitious or acquired from human usage, ⁹⁷⁶ is peculiar to itself, indigenous and uncontaminated by foreign ways, not by distinction of Nature, but of location. For their Nature is such as to welcome and retain such instruction as reaches them. This it is that renders many eels tractable, like those that are called sacred in Arethusa; and in many places there are fish which will respond to their own names, as the story goes of Crassus' moray, upon the death of which he wept. And once when Domitius said to him, "Isn't it true that you wept when a moray died?" he answered, "Isn't it true that you buried three wives and didn't weep?"

The priests' crocodiles not only recognize the voice of those who summon them and allow themselves to be handled, but open their mouths to let their teeth be cleaned by hand and wiped with towels. Recently our excellent Philinus came back from a trip to Egypt and told us that he had seen in Antaeopolis an old woman sleeping on a low bed beside a crocodile, which was stretched out beside her in a perfectly decorous way.

They have long been telling the tale that when King Ptolemy summoned the sacred crocodile and it would not heed him or obey in spite of his entreaties and requests, it seemed to the priests an omen of his death, which came about not long after; whence it appears that the race of water creatures is not wholly unendowed with your precious gift of divination. Indeed, I have heard that near Sura, a village in Lycia between

Phellus and Myra, men sit and watch the gyrations and flights and pursuits of fish and divine from them by a professional and rational system, as others do with birds.

241 But let these examples suffice to show that sea animals are not entirely unrelated to us or cut off from human fellowship. Of their uncontaminated and native intelligence their caution is strong evidence. For nothing that swims and does not merely stick or cling to rocks is easily taken or captured without trouble by man as are asses by wolves, bees by bee-eaters, cicadas by swallows, and snakes by deer, which easily attract them. This, in fact, is why deer are called elaphoi, not from their swiftness, but from their power of attracting snakes. So too the ram draws the wolf by stamping and they say that very many creatures, and particularly apes, are attracted to the panther by their pleasure in its scent. But in practically all sea-creatures any sensation is suspect and evokes an intelligently inspired defensive reaction against attack, so that fishing has been rendered no simple or trivial task, but needs all manner of implements and clever and deceitful tricks to use against the fish.

This is perfectly clear from ready examples: no one wants to have an angler's rod too thick, though it needs elasticity to withstand the thrashing of such fish as are caught; men select, rather, a slender rod so that it may not cast a broad shadow and arouse suspicion. In the next place, they do not thicken the line with many plies when they attach the loop and do not make it rough; for this, too, betrays the lure to the fish. They also contrive that the hairs which form the leader shall be as white as possible; for in this way they are less conspicuous in the sea because of the similarity of colour. The remark of the Poet:

Like lead she sank into the great sea depths,
Like lead infixed in horn of rustic ox
Which brings destruction to the ravenous fish —

some misunderstand this and imagine that the ancients used ox-hair for their lines, alleging that keras means "hair" and for this reason keirasthai means "to have one's hair cut" and koura is a "haircut" and the keroplastes in Archilochus is one who is fond⁹⁷⁷ of trimming and beautifying the hair. But this is not so: they use horse-hair which they take from males, for mares by wetting the hair with their urine make it weak. Aristarchus declares that there is nothing erudite or subtle in these lines; the fact is that a small piece of horn was attached to the line in front of the hook, since the fish, when they are confronted by anything else, chew the line in two. They use rounded hooks to catch mullets and bonitos, whose mouths are small; for they are wary of a broader hook. Often, indeed, the mullet suspects even a rounded hook and swims around it, flipping the bait with its tail and snatching up bits it has dislodged; or if it cannot do this, it closes its mouth and purses it up and with the tips of its lips nibbles away at the bait.

The sea-bass is braver than your elephant: it is not from another, but from himself without assistance, that he extracts the barb when he is caught by the hook; he swings his

head from side to side to widen the wound, enduring the pain of tearing his flesh until he can throw off the hook. The fox-shark does not often approach the hook and shuns the lure; but if he is caught, he immediately turns himself inside out, for by reason of the elasticity and flexibility of his body he can naturally shift and twist it about, so that when he is inside out, the hook falls away.

²⁵ 1 Now the examples I have given indicate intelligence and an ingenious, subtle use of it for opportune profit; but there are others that display, in combination with understanding, a social sense and mutual affection, as is the case with the barbier and the parrot-fish. For if one parrot-fish swallows the hook, the others present swarm upon the line and nibble it away; and the same fish, when any of their kind have fallen into the net, give them their tails from outside; when they eagerly fix their teeth in these, the others pull on them and bring them through in tow. And barbiers are even more strenuous in rescuing their fellows: getting under the line with their backs, they erect their sharp spines and try to saw the line through and cut it off with the rough edge.

Yet we know of no land animal that has the courage to assist another in danger — not bear or boar or lioness or panther. True it is that in the arena those of the same kind draw close together and huddle in a circle; yet they have neither knowledge nor desire to help each other. Instead, each one flees to get as far as possible from a wounded or dying fellow. That tale of the elephants carrying brushwood to the pits and giving their fallen comrade a ramp to mount is monstrous and far-fetched and dictates, as it were, that we are to believe it on a king's prescription — that is, on the writs of Juba. Suppose it to be true: it merely proves that many sea creatures are in no way inferior in community spirit and intelligence to the wisest of the land animals. As for their sociability, I shall soon make a special plea on that topic.

²⁶ 1 Now fishermen, observing that most fish evade the striking of the hook by such countermoves as wrestlers use, resorted, like the Persians, to force and used the dragnet, since for those caught in it there could be no escape with the help of reason or cleverness. For mullet and rainbow-wrasse are caught by casting-nets and round nets, as are also the bream and the sargue and the goby and the sea-bass. The so called net fish, that is surmullet and gilthead and sculpin, are caught in seines by trawling: accordingly it was quite correct for Homer to call this kind of net a "catch-all." Codfish, like bass, have devices even against these. For when the bass perceives that the trawl is approaching, it forces the mud apart and hammers a hollow in the bottom. When it has made room enough to allow the net to overrun it, it thrusts itself in and waits until the danger is past.

Now when the dolphin is caught and perceives itself to be trapped in the net, it bides its time, not at all disturbed but well pleased, for it feasts without stint on the fish that have been gathered with no trouble to itself. But as soon as it comes near the shore, it bites its way through the net and makes its escape. ⁹⁷⁸ Yet if it should not get away in time, on the first occasion it suffers no harm: the fishermen merely sew rushes to its crest and let it go. But if it is taken a second time, they recognize it from the seam and

punish it with a beating. This, however, rarely occurs: most dolphins are grateful for their pardon in the first instance and take care to do no harm in the future.

Further, among the many examples of wariness, precaution, or evasion, we must not pass over that of the cuttlefish: it has the so called mytis beside the neck full of black liquid, which they call "ink." When it is come upon, it discharges the liquid to the purpose that the sea shall be inked out and create darkness around it while it slips through and eludes the fisherman's gaze. In this it imitates Homer's gods who often "in a dark cloud" snatch up and smuggle away those whom they are pleased to save. But enough of this.

27 1 As for cleverness in attacking and catching prey, we may perceive subtle examples of it in many different species. The starfish, for example, knowing that everything with which it comes in contact dissolves and liquefies, offers its body and is indifferent to the contact of those that overtake or meet it. You know, of course, the property of the torpedo: not only does it paralyse all those who touch it, but even through the net creates a heavy numbness in the hands of the trawlers. And some who have experimented further with it report that if it is washed ashore alive and you pour water on it from above, you may perceive the numbness mounting to the hand and dulling your sense of touch by way of the water which, so it seems, suffers a change and is first infected. Having, therefore, an innate sense of this power, it never makes a frontal attack or endangers itself; rather, it swims in a circle around its prey and discharges its shocks as if they were darts, thus poisoning first the water, then through the water the creature which can neither defend itself nor escape, being held fast as if by chains and frozen stiff.

The so called fisherman is known to many; he gets his name from his actions. Aristotle says that the cuttlefish also makes use of this stratagem: he lets down, like a fishing line, a tentacle from his neck which is naturally designed to extend to a great length when it is released, or to be drawn to him when it is pulled in. So when he espies a little fish, he gives it the feeler to bite and then by degrees imperceptibly draws it back toward himself until the prey attached to the arm is within reach of his mouth.

As for the octopus' change of colour, Pindar has made it celebrated in the words

To all the cities to which you resort

Bring a mind like the changing skin of the seabeast;

and Theognis likewise:

Be minded like the octopus' hue:

The colour of its rock will meet the view.

The chameleon, to be sure, is metachromatic, but not from any design or desire to conceal itself; it changes colour uselessly from fear, being naturally timid and cowardly. And this is consistent with the abundance of air in it, as Theophrastus says; for nearly the whole body of the creature is occupied by its lungs, which shows it to be full of air and for this reason easily moved to change colour. But this same action on the part of the octopus is not an emotional response, but a deliberate change, since it uses this device

to escape what it fears and to capture what it feeds on: by this deceit it can both seize the latter, which does not try to escape, and avoid the former, which proceeds on its way. Now the story that it eats its own tentacles is a lie, but it is true that it fears the moray and the conger. It is, in fact, maltreated by them; for it cannot do them harm, since they slip from its grasp. ⁹⁷⁹ On the other hand, when the crawfish has once got them in its grasp, it wins the victory easily, for smoothness is no aid against roughness; yet when the octopus has once thrust its tentacles inside the crawfish, the latter succumbs. And so Nature has created this cycle and succession of mutual pursuit and flight as a field for the exercise and competitive practice of adroitness and intelligence.

²⁸ ¹ We have, to be sure, heard Aristotimus telling us about the hedgehog's foreknowledge of the winds; and our friend also admired the V shaped flight of cranes. I can produce no hedgehog of Cyzicus or Byzantium, but instead the whole body of sea-hedgehogs, which, when they perceive that storm and surf are coming, ballast themselves with little stones in order that they may not be capsized by reason of their lightness or be swept away by the swell, but may remain fixed in position through the weight of their little rocks.

Again, the cranes' change of flight against the wind is not merely the action of one species: all fish generally have the same notion and always swim against wave and current, taking care that a blast from the rear does not fold back their scales and expose and roughen their bodies. For this reason they always present the prow of their bodies to the waves, for in that way head first they cleave the sea, which depresses their gills and, flowing smoothly over the surface, keeps down, instead of ruffling up, the bristling kin. Now this, as I have said, is common to all fish except the sturgeon, which, they say, swims with wind and tide and does not fear the harrowing of its scales since the overlaps are not in the direction of the tail.

²⁹ ¹ The tunny is so sensitive to equinox and solstice that it teaches even men themselves without the need of astronomical tables; for wherever it may be when the winter solstice overtakes it, in that same place it stands and stays until the equinox. As for that clever device of the crane, the grasping of the stone by night so that if it falls, she may awake from sleep — how much cleverer, my friend, is the artifice of the dolphin, for whom it is illicit to stand still or to cease from motion. For its nature is to be ever active: the termination of its life and its movement is one and the same. When it needs sleep, it rises to the surface of the sea and allows itself to sink deeper and deeper on its back, lulled to rest by the swinging motion of the ground swell until it touches the bottom. Thus roused, it goes whizzing up, and when it reaches the surface, again goes slack, devising for itself a kind of rest combined with motion. And they say that tunnies do the same thing for the same reason.

Having just a moment ago given you an account of the tunny's mathematical foreknowledge of the reversal of the sun, of which Aristotle is a witness, I beg you to hear the tale of their arithmetical learning. But first, I swear, I must mention their

knowledge of optics, of which Aeschylus seems not to have been ignorant, for these are his words:

Squinting the left eye like a tunny fish.

They seem, indeed, to have poor sight in one eye. And it is for this reason that when they enter the Black Sea, they hug one bank on the right, and the other when they are going out, it being very prudent and sagacious of them always to entrust the protection of themselves to the better eye. Now since they apparently need arithmetic to preserve their consociation and affection for each other, they have attained such perfection of learning that, since they take great pleasure in feeding and schooling together, they always form the school into a cube, making it an altogether solid figure with a surface of six equal solid plane sides; then they swim on their way preserving their formation, a square that faces both ways. ⁹⁸⁰ Certainly a hooer watching for tunnies who counts the exact number of the surface at once makes known the total number of the shoal, since he knows that the depth is equal one to one with the breadth and the length.

³⁰¹ Schooling together has also given the bonitos their name of amia and I think this is true of year-old tunnies as well. As for the other kinds which are observed to live in shoals in mutual society, it is impossible to state their number. Let us rather, therefore, proceed to examine those that have a special partnership, that is, symbiosis. One of these is the pinna-guard, over which Chrysippus spilled a very great deal of ink; indeed it has a reserved seat in every single book of his, whether ethical or physical. Chrysippus has obviously not investigated the sponge-guard; otherwise he could hardly have left it out. Now the pinna-guard is a crab-like creature, so they say, who lives with the pinna and sits in front of the shell guarding the entrance. It allows the pinna to remain wide open and agape until one of the little fish that are their prey gets within; then the guard nips the flesh of the pinna and slips inside; the shell is closed and together they feast on the imprisoned prey.

The sponge is governed by a little creature not resembling a crab, but much like a spider. Now the sponge is no lifeless, insensitive, bloodless thing; though it clings to the rocks, as many other animals do, Cit has a peculiar movement outward and inward which needs, as it were, admonition and supervision. In any case it is loose in texture and its pores are relaxed because of its sloth and dullness; but when anything edible enters, the guard gives the signal, and it closes up and consumes the prey. Even more, if a man approaches or touches it, informed by the scratching of the guard, it shudders, as it were, and so closes itself up by stiffening and contracting that it is not an easy, but a very difficult, matter for the hunters to undercut it.

The purplefish lives in colonies which build up a comb together, like bees. In this the species is said to propagate; they catch at edible bits of oyster-green and seaweed that stick to shells, and furnish each other with a sort of periodic rotating banquet, as they feed one after another in series.

³¹¹ And why should anyone be surprised at the community life of these when the most unsociable and brutal of all creatures bred in river, lake, or sea, the crocodile, shows

himself marvellously proficient at partnership and goodwill in his dealings with the Egyptian plover? The plover is a bird of the swamps and river banks and it guards the crocodile, not supplying its own food, but as a boarder making a meal of the crocodile's scraps. Now when it perceives that, during the crocodile's sleep, the ichneumon is planning to attack it, smearing itself with mud like an athlete dusting himself for the fray, the bird awakes the crocodile by crying and pecking at it. And the crocodile becomes so gentle with it that it will open its mouth and let it in and is pleased that the bird quietly pecks out, with its bill, bits of flesh which are caught in the teeth and cleans them up. When it is now satisfied and wants to close its mouth, it tilts its snout upward as an indication of its desire and does not let it down until the plover, at once perceiving the intention, flies out.

The so called "guide" is a small fish, in size and shape like a goby; but by reason of the roughness of its scales it is said to resemble a ruffled bird. It always accompanies one of the great whales, swimming in front of it and directing its course so that it may not go aground in shallows or be cut off in some lagoon or strait from which exit may be difficult. ⁹⁸¹ The whale follows it, as a ship obeys the helm, changing course with great docility. And whatever else, creature or boat or stone, it embraces in its gaping jaws is at once destroyed and goes to its ruin completely engulfed; but the little fish it knows and receives inside its mouth as in a haven. While the fish sleeps within, the whale remains motionless and lies by; but when it comes out again, the beast accompanies it and does not depart from it day or night; or, if it does, it gets lost and wanders at random. Many, indeed, have been cast up on the land and perished, being, as it were, without a pilot. We, in fact, were witnesses of such a mishap near Anticyra not long ago; and they relate that some time ago, when a whale came aground not far from Boulis and rotted, a plague ensued.

Is it, then, justifiable to compare with these associations and companionships those friendships which Aristotle says exist between foxes and snakes because of their common hostility to the eagle; or those between bustards and horses because the former like to approach and pick over the dung? As for me, I perceive even in ants or bees no such concern for each other. It is true that every one of them promotes the common task, yet none of them has any interest in or regard for his fellow individually.

^{32 1} And we shall observe this difference even more clearly when we turn our attention to the oldest and most important of social institutions and duties, those concerned with generation and procreation. Now in the first place those fish that inhabit a sea that borders on lagoons or receives rivers resort to these when they are ready to deposit their eggs, seeking the tranquillity and smoothness of fresh water, since calm is a good midwife. Besides, lagoons and rivers are devoid of sea monsters, so that the eggs and fry may survive. This is the reason why the Black Sea is most favoured for spawning by very many fish. It breeds no large sea beasts at all except an infrequent seal and a small dolphin; besides, the influx of rivers — and those which empty into the Black Sea are numerous and very large — creates a gentle blend conducive to the

production of offspring. The most wonderful tale is told about the anthias, which Homer calls "Sacred Fish." Yet some think that "sacred" means "important," just as we call the important bone os sacrum and epilepsy, an important disease, the sacred disease. Others interpret it in the ordinary sense as meaning "dedicated" or "consecrated." Eratosthenes seems to refer to the gilthead when he says

Swift courser golden-browed, the sacred fish.

Many say that this is the sturgeon, which is rare and hard to catch, though it is often seen off the coast of Pamphylia. If any ever do succeed in catching it, they put on wreaths themselves and wreath their boats; and, as they sail past, they are welcomed and honoured with shouts and applause. But most authorities hold that it is the anthias that is and is called "sacred," for wherever this fish appears there are no sea monsters. Sponge-fishers may dive in confidence and fish may spawn without fear, as though they had a guarantor of their immunity. The reason for this is a puzzle: whether the monsters avoid the anthias as elephants do a pig and lions a cock, or whether there are indications of places free from monsters, which the fish comes to know and frequents, being an intelligent creature with a good memory.

^{33 1} Then again the care of the young is shared by both parents: the males do not eat their own young, but stand by the spawn to guard the eggs, as Aristotle relates. Some follow the female and sprinkle the eggs gradually with milt, for otherwise the spawn will not grow, but remains imperfect and undeveloped. In particular the wrasse makes a sort of nest of seaweed, envelops the spawn in it, and shelters it from the waves.

⁹⁸² The affection of the dogfish for its young is not inferior in warmth and kindness to that of any of the tamest animals; for they lay the egg, then sustain and carry the newly hatched young, not without, but within themselves, as if from a second birth. When the young grow larger, the parents let them out and teach them to swim close by; then again they collect them through their mouths and allow their bodies to be used as dwelling-places, affording at once room and board and sanctuary until the young become strong enough to shift for themselves.

Wonderful also is the care of the tortoise for the birth and preservation of her young. To bear them she comes out of the sea to the shore near at hand; but since she is unable to incubate the eggs or to remain on dry land for long, she deposits them on the strand and heaps over them the smoothest and softest part of the sand. When she has buried and concealed them securely, some say that she scratches and scribbles the place with her feet, making it easy for her to recognize; others affirm that it is because she has been turned on her back by the male that she leaves peculiar marks and impressions about the place. But what is more remarkable than this, she waits for the fortieth day (for that is the number required to develop and hatch out the eggs) and then approaches. And each tortoise recognizes her own treasure and opens it more joyously and eagerly than a man does a deposit of gold.

^{34 1} The accounts given of the crocodile are similar in other respects, but the animal's ability to estimate the right place goes beyond man's power to guess or calculate the

cause. Hence they affirm that this creature's foreknowledge is divine and not rational. For neither to a greater or a less distance, but just so far as the Nile will spread that season and cover the land in flood, just so far does she go to deposit her eggs, with such accuracy that any farmer finding the eggs may know himself and predict to others how far the river will advance. And her purpose in being so exact is to prevent either herself or her eggs getting wet when she sits on them. When they are hatched, the one which, upon emerging, does not immediately seize in its mouth anything that comes along, fly or midge or worm or straw or plant, the mother tears to pieces and bites to death; but those that are bold and active she loves and tends, thus bestowing her affection by judgement, as the wisest of men think right, not by emotion.

Furthermore, seals too bear their young on dry land and little by little induce their offspring to try the sea, then quickly take them out again. This they do often at intervals until the young are conditioned in this way to feel confidence and enjoy life in the sea.

Frogs in their coupling use a call, the so called ololygon, a cry of wooing and mating. When the male has thus attracted the female, they wait for the night together, for they cannot consort in the water and during the day they are afraid to do so on land; but when the darkness falls, they come out and embrace with impunity. On other occasions when their cry is shrill, it is because they expect rain. And this is among the surest of signs.

³⁵¹ But, dear Poseidon! What an absurd and ridiculous error I have almost fallen into: while I am spending my time on seals and frogs, I have neglected and omitted the wisest of sea creatures, the most beloved of the gods! For what nightingales are to be compared with the halcyon for its love of sweet sound, or what swallows for its love of offspring, or what doves for its love of its mate, or what bees for its skill in construction? What creature's procreation and birth pangs has the god so honoured? For Leto's parturition, so they say, only one island was made firm to receive her; but when the halcyon lays her eggs, about the time of the winter solstice, the god brings the whole sea to rest, without a wave, without a swell. ⁹⁸³ And this is the reason why there is no other creature that men love more. Thanks to her they sail the sea without a fear in the dead of winter for seven days and seven nights. For the moment, journey by sea is safer for them than by land. If it is proper to speak briefly of her several virtues, she is so devoted to her mate that she keeps him company, not for a single season, but throughout the year. Yet it is not through wantonness that she admits him to her company, for she never consorts at all with any other male; it is through friendship and affection, as with any lawful wife. When by reason of old age the male becomes too weak and sluggish to keep up with her, she takes the burden on herself, carries him and feeds him, never forsaking, never abandoning him; but mounting him on her own shoulders, she conveys him everywhere she goes and looks after him, abiding with him until the end.

As for love of her offspring and care for their preservation, as soon as she perceives herself to be pregnant, she applies herself to building the nest, not making pats of mud or cementing it on walls and roofs like the house-martin; nor does she use the activity of

many different members of her body, as when the bee employs its whole frame to enter and open the wax, with all six feet pressing at the same time to fashion the whole mass into hexagonal cells. But the halcyon, having but one simple instrument, one piece of equipment, one tool — her bill and nothing else, co operating with her industry and ingenuity — what she contrives and constructs would be hard to believe without ocular evidence, seeing the object that she moulds — or rather the ship that she builds. Of many possible forms, this alone cannot be capsized or even wet its cargo. She collects the spines of garfish and binds and weaves them together, some straight, others transverse, as if she were thrusting woven threads through the warp, adding such bends and knots of one with another that a compact, round unit is formed, slightly prolate in shape, like a fisherman's weel. When it is finished, she brings and deposits it beside the surging waves, where the sea beats gently upon it and instructs her how to mend and strengthen whatever is not yet good and tight, as she observes it loosened by the blows. She so tautens and secures the joints that it is difficult even for stones or iron to break or pierce it. The proportions and shape of the hollow interior are as admirable as anything about it; for it is so constructed as to admit herself only, while the entrance remains wholly hidden and invisible to others — with the result that not even a drop of water can get in. Now I presume that all of you have seen this nest; as for me, since I have often seen and touched it, it comes to my mind to chant the words

Once such a thing in Delos near Apollo's shrine

I saw, the Altar of Horn, celebrated as one of the Seven Wonders of the World because it needs no glue or any other binding, but is joined and fastened together, made entirely of horns taken from the right side of the head. Now may the god be propitious to me while I sing of the Sea Siren — and indeed, being both a musician and an islander, he should laugh good-naturedly at my opponents' scoffing questions. Why should he not be called a "conger-slayer" or Artemis be termed a "surmullet-slayer"? Since he well knows that Aphrodite, born of the sea, regards practically all sea creatures as sacred and related to herself and relishes the slaughter of none of them. In Leptis, you know, the priests of Poseidon refrain entirely from any sea food, and those initiated into the mysteries at Eleusis hold the surmullet in veneration, while the priestess of Hera at Argos abstains from this fish to pay it honour. For surmullets are particularly good at killing and eating the sea-hare, which is lethal to man. It is for this reason that surmullets possess this immunity, as being friendly and life-saving creatures.

36 1 984 Furthermore, many of the Greeks have temples and altars to Artemis Dictynna and Apollo Delphinios; and that place which the god had chosen for himself the poet says was settled by Cretans under the guidance of a dolphin. It was not, however, the god who changed his shape and swam in front of the expedition, as tellers of tales relate; instead, he sent a dolphin to guide the men and bring them to Cirrha. They also relate that Soteles and Dionysius, the men sent by Ptolemy Soter to Sinope to bring back

Serapis, were driven against their will by a violent wind out of their course beyond Malea, Bwith the Peloponnesus on their right. When they were lost and discouraged, a dolphin appeared by the prow and, as it were, invited them to follow and led them into such parts as had safe roadsteads with but a gentle swell until, by conducting and escorting the vessel in this manner, it brought them to Cirrha. Whence it came about that when they had offered thanksgiving for their safe landing, they came to see that of the two statues they should take away the one of Pluto, but should merely take an impress of that of Persephone and leave it behind.

Well might the god be fond of the music-loving character of the dolphin, to which Pindar likens himself, saying that he is roused

Like a dolphin of the sea
Who on the waveless deep of ocean
Is moved by the lovely sound of flutes.

Yet it is even more likely that its affection for men renders it dear to the gods; for it is the only creature who loves man for his own sake. Of the land animals, some avoid man altogether, others, the tamest kind, pay court for utilitarian reasons only to those who feed them, as do dogs and horses and elephants to their familiars. Martins take to houses to get what they need, darkness and a minimum of security, but avoid and fear man as a dangerous wild beast. To the dolphin alone, beyond all others, nature has granted what the best philosophers seek: friendship for no advantage. Though it has no need at all of any man, yet it is a genial friend to all and has helped many. The story of Arion is familiar to all and widely known; and you, my friend, opportunely put us in mind of the tale of Hesiod,

But you failed to reach the end of the tale.

When you told of the dog, you should not have left out the dolphins, for the information of the dog that barked and rushed with a snarl on the murderers would have been meaningless if the dolphins had not taken up the corpse as it was floating on the sea near the Nemeon and zealously passed it from group to group until they put it ashore at Rhium and so made it clear that the man had been stabbed.

Myrsilus of Lesbos tells the tale of Enalus the Aeolian who was in love with that daughter of Smintheus who, in accordance with the oracle of Amphitrite, was cast into the sea by the Penthilidae, whereupon Enalus himself leaped into the sea and was brought out safe on Lesbos by a dolphin.

And the goodwill and friendship of the dolphin for the lad of Iasus was thought by reason of its greatness to be true love. For it used to swim and play with him during the day, allowing itself to be touched; and when the boy mounted upon its back, it was not reluctant, but used to carry him with pleasure wherever he directed it to go, while all the inhabitants of Iasus flocked to the shore each time this happened. Once a violent storm of rain and hail occurred and the boy slipped off and was drowned. The dolphin

took the body and threw both it and itself together on the land and would not leave until it too had died, thinking it right to share a death for which it imagined that it shared the responsibility. And in memory of this calamity the inhabitants of Iasus have minted their coins with the figure of a boy riding a dolphin.

From this the wild tales about Coeranus gained credence.⁹⁸⁵ He was a Parian by birth who, at Byzantium, bought a draught of dolphins which had been caught in a net and were in danger of slaughter, and set them all free. A little later he was on a sea voyage in a penteconter, so they say, with fifty pirates aboard; in the strait between Naxos and Paros the ship capsized and all the others were lost, while Coeranus, they relate, because a dolphin sped beneath him and buoyed him up, was put ashore at Sicinus, near a cave which is pointed out to this day and bears the name of Coeraneum. It is on this man that Archilochus is said to have written the line

Out of fifty, kindly Poseidon left only Coeranus.

When later he died, his relatives were burning the body near the sea when a large shoal of dolphins appeared off shore as though they were making it plain that they had come for the funeral, and they waited until it was completed.

That the shield of Odysseus had a dolphin emblazoned on it, Stesichorus also has related; and the Zacynthians perpetuate the reason for it, as Critheus testifies. For when Telemachus was a small boy, so they say, he fell into the deep inshore water and was saved by dolphins who came to his aid and swam with him to the beach; and that was the reason why his father had a dolphin engraved on his ring and emblazoned on his shield, making this requital to the animal.

Yet since I began by saying that I would not tell you any tall tales and since, without observing what I was up to, I have now, besides the dolphins, run aground on both Odysseus and Coeranus to a point beyond belief, I lay this penalty upon myself: to conclude here and now.

³⁷¹ Aristotimus. So, gentlemen of the jury, you may now cast your votes.

Soclarus. As for us, we have for some time held the view of Sophocles:

It is a marvel how of rival sides

The strife of tongues welds both so close together.

For by combining what you have said against each other, you will together put up a good fight against those who would deprive animals of reason and understanding.

Beasts are rational



⁹⁸⁵ 1 1 Odysseus. These facts, Circe, I believe I have learned and shall not forget them; yet I should be happy to learn from you further whether there are any Greeks among those whom you have changed from the shape of men into wolves and lions.

Circe. Quite a few, beloved Odysseus. But what is your reason for asking this question?

Odysseus. It is, I swear, because it would bring me noble glory among the Greeks if by your favour I should restore comrades of mine to their original humanity and not allow them to grow old in the unnatural guise of beasts, leading an existence that is so piteous and shameful

Circe. Here's a lad who finds it appropriate that not only himself and his companions, but even total strangers should, through his stupidity, find his ambition their ruin.

Odysseus. This is a new potion of words that you are stirring and drugging for me, Circe. It will certainly transform me literally into a beast if I am to take your word for it that changing from beast to man spells ruin.

Circe. Haven't you already worked a stranger magic than this on yourself? You who refused an ageless, immortal life at my side and would struggle through a thousand new dangers to a woman who is mortal and, I can assure you, no longer so very young — ⁹⁸⁶ and this for no other object than to make yourself more gaped at and renowned than you already are, pursuing an empty phantom instead of what is truly good.

Odysseus. All right, let it be as you say, Circe. Why must we quarrel again and again about the same matters? Now please just grant me the favour of letting the men go free.

Circe. By the Black Goddess, it's not so simple as that. These creatures are no run of the mill. You must ask them first if they are willing. If they say no, my hero, you'll have to argue with them and convince them. And if you don't, and they win the argument, then you must be content with having exercised poor judgement about yourself and your friends.

Odysseus. Dear lady, why are you making fun of me? How can they argue with me or I with them so long as they are asses and hogs and lions?

Circe. Courage, courage, my ambitious friend. I'll see to it that you shall find them both receptive and responsive. Or rather, one of the number will be enough to thrust and parry for them all. Presto! You may talk with this one.

Odysseus. And how am I to address him, Circe? Who in the world was he?

Circe. What's that to do with the issue? Call him Gryllus, if you like. I'll retire now to avoid any suggestion that he is arguing against his own convictions to curry favour

with me.

² ¹ Gryllus. Hello, Odysseus.

Odysseus. And you too, Gryllus, for heaven's sake!

Gryllus. What do you want to ask?

Odysseus. Since I am aware that you have been men, I feel sorry for all of you in your present plight; yet it is only natural that I should be more concerned for those of you who were Greeks before you fell into this misfortune. So now I have asked Circe to remove the spell from any Greek who chooses and restore him to his original shape and let him go back home with us.

Gryllus. Stop, Odysseus! Not a word more! You see, we don't any of us think much of you either, for evidently it was a farce, that talk of your cleverness and your fame as one whose intelligence far surpassed the rest — a man who boggles at the simple matter of changing from worse to better because he hasn't considered the matter. For just as children dread the doctor's doses and run from lessons, the very things that, by changing them from invalids and fools, will make them healthier and wiser, just so you have shied away from the change from one shape to another. At this very moment you are not only living in fear and trembling as a companion of Circe, frightened that she may, before you know it, turn you into a pig or a wolf, but you are also trying to persuade us, who live in an abundance of good things, to abandon them, and with them the lady who provides them, and sail away with you, when we have again become men, the most unfortunate of all creatures!

Odysseus. To me, Gryllus, you seem to have lost not only your shape, but your intelligence also under the influence of that drug. You have become infected with strange and completely perverted notions. Or was it rather an inclination to swinishness that conjured you into this shape?

Gryllus. Neither of these, king of Cephallenians. But if it is your pleasure to discuss the matter instead of hurling abuse, I shall quickly make you see that we are right to prefer our present life in place of the former one, now that we have tried both.

Odysseus. Go on. I should like to hear you.

³ ¹ Gryllus. And I, in that case, to instruct you. Let us begin with the virtues, which, we note, inspire you with pride; for you rate yourselves as far superior to animals in justice and wisdom and courage and all the rest of them. But answer me this, wisest of men! Once I heard you telling Circe about the land of Cyclopes, that though it is not ploughed at all nor does anyone sow there, yet it is naturally so fertile and fecund that it produces spontaneously every kind of crops. ⁹⁸⁷ Do you, then, rate this land higher than rugged, goat-pasturing Ithaca, which barely yields the tiller a meagre, churlish, trifling crop after great efforts and much toil? And see that you don't lose your temper and give me a patriotic answer that isn't what you really believe.

Odysseus. I have no need to lie; for though I love and cherish my native soil more, the other wins my approval and admiration.

Gryllus. Then this, we shall say, is the situation: the wisest of men thinks fit to commend and approve one thing while he loves and prefers another. Now I assume that your answer applies to the spiritual field also, for the situation is the same as with the land: that spiritual soil is better which produces a harvest of virtue as a spontaneous crop without toil.

Odysseus. Yes, this too you may assume.

Gryllus. At this moment, then, you are conceding the point that the soul of beasts has a greater natural capacity and perfection for the generation of virtue; for without command or instruction, “unsown and unploughed,” as it were, it naturally brings forth and develops such virtue as is proper in each case.

Odysseus. And what sort of virtue, Gryllus, is ever found in beasts?

41 Gryllus. Ask rather what sort of virtue is not found in them more than in the wisest of men? Take first, if you please, courage, in which you take great pride, not even pretending to blush when you are called “valiant” and “sacker of cities.” Yet you, you villain, are the man who by tricks and frauds have led astray men who knew only a straightforward, noble style of war and were unversed in deceit and lies; while on your freedom from scruple you confer the name of the virtue that is least compatible with such nefariousness. Wild beasts, however, you will observe, are guileless and artless in their struggles, whether against one another or against you, and conduct their battles with unmistakably naked courage under the impulse of genuine valour. No edict summons them, nor do they fear a writ of desertion. No, it is their nature to flee subjection; with a stout heart they maintain an indomitable spirit to the very end. Nor are they conquered even when physically overpowered; they never give up in their hearts, even while perishing in the fray. In many cases, when beasts are dying, their valour withdraws together with the fighting spirit to some point where it is concentrated in one member and resists the slayer with convulsive movements and fierce anger until, like a fire, it is completely extinguished and departs.

Beasts never beg or sue for pity or acknowledge defeat: lion is never slave to lion, or horse to horse through cowardice, as man is to man when he unprotestingly accepts the name whose root is cowardice. And when men have subdued beasts by snares and tricks, such of them as are full grown refuse food and endure the pangs of thirst until they induce and embrace death in place of slavery. But nestlings and cubs, which by reason of age are tender and docile, are offered many beguiling allurements and enticements that act as drugs. These give them a taste for unnatural pleasures and modes of life, and in time make them spiritless to the point where they accept and submit to their so called “taming,” which is really Fan emasculation of their fighting spirit.

These facts make it perfectly obvious that bravery is an innate characteristic of beasts, while in human beings an independent spirit is actually contrary to nature. The point that best proves this, gentle Odysseus, is the fact that in beasts valour is naturally equal in both sexes and the female is in no way inferior to the male. She takes her part both in the struggle for existence and in the defence of her brood. You have heard, I

suppose, of the sow of Crommyon which, though a female beast, caused so much trouble to Theseus. ⁹⁸⁸ That famous Sphinx would have got no good of her wisdom as she sat on the heights of Mt. Phicium, weaving her riddles and puzzles, if she had not continued to surpass the Thebans greatly in power and courage. Somewhere thereabouts lived also the Teumesian vixen, a “thing atrocious”; and not far away, they say, was the Pythoness who fought with Apollo for the oracle at Delphi. Your king received Aethe from the Sicyusn as a recompense for excusing him from military service, making a very wise choice when he preferred a fine, spirited mare to a cowardly man. You yourself have often observed in panthers and lionesses that the female in no way yields to the male in spirit and valour. Yet, while you are off at the wars, your wife sits at home by the fire and troubles herself not so much as a swallow to ward off those who come against herself and her home — and this though she is a Spartan born and bred. So why should I go on to mention Carian or Maeonian women? Surely from what has been said it is perfectly obvious that men have no natural claim to courage; if they did, women would have just as great a portion of valour. It follows that your practice of courage is brought about by legal compulsion, which is neither voluntary nor intentional, but in subservience to custom and censure and moulded by extraneous beliefs and arguments. When you face toils and dangers, you do so not because you are courageous, but because you are more afraid of some alternative. For just as that one of your companions who is the first to board ship stands up to the light oar, not because he thinks nothing of it, but because he fears and shuns the heavier one; just so he who accepts the lash to escape the sword, or meets a foe in battle rather than be tortured or killed, does so not from courage to face the one situation, but from fear of the other. So it is clear that all your courage is merely the cowardice of prudence and all your valour merely fear that has the good sense to escape one course by taking another. And, to sum up, if you think that you are better in courage than beasts, why do your poets call the doughtiest fighters “wolf-minded” and “lion-hearted” and “like a boar in valour,” though no poet ever called a lion “man-hearted” or a boar “like a man in valour”? But, I imagine, just as when those who are swift are called “wind-footed” and those who are handsome are called “godlike,” there is exaggeration in the imagery; just so the poets bring in a higher ideal when they compare mighty warriors to something else. And the reason is that the spirit of anger is, as it were, the tempering or the cutting edge of courage. Now beasts use this undiluted in their contests, whereas you men have it mixed with calculation, as wine with water, so that it is displaced in the presence of danger and fails you when you need it most. Some of you even declare that anger should not enter at all into fighting, but be dismissed in order to make use of sober calculation; their contention is correct so far as self-preservation goes, but is disgracefully false as regards valorous defence. For surely it is absurd for you to find fault with Nature because she did not equip your bodies with natural stings, or place fighting tusks among your teeth, or give you nails like curved claws, while you yourselves remove or curb the emotional instrument that Nature has given.

51 Odysseus. Bless me, Gryllus, you must once have been a very clever sophist, one may judge, since even as things are, and speaking from your swinishness, you can attack the subject with such fervent ardour. But why have you failed to discuss temperance, the next in order?

Gryllus. Because I thought that you would first wish to take exception to what I have said. But you are eager to hear about temperance since you are the husband of a model of chastity and believe that you yourself have given a proof of self-control by rejecting the embraces of Circe. And in this you are no more continent than any of the beasts; for neither do they desire to consort with their betters, 989 but pursue both pleasure and love with mates of like species. So it is no wonder that, like the Mendesian goat in Egypt which, when shut up with many beautiful women, is said not to be eager to consort with them, but is far more excited about nannies, you likewise are contented with the kind of love that is familiar to you and, being a mortal, are not eager to sleep with a goddess. As for the chastity of Penelope, the cawing of countless crows will pour laughter and contempt upon it; for every crow, if her mate dies, remains a widow, not merely for a short time, but for nine generations of men. It follows that your fair Penelope is nine times inferior in chastity to any crow you please.

61 Now since you are not unaware that I am a sophist, let me marshal my arguments in some order by defining temperance and analysing the desires according to their kinds. Temperance, then, is a curtailment and an ordering of the desires that eliminate those that are extraneous or superfluous and discipline in modest and timely fashion those that are essential. You can, of course, observe countless differences in the desires . . . and the desire to eat and drink is at once natural and essential, while the pleasures of love, which, though they find their origin in nature, yet may be forgone and discarded without much inconvenience, have been called natural, but not essential. But there are desires of another kind, neither essential nor natural, that are imported in a deluge from without as a result of your inane illusions and because you lack true culture. So great is their multitude that the natural desires are, every one of them, all but overwhelmed, as though an alien rabble were overpowering the native citizenry. But beasts have souls completely inaccessible and closed to these adventitious passions and live their lives as free from empty illusions as though they dwelt far from the sea. They fall short in the matter of delicate and luxurious living, but solidly protect their sobriety and the better regulation of their desires since those that dwell within them are neither numerous nor alien.

Certainly there was a time when I myself, no less than you now, was dazzled by gold and held it to be an incomparable possession; so likewise I was caught by the lure of silver and ivory and the man who had most property of this sort seemed to me to be a blissful favourite of the gods, whether he was a Phrygian or a Carian, one more villainous than Dolon or more unfortunate than Priam. In that situation, constantly activated by these desires, I reaped no joy or pleasure from the other things of life, which I had sufficiently and to spare. El grumbled at my life, finding myself destitute of

the most important things and a loser in the lottery of fortune. This is the reason why, as I recall, when I saw you once in Crete tricked out in holiday attire, it was not your intellect or your virtue that I envied, but the softness of the elegantly woven garment and the beautiful wool of your purple cloak that I admired and gaped at (the clasp, I believe, was of gold and had some frivolity worked on it in exquisitely fine intaglio). I followed you about as enchanted as a woman. But now I am rid and purified of all those empty illusions. I have no eyes for gold and silver and can pass them by just like any common stone; and as for your fine robes and tapestries, I swear there's nothing sweeter for me to rest in when I'm full than deep, soft mud. None, then, of such adventitious desires has a place in our souls; our life for the most part is controlled by the essential desires and pleasures. As for those that are non-essential, but merely natural, we resort to them without either irregularity or excess.

7 1 990 Let us, in fact, first describe these pleasures. Our pleasure in fragrant substances, those that by their nature stimulate our sense of smell, besides the fact that our enjoyment of this is simple and costs nothing, also contributes to utility by providing a way for us to tell good food from bad. For the tongue is said to be, and is, a judge of what is sweet or bitter or sour, when liquid flavours combine and fuse with the organ of taste; but our sense of smell, even before we taste, is a judge that can much more critically distinguish the quality of each article of food than any royal taster in the world. It admits what is proper, rejects what is alien, and will not let it touch or give pain to the taste, but informs on and denounces what is bad before any harm is done. And in other respects smell is no nuisance to us, as it is to you, forcing to collect and mix together incense of one kind or another and cinnamon and nard and malobathrum and Arabian aromatic reeds, with the aid of a formidable dyer's or witch's art, of the sort to which you give the name of unguentation, thus buying great price an effeminate, emasculating luxury which has absolutely no real use. Yet, though such is its nature, it has depraved not only every woman, but lately the greater part of men as well, so that they refuse to sleep even with their own wives unless they come to bed Creeking with myrrh and scented powders. But sows attract boars and nannies bucks and other female creatures their consorts by means of their own special odours; scented, as they are, with pure dew and grassy meadows, they are attracted to the nuptial union by mutual affection. The females are not coy and do not cloak their desires with deceits or trickeries or denials; nor do the males, driven on by the sting of mad lust, purchase the act of procreation by money or toil or servitude. No! Both parties celebrate at the proper time a love without deceit or hire, a love which in the season of spring awakens, like the burgeoning of plants and trees, the desire of animals, and then immediately extinguishes it. Neither does the female continue to receive the male after she has conceived, nor does the male attempt her. So slight and feeble is the regard we have for pleasure: our whole concern is with Nature. Whence it comes about that to this very day the desires of beasts have encompassed no homosexual mating. But you have a fair amount of such trafficking among your high and mighty nobility, to say nothing of the

baser sort. Agamemnon came to Boeotia hunting for Argynnus, who tried to elude him, and slandering the sea and winds . . . then he gave his noble self a noble bath in Lake Copaïs to drown his passion there and get rid of his desire. Just so Heracles, pursuing a beardless lad, lagged behind the other heroes and deserted the expedition. On the Rotunda of Ptoian Apollo one of your men secretly inscribed fair is Achilles — when Achilles already had a son. And I hear that the inscription is still in place. But a cock that mounts another for the lack of a female is burned alive because some prophet or seer declares that such an event is an important and terrible omen. On this basis even men themselves acknowledge that beasts have a better claim to temperance and the non-violation of nature in their pleasures. Not even Nature, with Law for her ally, can keep within bounds the unchastened vice of your hearts; but as though swept by the current of their lusts beyond the barrier at many points, men do such deeds as wantonly outrage Nature, upset her order, and confuse her distinctions. For men have, in fact, attempted to consort with goats and sows and mares, and women have gone mad with lust for male beasts. ⁹⁹¹ From such unions your Minotaurs and Aegipans, and, I suppose, your Sphinxes and Centaurs have arisen. Yet it is through hunger that dogs have occasionally eaten a man; and birds have tasted of human flesh through necessity; but no beast has ever attempted a human body for lustful reasons. But the beasts I have mentioned and many others have been victims of the violent and lawless lusts of man.

⁸¹ Though men are so vile and incontinent where the desires I have spoken of are concerned, they can be proved to be even more so in the case of essential desires, being here far inferior to animals in temperance. These are the desires for food and drink, in which we beasts always take our pleasure along with some sort of utility; whereas you, in your pursuit of pleasure rather than natural nourishment, are punished by many serious ailments which, welling up from one single source, the surfeit of your bodies, fill you with all manner of flatulence that is difficult to purge. In the first place each species of animal has one single food proper to it, grass or some root or fruit. Those that are carnivorous resort to no other kind of nourishment, nor do they deprive those weaker than themselves of sustenance; but the lion lets the deer, and the wolf lets the sheep, feed in its natural pasture. But man in his pleasures is led astray by gluttony to everything edible; he tries and tastes everything as if he had not yet come to recognize what is suitable and proper for him; alone of all creatures he is omnivorous.

In the first place his eating of flesh is caused by no lack of means or methods, for he can always in season harvest and garner and gather in such a succession of plants and grains as will all but tire him out with their abundance; but driven on by luxurious desires and satiety with merely essential nourishment, he pursues illicit food, made unclean by the slaughter of beasts; and he does this in a much more cruel way than the most savage beasts of prey. Blood and gore and raw flesh are the proper diet of kite and wolf and snake; to man they are an appetizer. Then, too, man makes use of every kind of food and does not, like beasts abstain from most kinds and consequently make war on a

few only that he must have for food. In a word, nothing that flies or swims or moves on land has escaped your so called civilized and hospitable tables.

91 Well, then. It is admitted that you use animals as appetizers to sweeten your fare. Why, therefore . . . Animal intelligence, on the contrary, allows no room for useless and pointless arts; and in the case of essential ones, Ewe do not make one man with constant study cling to one department of knowledge and rivet him jealously to that; nor do we receive our arts as alien products or pay to be taught them. Our intelligence produces them on the spot unaided, as its own congenital and legitimate skills. I have heard that in Egypt everyone is a physician; and in the case of beasts each one is not only his own specialist in medicine, but also in the providing of food, in warfare and hunting as well as in self-defence and music, in so far as any kind of animal has a natural gift for it. From whom have we swine learned, when we are sick, to resort to rivers to catch crabs? Who taught tortoises to devour marjoram after eating the snake? And who instructed Cretan goats, when they are pierced by an arrow, to look for dittany, after eating which the arrowhead falls out? For if you speak the truth and say that Nature is their teacher, you are elevating the intelligence of animals to the most sovereign and wisest of first principles. If you do not think that it should be called either reason or intelligence, it is high time for you to cast about for some fairer and even more honourable term to describe it, since certainly the faculty that it brings to bear in action is better and more remarkable. 992 It is no uninstructed or untrained faculty, but rather self-taught and self-sufficient — and not for lack of strength. It is just because of the health and completeness of its native virtue that it is indifferent to the contributions to its intelligence supplied by the lore of others. Such animals, at any rate, as man for amusement or easy living induces to accept instruction and training have understanding to grasp what they are taught even when it goes contrary to their physical endowment, so superior are their mental powers. I say nothing of puppies that are trained as hunters, or colts schooled to keep time in their gait, or crows that are taught to talk, or dogs, to jump through revolving hoops. In the theatres horses and steers go through an exact routine in which they lie down or dance or hold a precarious pose or perform movements not at all easy even for men; and they remember what they have been taught, these exhibitions of docility which are not in the least useful for anything else. If you are doubtful that we can learn arts, then let me tell you that we can even teach them. When partridges are making their escape, they accustom their fledglings to hide by falling on their backs and holding a lump of earth over themselves with their claws. You can observe storks on the roof, the adults showing the art of flying to the young as they make their trial flights. Nightingales set the example for their young to sing; while nestlings that are caught young and brought up by human care are poorer singers, as though they had left the care of their teacher too early. . . . and since I have entered into this new body of mine, I marvel at those arguments by which the sophists brought me to consider all creatures except man irrational and senseless.

Odysseus. So now, Gryllus, you are transformed. Do you attribute reason even to the sheep and the ass?

Gryllus. From even these, dearest Odysseus, it is perfectly possible to gather that animals have a natural endowment of reason and intellect. For just as one tree is not more nor less inanimate than another, but they are all in the same state of insensibility, since none is endowed with soul, in the same way one animal would not be thought to be more sluggish or indocile mentally than another if they did not all possess reason and intellect to some degree — though some have a greater or less proportion than others. Please note that cases of dullness and stupidity in some animals are demonstrated by the cleverness and sharpness of others — as when you compare an ass and a sheep with a fox or a wolf or a bee. It is like comparing Polyphemus to you or that dunce Coroebus to your grandfather Autolycus. I scarcely believe that there is such a spread between one animal and another as there is between man and man in the matter of judgement and reasoning and memory.

Odysseus. But consider, Gryllus: it is not a fearful piece of violence to grant reason to creatures that have no inherent knowledge of God?

Gryllus. Then shall we deny, Odysseus, that so wise and remarkable a man as you had Sisyphus for a father?

On the eating of flesh



I

993 1 1 Can you really ask what reason Pythagoras had for abstaining from flesh? For my part I rather wonder both by what accident and in what state of soul or mind the first man who did so, touched his mouth to gore and brought his lips to the flesh of a dead creature, he who set forth tables of dead, stale bodies and ventured to call food and nourishment the parts that had a little before bellowed and cried, moved and lived. How could his eyes endure the slaughter when throats were slit and hides flayed and limbs torn from limb? How could his nose endure the stench? How was it that the pollution did not turn away his taste, which made contact with the sores of others and sucked juices and serums from mortal wounds?

The skins shivered; and upon the spits the flesh bellowed,
Both cooked and raw; the voice of kine was heard.

Though this is an invention and a myth, yet that sort of dinner is really portentous — when a man craves the meat that is still bellowing, giving instructions which tell us on what animals we are to feed while they are still alive uttering their cries, and organizing various methods of seasoning and roasting and serving. It is the man who first began these practices that one should seek out, not him who all too late desisted.

2 1 Or would everyone declare that the reason for those who first instituted flesh-eating was the necessity of their poverty? It was not while they passed their time in lawful desires nor when they had necessities in abundance that after indulgence in unnatural and antisocial pleasures they resorted to such a practice. If, at this moment, they could recover feelings and voice, they might, indeed, remark: “Oh blessed and beloved of the gods, you who live now, what an age has fallen to your lot wherein you enjoy and assimilate a heritage abounding in good things! How many plants grow for you! What vintages you gather! What wealth you may draw from the plains and what pleasant sustenance from trees! Why, you may even live luxuriously without the stain of blood. But as for us, it was a most dismal and fearful portion of the world’s history the confronted us, falling as we did into great and unbearable poverty brought on by our first appearance among the living. As yet the heavens and the stars were concealed by dense air that was contaminated with turbid moisture, not easily to be penetrated, and fire and furious wind. Not yet was the sun established undeviating

In his firm course,

Dividing day and night; he brought them back
Again and crowned them with the fruitful hours
All wreathed with bloom, while violence

had been done to earth by rivers pouring forth their floods at random and most parts were deformed by pools. Earth was made a wilderness by deep quagmires and the unfruitful growth of thickets and forests; nor was there as yet any agricultural production or professional tool or any resource of skill. Our hunger gave us no respite nor was there any seed at that time awaiting the annual season of sowing. What wonder if, contrary to nature, we made use of the flesh of beasts when even mud was eaten and the bark of trees devoured, and to light on sprouting grass or the root of a rush was a piece of luck? When we had tasted and eaten acorns we danced for joy around some oak, calling it “life-giving” and “mother” and “nurse.”⁹⁹⁴ This was the only festival that those times had discovered; all else was a medley of anguish and gloom. But you who live now, what madness, what frenzy drives you to the pollution of shedding blood, you who have such a superfluity of necessities? Why slander the earth by implying that she cannot support you? Why impiously offend law-giving Demeter and bring shame upon Dionysus, lord of the cultivated vine, the gracious one, as if you did not receive enough from their hands? Are you not ashamed to mingle domestic crops with blood and gore? You call serpents and panthers and lions savage, but you yourselves, by your own foul slaughters, leave them no room to outdo you in cruelty; for their slaughter is their living, yours is a mere appetizer.”

³¹ It is certainly not lions and wolves that we eat out of self-defence; on the contrary, we ignore these and slaughter harmless, tame creatures without stings or teeth to harm us, creatures that, I swear, Nature appears to have produced for the sake of their beauty and grace. . . .

[It is as though one, seeing the Nile overflow its banks, filling the landscape with its fertile and productive stream, should not marvel at this, its nourishing of plants and its fruitfulness in such crops as are most to be cultivated and contribute most to the support of life, but should espy a crocodile swimming there somewhere or an asp being swept along or a thousand other savage creatures and should cite them as the reasons for his censure and this compulsion to do as he does. Or, I swear, it is as though one fixed one's gaze on this land and its soil covered with cultivated crops and heavy with ears of wheat, and then, looking beneath these rich harvests, one were to catch sight somewhere of a growth of darnel or broom-rage and, without more ado, ceasing to reap the benefit and claim the booty of the good crops, burst into a tirade about the weeds. Another example: if one should see an orator making a speech at some trial where he was advocate, a speech in which his eloquence in full flood was advancing to the succour of someone in jeopardy or (so help me) to the conviction or denunciation of rash acts or defaults — a flood of eloquence not simple or jejune, but charged with many (or rather all kinds of) emotional appeals for the simultaneous influencing of the many different

kinds of minds in the audience or jury, which must either be roused and won over or (by heaven!) soothed and made gentle and calm — then if one neglected to observe and take into account this main point and issue of the matter, but merely picked out flaws of style that the flood of oratory, as it moved to its goal, had swept along by the momentum of its current, flaws that came rushing out and slipped by with the rest — and seeing . . . of some popular leader. . . .]

4 1 But nothing abashed us, not the flower-like tinting of the flesh, not the persuasiveness of the harmonious voice, not the cleanliness of their habits or the unusual intelligence that may be found in the poor wretches. No, for the sake of a little flesh we deprive them of sun, of light, of the duration of life to which they are entitled by birth and being. Then we go on to assume that when they utter cries and squeaks their speech is inarticulate, that they do not, begging for mercy, entreating, seeking justice, each one of them say, “I do not ask to be spared in case of necessity; only spare me your arrogance! Kill me to eat, but not to please your palate!” Oh, the cruelty of it! What a terrible thing it is to look on when the tables of the rich are spread, men who employ cooks and spicers to groom the dead! And it is even more terrible to look on when they are taken away, for more is left than has been eaten. So the beasts died for nothing! There are others who refuse when the dishes are already set before them and will not have them cut into or sliced. Though they bid spare the dead, they did not spare the living.

5 1 We declare, then, that it is absurd for them to say that the practice of flesh-eating is based on Nature. For that man is not naturally carnivorous is, in the first place, obvious from the structure of his body. A man’s frame is in no way similar to those creatures who were made for flesh-eating: he has no hooked beak or sharp nails or jagged teeth, 995 no strong stomach or warmth of vital fluids able to digest and assimilate a heavy diet of flesh. It is from this very fact, the evenness of our teeth, the smallness of our mouths, the softness of our tongues, our possession of vital fluids too inert to digest meat that Nature disavows our eating of flesh. If you declare that you are naturally designed for such a diet, then first kill for yourself what you want to eat. Do it, however, only through your own resources, unaided by cleaver or cudgel or any kind of axe. Rather, just as wolves and bears and lions themselves slay what they eat, so you are to fell an ox with your fangs or a boar with your jaws, or tear a lamb or hare in bits. Fall upon it and eat it still living, as animals do. But if you wait for what you eat to be dead, if you have qualms about enjoying the flesh while life is still present, why do you continue, contrary to nature, to eat what possesses life? Even when it is lifeless and dead, however, no one eats the flesh just as it is; men boil it and roast it, altering it by fire and drugs, recasting and diverting and smothering with countless condiments the taste of gore so that the palate may be deceived and accept what is foreign to it.

It was, indeed, a witty remark of the Spartan who bought a little fish in an inn and gave it to the innkeeper to prepare. When the latter asked for cheese and vinegar and oil, the Spartan said, “If I had those, I should not have bought a fish. “But we are so

refined in our blood-letting that we term flesh a supplementary food; and then we need “supplements” for the flesh itself, mixing oil, wine, honey, fish paste, vinegar, with Syrian and Arabian spices, as though we were really embalming a corpse for burial. The fact is that meat is so softened and dissolved and, in a way, predigested that it is hard for digestion to cope with it; and if digestion loses the battle, the meats affect us with dreadful pains and malignant forms of indigestion.

6 1 Diogenes ventured to eat a raw octopus in order to put an end to the inconvenience of preparing cooked food. In the midst of a large throng he veiled his head and, as he brought the flesh to his mouth, said, “It is for you that I am risking my life.” Good heavens, a wondrous fine risk! Just like Pelopidas for the liberty of the Thebans or Harmodius and Aristogiton for that of the Athenians, this philosopher risked his life struggling with a raw octopus — in order to brutalize our lives!

Note that the eating of flesh is not only physically against nature, but it also makes us spiritually coarse and gross by reason of satiety and surfeit. “For wine and indulgence in meat make at body strong and vigorous, but the soul weak.” And in order that I may not offend athletes, I shall take my own people as examples. It is a fact that the Athenians used to call us Boeotians beef-witted and insensitive and foolish, precisely because we stuffed ourselves. “These men are swine”; . . . and Menander says, “Who have jaws; and Pindar “And then to learn . . .”; “A dry soul is wisest” according to Heraclitus. Empty jars make a noise when struck, but full ones do not resound to blows. Thin bronze objects will pass the sounds from one to another in a circle until you dampen and deaden the noise with your hand as the beat goes round. The eye when it is flooded by an excess of moisture grows dim and weakened for its proper task. When we examine the sun through dank atmosphere and a fog of gross vapours, we do not see it clear and bright, but submerged and misty, with elusive rays. In just the same way, then, when the body is turbulent and surfeited and burdened with improper food, the lustre and light of the soul inevitably come through it blurred and confused, aberrant and inconstant, since the soul lacks the brilliance and intensity to penetrate to the minute and obscure issues of active life.

7 1 But apart from these considerations, do you not find here a wonderful means of training in social responsibility? Who could wrong a human being ⁹⁹⁶ when he found himself so gently and humanely disposed ^atoward other non-human creatures? Two days ago in a discussion I quoted the remark of Xenocrates, that the Athenians punished the man who had flayed a ram while it was still alive; yet, as I think, he who tortures a living creature is no worse than he who slaughters it outright. But it seems that we are more observant of acts contrary to convention than of those that are contrary to nature. In that place, then, I made my remarks in a popular vein. I still hesitate, however, to attempt a discussion of the principle underlying my opinion, great as it is, and mysterious and incredible, as Plato says, with merely clever men of mortal opinions, just as a steersman hesitates to shift his course in the midst of a storm, or a playwright to raise his god from the machine in the midst of a play. Yet perhaps it is not unsuitable to

set the pitch and announce the theme by quoting some verses of Empedocles. . . . By these lines he means, though he does not say so directly, that human souls are imprisoned in mortal bodies as a punishment for murder, the eating of animal flesh, and cannibalism. This doctrine, however, seems to be even older, for the stories told about the sufferings and dismemberment of Dionysus and the outrageous assaults of the Titans upon him, and their punishment and blasting by thunderbolt after they had tasted his blood — all this is a myth which in its inner meaning has to do with rebirth. For to that faculty in us which is unreasonable and disordered and violent, and does not come from the gods, but from evil spirits, the ancients gave the name Titans, that is to say, those that are punished and subjected to correction. . . .

II

⁹⁹⁶ 1 1 Reason urges us with fresh ideas and fresh zeal to attack again our yesterday's discourse on the eating of flesh. It is indeed difficult, as Cato remarked, to talk to bellies which have no ears. And the potion of familiarity has been drunk, like that of Circe

Commingling pains and pangs, tricks and tears;

nor is it easy to extract the hook of flesh-eating, entangled as it is and embedded in the love of pleasure. And, like the Egyptians who extract the viscera of the dead and cut them open in view of the sun, then throw them away as being the cause of every single sin that the man had committed, it would be well for us to excise our own gluttony and lust to kill and become pure for the remainder of our lives, since it is not so much our belly that drives us to the pollution of slaughter; it is itself polluted by our incontinence. Yet if, for heavens sake, it is really impossible for us to be far from error because we are on such terms of familiarity with it, let us at least be ashamed of our ill doing and resort to it only in reason. We shall eat flesh, but from hunger, not as a luxury. We shall kill an animal, but in pity and sorrow, not degrading or torturing it — which is the current practice in many cases, some thrusting red-hot spits into the throats of swine ⁹⁹⁷ so that by the plunging in of the iron the blood may be emulsified and, as it circulates through the body, may make the flesh tender and delicate. Others jump upon the udders of sows about to give birth and kick them so that, when they have blended together blood and milk and gore (Zeus the Purifier!) and the unborn young have at the same time been destroyed at the moment of birth, they may eat the most inflamed part of the creature. Still others sew up the eyes of cranes and swans, shut them up in darkness and fatten them, making flesh appetizing with strange compounds and spicy mixtures.

² 1 From these practices it is perfectly evident that it is not for nourishment or need or necessity, but out of satiety and insolence and luxury that they have turned this lawless custom into a pleasure. Then, just as with women who are insatiable in seeking pleasure, their lust tries everything, goes astray, and explores the gamut of profligacy

until at last it ends in unspeakable practices; so intemperance in eating passes beyond the necessary ends of nature and resorts to cruelty and lawlessness to give variety to appetite. For it is in their own company that organs of sense are infected and won over and become licentious when they do not keep to natural standards. Just so the art of hearing has fallen sick, corrupting musical taste. From this our luxury and debauchery conceives a desire for shameful caresses and effeminate titillations. These taught the sight not to take pleasure in warlike exercises or gesticulations or refined dances or statues and paintings, but to regard the slaughter and death of men, their wounds and combats, as the most precious sort of spectacle. Just so intemperate intercourse follows a lawless meal, inharmonious music follows a shameful debauch, barbarous spectacles follow shameless songs and sounds, insensitivity and cruelty toward human kind follow savage exhibitions in the theatre. It was for this reason that the godlike Lycurgus gave directions in certain rhetrae that the doors and roofs of houses should be fashioned by saw and axe alone and no other tool should be used — not of course because he had a quarrel with gimlets and adzes and other instruments for delicate work. It was because he knew that through such rough-hewn work you will not be introducing a gilded couch, nor will you be so rash as to bring silver tables and purple rugs and precious stones into a simple house. The corollary of such a house and couch and table and cup is a dinner which is unpretentious and a lunch which is truly democratic; but all manner of luxury and extravagance follow the lead of an evil way of life

As new-weaned foal beside his mother runs.

31 For what sort of dinner is not costly for which a living creature loses its life? Do we hold a life cheap? I do not yet go so far as to say that it may well be the life of your mother or father or some friend or child, as Empedocles declared. Yet it does, at least, possess some perception, hearing, seeing, imagination, intelligence, which last every creature receives from Nature to enable it to acquire what is proper for it and to evade what is not. Do but consider which are the philosophers who serve the better to humanize us: those who bid us eat our children and friends and fathers and wives after their death, or Pythagoras and Empedocles who try to accustom us to act justly towards other creatures also? You ridicule a man who abstains from eating mutton. But are we, they will say, to refrain from laughter when we see you slicing off portions from a dead father or mother and sending them to absent friends and inviting those who are at hand, heaping their plates with flesh? But as it is, perhaps we commit a sin when we touch these books of theirs without cleansing our hands and faces, our feet and ears — unless, by Heaven, it is a purification of those members to speak on such a subject as this, “washing,” as Plato says, 998 “the brine from one’s ears with the fresh water of discourse.” If one should compare these two sets of books and doctrines, the former may serve as philosophy for the Scyths and Sogdians and the Black Cloaks, whose story as told by Herodotus gains no credit; but the precepts of Pythagoras and Empedocles were the laws for the ancient Greeks, along with their diet of wheat. . . . [Because there is no question of justice between us and the irrational animals.]

41 Who, then, were they who later decreed this?

The first to forge the highway's murderous sword,
And first to eat the flesh of ploughing ox.

This is the way, you may be sure, in which tyrants begin their course of bloody slaughters. Just as, for instance, at Athens they put to death initially the worst of the sycophants, and likewise in the second and third instances; but next, having become accustomed to bloodshed they allowed Niceratus, the son of Nicias, to be killed and the general Theramenes and the philosopher Polemarchus. Just so, at the beginning it was some wild and harmful animal that was eaten, then a bird or fish that had its flesh torn. And so when our murderous instincts had tasted blood and grew practised on wild animals, they advanced to the labouring ox and the well-behaved sheep and the housewarding cock; thus, little by little giving a hard edge to our insatiable appetite, we have advanced to wars and the slaughter and murder of human beings. Yet if someone once demonstrates that souls in their rebirths make use of common bodies and that what is now rational reverts to the irrational, and again what is now wild becomes tame, and that Nature changes everything and assigns new dwellings

Clothing souls with unfamiliar coat of flesh;

will not this deter the unruly element in those who have adopted the doctrine from implanting disease and indigestion in our bodies and perverting our souls to an ever more cruel lawlessness, as soon as we are broken of the habit of not entertaining a guest or celebrating a marriage or consorting with our friends without bloodshed and murder?

51 Yet even if the argument of the migration of souls from body to body is not demonstrated to the point of complete belief, there is enough doubt to make us quite cautious and fearful. It is as though in a clash of armies by night you had drawn your sword and were rushing at a man whose fallen body was hidden by his armour and should hear someone remarking that he wasn't quite sure, but that he thought and believed that the prostrate figure was that of your son or brother or father or tent-mate — which would be the better course: to approve a false suspicion and spare your enemy as your friend, or to disregard an uncertain authority and kill your friend as your foe? The latter course you will declare to be shocking. Consider also Merope in the play raising her axe against her son himself because she believes him to be that son's murderer and saying

This blow I give you is more costly yet —

what a stir she rouses in the theatre as she brings them to their feet in terror lest she wound the youth before the old man can stop her! Now suppose one old man stood beside her saying, "Hit him! He's your enemy," and another who said, "Don't strike! He is your son": which would be the greater misdeed, to omit the punishment of an enemy because of the son, or to slay a child under the impulse of anger against an enemy? In a case, then, where it is not hate or anger or self-defence or fear for ourselves that induces us to murder, but the motive of pleasure, and the victim stands there under our power with its head bent back and one of our philosophers says, "Kill it! It's only a

brute beast”; but the other says, “Stop! What if the soul of some relative or friend has found its way into this body?” — Good God! Of course the risk is equal or much the same in the two cases — if I refuse to eat flesh, or if I, disbelieving, kill my child or some other relative!

999 6 1 There remains yet another contention with the Stoics about flesh-eating, and this is not “equal,” either. For what is this great “tension” on the belly and the kitchen? Why, when they count pleasure effeminate and denounce it as being neither a good nor an “advanced principle “ nor “commensurate with Nature,” are they so concerned with these pleasures? It would certainly be consistent for them, since they banish perfume and cakes from their banquets, to be more squeamish about blood and flesh. But as it is, confining as it were their philosophy to their ledgers, they economize on their dinners in trivial and needless details while they do not deprecate this inhuman and murderous item of expense. “Of course,” they say, “we human beings have no compact of justice with irrational animals.” Nor, one might reply, have you with perfume or exotic sweetmeats either. Refrain from animals also, if you are expelling the useless and unnecessary element in pleasure from all its lurking-places.

7 1 Let us, however, now examine the point whether we really have no compact of justice with animals; and let us do so in no artificial or sophistical manner, but fixing our attention on our own emotions and conversing like human beings with ourselves and weighing . . .